Categories

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This chapter contains some elementary material about categories, functors, and natural transformations. Notably, we discuss and explore:

1. Categories (Section 11.1).
2. Examples of categories (Section 11.2).
3. The quadruple adjunction $\pi_0 \dashv (-)_{\mathrm{disc}} \dashv \mathrm{Obj} \dashv (-)_{\mathrm{indisc}}$ between the category of categories and the category of sets (Section 11.3).
4. Groupoids, categories in which all morphisms admit inverses (Section 11.4).
5. Functors (Section 11.5).
6. The conditions one may impose on functors in decreasing order of importance:
   (a) Section 11.6 introduces the foundationally important conditions one may impose on functors, such as faithfulness, conservativity, essential surjectivity, etc.
   (b) Section 11.7 introduces more conditions one may impose on functors that are still important but less omni-present than those of Section 11.6, such as being dominant, being a monomorphism, being pseudomonic, etc.
   (c) Section 11.8 introduces some rather rare or uncommon conditions one may impose on functors that are nevertheless still useful to explicit record in this chapter.
7. Natural transformations (Section 11.9).

8. The various categorical and 2-categorical structures formed by categories, functors, and natural transformations (Section 11.10).

Contents

11.1 Categories **4**
11.1.1 Foundations 4
11.1.2 Subcategories 6
11.1.3 Skeletons of Categories 8
11.1.4 Precomposition and Postcomposition 9

11.2 Examples of Categories **12**
11.2.1 The Empty Category 12
11.2.2 The Punctual Category 12
11.2.3 Monoids as One-Object Categories 13
11.2.4 Ordinal Categories 14
11.2.5 The Walking Arrow 15
11.2.6 More Examples of Categories 16
11.2.7 Posetal Categories 16

11.3 The Quadruple Adjunction With Sets **18**
11.3.1 Statement 18
11.3.2 Connected Components and Connected Categories 19
11.3.3 Discrete Categories 22
11.3.4 Indiscrete Categories 25

11.4 Groupoids **28**
11.4.1 Isomorphisms 28
11.4.2 Groupoids 28
11.4.3 The Groupoid Completion of a Category 29
11.4.4 The Core of a Category 33

11.5 Functors **37**
11.5.1 Foundations 37
11.5.2 Contravariant Functors 42
11.5.3 Forgetful Functors 44
11.5.4 The Natural Transformation Associated to a Functor 46

11.6 Conditions on Functors **49**
11.6.1 Faithful Functors 49
11.6.2 Full Functors 53
11.6.3 Fully Faithful Functors 61
11.6.4 Conservative Functors 66
11.6.5 Essentially Injective Functors 67
11.6.6 Essentially Surjective Functors 68
11.6.7 Equivalences of Categories 69
11.6.8 Isomorphisms of Categories 72

11.7 More Conditions on Functors **74**
11.7.1 Dominant Functors 74
11.7.2 Monomorphisms of Categories 75
11.7.3 Epimorphisms of Categories 77
11.7.4 Pseudomonic Functors 78
11.7.5 Pseudoepic Functors 81

11.8 Even More Conditions on Functors **84**
11.8.1 Injective on Objects Functors 84
11.8.2 Surjective on Objects Functors 85
11.8.3 Bijective on Objects Functors 86
11.8.4 Functors Representably Faithful on Cores 86
11.8.5 Functors Representably Full on Cores 87
11.8.6 Functors Representably Fully Faithful on Cores 88
11.8.7 Functors Corepresentably Faithful on Cores 90
11.8.8 Functors Corepresentably Full on Cores 90
11.8.9 Functors Corepresentably Fully Faithful on Cores 92

11.9 Natural Transformations **93**
11.9.1 Transformations 93
11.9.2 Natural Transformations 94
11.9.3 Examples of Natural Transformations 95
11.9.4 Vertical Composition of Natural Transformations 97
11.9.5 Horizontal Composition of Natural Transformations 101
11.9.6 Properties of Natural Transformations 108
11.9.7 Natural Isomorphisms 109

11.10 Categories of Categories **111**
11.10.1 Functor Categories 111

11.10.2 The Category of Categories and Functors 115
11.10.3 The 2-Category of Categories, Functors, and Natural Transformations 117
11.10.4 The Category of Groupoids 119
11.10.5 The 2-Category of Groupoids 119

A Other Chapters 120

11.1 Categories

11.1.1 Foundations

Definition 11.1.1.1.1 ▶ Categories

A **category** $(\mathcal{C}, \circ^{\mathcal{C}}, \mathbb{1}^{\mathcal{C}})$ consists of:

- *Objects.* A class $\mathrm{Obj}(\mathcal{C})$ of **objects**.
- *Morphisms.* For each $A, B \in \mathrm{Obj}(\mathcal{C})$, a class $\mathrm{Hom}_{\mathcal{C}}(A, B)$, called the **class of morphisms of $\mathcal{C}$ from A to B**.
- *Identities.* For each $A \in \mathrm{Obj}(\mathcal{C})$, a map of sets

$$\mathbb{1}^{\mathcal{C}}_A \colon \mathrm{pt} \longrightarrow \mathrm{Hom}_{\mathcal{C}}(A, A),$$

  called the **unit map of $\mathcal{C}$ at A**, determining a morphism

$$\mathrm{id}_A \colon A \longrightarrow A$$

  of $\mathcal{C}$, called the **identity morphism of A**.
- *Composition.* For each $A, B, C \in \mathrm{Obj}(\mathcal{C})$, a map of sets

$$\circ^{\mathcal{C}}_{A,B,C} \colon \mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(A, C),$$

  called the **composition map of $\mathcal{C}$ at (A, B, C)**.

such that the following conditions are satisfied:

1. *Associativity.* The diagram

$$\begin{tikzcd}
 & \mathrm{Hom}_{\mathcal{C}}(C,D) \times (\mathrm{Hom}_{\mathcal{C}}(B,C) \times \mathrm{Hom}_{\mathcal{C}}(A,B)) \arrow[dr, "\mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(C,D)} \times \circ^{\mathcal{C}}_{A,B,C}"] & \\
(\mathrm{Hom}_{\mathcal{C}}(C,D) \times \mathrm{Hom}_{\mathcal{C}}(B,C)) \times \mathrm{Hom}_{\mathcal{C}}(A,B) \arrow[ur, dashed, "\alpha^{\mathsf{Sets}}_{\mathrm{Hom}_{\mathcal{C}}(C,D),\mathrm{Hom}_{\mathcal{C}}(B,C),\mathrm{Hom}_{\mathcal{C}}(A,B)}", "\sim"'] \arrow[d, "\circ^{\mathcal{C}}_{B,C,D} \times \mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(A,B)}"'] & & \mathrm{Hom}_{\mathcal{C}}(C,D) \times \mathrm{Hom}_{\mathcal{C}}(A,C) \arrow[d, "\circ^{\mathcal{C}}_{A,C,D}"] \\
\mathrm{Hom}_{\mathcal{C}}(B,D) \times \mathrm{Hom}_{\mathcal{C}}(A,B) \arrow[rr, "\circ^{\mathcal{C}}_{A,B,D}"'] & & \mathrm{Hom}_{\mathcal{C}}(A,D)
\end{tikzcd}$$

commutes, i.e. for each composable triple (f, g, h) of morphisms of $\mathcal{C}$, we have

$$(f \circ g) \circ h = f \circ (g \circ h).$$

2. *Left Unitality.* The diagram

$$\begin{tikzcd}
\mathrm{pt} \times \mathrm{Hom}_{\mathcal{C}}(A,B) \arrow[d, "\mathbb{1}^{\mathcal{C}}_{B} \times \mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(A,B)}"'] \arrow[dr, dashed, "\lambda^{\mathsf{Sets}}_{\mathrm{Hom}_{\mathcal{C}}(A,B)}", "\sim"'] & \\
\mathrm{Hom}_{\mathcal{C}}(B,B) \times \mathrm{Hom}_{\mathcal{C}}(A,B) \arrow[r, "\circ^{\mathcal{C}}_{A,B,B}"'] & \mathrm{Hom}_{\mathcal{C}}(A,B)
\end{tikzcd}$$

commutes, i.e. for each morphism $f\colon A \longrightarrow B$ of $\mathcal{C}$, we have

$$\mathrm{id}_B \circ f = f.$$

3. *Right Unitality.* The diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\mathcal{C}}(A,B) \times \mathrm{pt} \arrow[d, "\mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(A,B)} \times \mathbb{1}^{\mathcal{C}}_{A}"'] \arrow[dr, dashed, "\rho^{\mathsf{Sets}}_{\mathrm{Hom}_{\mathcal{C}}(A,B)}", "\sim"'] & \\
\mathrm{Hom}_{\mathcal{C}}(A,B) \times \mathrm{Hom}_{\mathcal{C}}(A,A) \arrow[r, "\circ^{\mathcal{C}}_{A,A,B}"'] & \mathrm{Hom}_{\mathcal{C}}(A,B)
\end{tikzcd}$$

commutes, i.e. for each morphism $f\colon A \longrightarrow B$ of $\mathcal{C}$, we have

$$f \circ \mathrm{id}_A = f.$$

Notation 11.1.1.1.2 ► Further Notation for Morphisms in Categories

Let $\mathcal{C}$ be a category.

1. We also write $\mathcal{C}(A, B)$ for $\mathrm{Hom}_{\mathcal{C}}(A, B)$.
2. We write $\mathrm{Mor}(\mathcal{C})$ for the class of all morphisms of $\mathcal{C}$.

Definition 11.1.1.1.3 ► Size Conditions on Categories

Let κ be a regular cardinal. A category $\mathcal{C}$ is

1. **Locally small** if, for each $A, B \in \mathrm{Obj}(\mathcal{C})$, the class $\mathrm{Hom}_{\mathcal{C}}(A, B)$ is a set.
2. **Locally essentially small** if, for each $A, B \in \mathrm{Obj}(\mathcal{C})$, the class

$$\mathrm{Hom}_{\mathcal{C}}(A, B)/\{\text{isomorphisms}\}$$

   is a set.
3. **Small** if $\mathcal{C}$ is locally small and $\mathrm{Obj}(\mathcal{C})$ is a set.
4. κ**-Small** if $\mathcal{C}$ is locally small, $\mathrm{Obj}(\mathcal{C})$ is a set, and we have $\#\mathrm{Obj}(\mathcal{C}) < \kappa$.

11.1.2 Subcategories

Let $\mathcal{C}$ be a category.

Definition 11.1.2.1.1 ► Subcategories

A **subcategory** of $\mathcal{C}$ is a category $\mathcal{A}$ satisfying the following conditions:

1. *Objects.* We have $\mathrm{Obj}(\mathcal{A}) \subset \mathrm{Obj}(\mathcal{C})$.
2. *Morphisms.* For each $A, B \in \mathrm{Obj}(\mathcal{A})$, we have

$$\mathrm{Hom}_{\mathcal{A}}(A, B) \subset \mathrm{Hom}_{\mathcal{C}}(A, B).$$

3. *Identities.* For each $A \in \mathrm{Obj}(\mathcal{A})$, we have

$$\mathbb{1}_A^{\mathcal{A}} = \mathbb{1}_A^{\mathcal{C}}.$$

4. *Composition.* For each $A, B, C \in \mathrm{Obj}(\mathcal{A})$, we have

$$\circ_{A,B,C}^{\mathcal{A}} = \circ_{A,B,C}^{\mathcal{C}}.$$

Definition 11.1.2.1.2 ▶ Full Subcategories

A subcategory $\mathcal{A}$ of $\mathcal{C}$ is **full** if the canonical inclusion functor $\mathcal{A} \to \mathcal{C}$ is full, i.e. if, for each $A, B \in \mathrm{Obj}(\mathcal{A})$, the inclusion

$$\iota_{A,B} \colon \mathrm{Hom}_{\mathcal{A}}(A, B) \hookrightarrow \mathrm{Hom}_{\mathcal{C}}(A, B)$$

is surjective (and thus bijective).

Definition 11.1.2.1.3 ▶ Strictly Full Subcategories

A subcategory $\mathcal{A}$ of a category $\mathcal{C}$ is **strictly full** if it satisfies the following conditions:

1. *Fullness.* The subcategory $\mathcal{A}$ is full.
2. *Closedness Under Isomorphisms.* The class $\mathrm{Obj}(\mathcal{A})$ is closed under isomorphisms.[1]

[1] That is, given $A \in \mathrm{Obj}(\mathcal{A})$ and $C \in \mathrm{Obj}(\mathcal{C})$, if $C \cong A$, then $C \in \mathrm{Obj}(\mathcal{A})$.

Definition 11.1.2.1.4 ▶ Wide Subcategories

A subcategory $\mathcal{A}$ of $\mathcal{C}$ is **wide**[1] if $\mathrm{Obj}(\mathcal{A}) = \mathrm{Obj}(\mathcal{C})$.

[1] *Further Terminology:* Also called **lluf**.

11.1.3 Skeletons of Categories

Definition 11.1.3.1.1 ▶ Skeletons of Categories

A[1] **skeleton** of a category $\mathcal{C}$ is a full subcategory $\mathsf{Sk}(\mathcal{C})$ with one object from each isomorphism class of objects of $\mathcal{C}$.

[1] Due to Item 3 of Proposition 11.1.3.1.3, which states that any two skeletons of a category are equivalent, we often refer to any such full subcategory $\mathsf{Sk}(\mathcal{C})$ of $\mathcal{C}$ as *the* skeleton of $\mathcal{C}$.

Definition 11.1.3.1.2 ▶ Skeletal Categories

A category $\mathcal{C}$ is **skeletal** if $\mathcal{C} \cong \mathsf{Sk}(\mathcal{C})$.[1]

[1] That is, $\mathcal{C}$ is **skeletal** if isomorphic objects of $\mathcal{C}$ are equal.

Proposition 11.1.3.1.3 ▶ Properties of Skeletons of Categories

Let $\mathcal{C}$ be a category.

1. *Existence.* Assuming the axiom of choice, $\mathsf{Sk}(\mathcal{C})$ always exists.
2. *Pseudofunctoriality.* The assignment $\mathcal{C} \mapsto \mathsf{Sk}(\mathcal{C})$ defines a pseudofunctor
$$\mathsf{Sk}\colon \mathsf{Cats}_2 \to \mathsf{Cats}_2.$$
3. *Uniqueness Up to Equivalence.* Any two skeletons of $\mathcal{C}$ are equivalent.
4. *Inclusions of Skeletons Are Equivalences.* The inclusion
$$\iota_{\mathcal{C}}\colon \mathsf{Sk}(\mathcal{C}) \hookrightarrow \mathcal{C}$$
of a skeleton of $\mathcal{C}$ into $\mathcal{C}$ is an equivalence of categories.

Proof 11.1.3.1.4 ▶ Proof of Proposition 11.1.3.1.3

Item 1: Existence

See [nLab23, Section "Existence of Skeletons of Categories‘].

Item 2: Pseudofunctoriality

See [nLab23, Section "Skeletons as an Endo-Pseudofunctor on $\mathfrak{Cat}$"].

Item 3: Uniqueness Up to Equivalence

Clear.

Item 4: Inclusions of Skeletons Are Equivalences

Clear. ∎

11.1.4 Precomposition and Postcomposition

Let $\mathcal{C}$ be a category and let $A, B, C \in \mathrm{Obj}(\mathcal{C})$.

Definition 11.1.4.1.1 ► Precomposition and Postcomposition Functions

Let $f\colon A \to B$ and $g\colon B \to C$ be morphisms of $\mathcal{C}$.

1. The **precomposition function associated to** f is the function

$$f^*\colon \mathrm{Hom}_{\mathcal{C}}(B, C) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$$

defined by

$$f^*(\phi) \overset{\text{def}}{=} \phi \circ f$$

for each $\phi \in \mathrm{Hom}_{\mathcal{C}}(B, C)$.

2. The **postcomposition function associated to** g is the function

$$g_*\colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$$

defined by

$$g_*(\phi) \overset{\text{def}}{=} g \circ \phi$$

for each $\phi \in \mathrm{Hom}_{\mathcal{C}}(A, B)$.

Proposition 11.1.4.1.2 ► Properties of Pre/Postcomposition

Let $A, B, C, D \in \mathrm{Obj}(\mathcal{C})$ and let $f\colon A \to B$ and $g\colon B \to C$ be morphisms of $\mathcal{C}$.

1. *Interaction Between Precomposition and Postcomposition.* We have

$$g_* \circ f^* = f^* \circ g_*, \qquad \begin{tikzcd} \mathrm{Hom}_C(B,C) \arrow[r, "g_*"] \arrow[d, "f^*"'] & \mathrm{Hom}_C(B,D) \arrow[d, "f^*"] \\ \mathrm{Hom}_C(A,C) \arrow[r, "g_*"'] & \mathrm{Hom}_C(A,D). \end{tikzcd}$$

2. *Interaction With Composition I.* We have

$$(g \circ f)^* = f^* \circ g^*, \qquad \begin{tikzcd} \mathrm{Hom}_C(X,A) \arrow[r, "f_*"] \arrow[dr, "(g\circ f)_*"'] & \mathrm{Hom}_C(X,B) \arrow[d, "g_*"] \\ & \mathrm{Hom}_C(X,C), \end{tikzcd}$$

$$(g \circ f)_* = g_* \circ f_*, \qquad \begin{tikzcd} \mathrm{Hom}_C(C,X) \arrow[r, "g^*"] \arrow[dr, "(g\circ f)^*"'] & \mathrm{Hom}_C(B,X) \arrow[d, "f^*"] \\ & \mathrm{Hom}_C(A,X). \end{tikzcd}$$

3. *Interaction With Composition II.* We have

$$\begin{tikzcd} \mathrm{pt} \arrow[r, "{[f]}"] \arrow[dr, "{[g\circ f]}"'] & \mathrm{Hom}_C(A,B) \arrow[d, "g_*"] \\ & \mathrm{Hom}_C(A,C) \end{tikzcd} \qquad \begin{aligned} [g \circ f] &= g_* \circ [f], \\ [g \circ f] &= f^* \circ [g], \end{aligned} \qquad \begin{tikzcd} \mathrm{pt} \arrow[r, "{[g]}"] \arrow[dr, "{[g\circ f]}"'] & \mathrm{Hom}_C(B,C) \arrow[d, "f^*"] \\ & \mathrm{Hom}_C(A,C). \end{tikzcd}$$

4. *Interaction With Composition III.* We have

$$f^* \circ \circ^C_{A,B,C} = \circ^C_{X,B,C} \circ (f^* \times \mathrm{id}),$$

$$\begin{tikzcd}
\mathrm{Hom}_C(B,C) \times \mathrm{Hom}_C(A,B) \arrow[r, "\circ^C_{A,B,C}"] \arrow[d, "\mathrm{id}\times f^*"'] & \mathrm{Hom}_C(A,C) \arrow[d, "f^*"] \\
\mathrm{Hom}_C(B,C) \times \mathrm{Hom}_C(X,B) \arrow[r, "\circ^C_{X,B,C}"'] & \mathrm{Hom}_C(X,C),
\end{tikzcd}$$

$$g_* \circ \circ^C_{A,B,C} = \circ^C_{A,B,D} \circ (\mathrm{id} \times g_*),$$

$$\begin{tikzcd}
\mathrm{Hom}_C(B,C) \times \mathrm{Hom}_C(A,B) \arrow[r, "\circ^C_{A,B,C}"] \arrow[d, "g_*\times\mathrm{id}"'] & \mathrm{Hom}_C(A,C) \arrow[d, "g^*"] \\
\mathrm{Hom}_C(B,D) \times \mathrm{Hom}_C(A,B) \arrow[r, "\circ^C_{A,B,D}"'] & \mathrm{Hom}_C(A,D).
\end{tikzcd}$$

5. *Interaction With Identities.* We have

$$\begin{aligned}
(\mathrm{id}_A)^* &= \mathrm{id}_{\mathrm{Hom}_C(A,B)}, \\
(\mathrm{id}_B)_* &= \mathrm{id}_{\mathrm{Hom}_C(A,B)}.
\end{aligned}$$

Proof 11.1.4.1.3 ▶ Proof of Proposition 11.1.4.1.2

Item 1: Interaction Between Precomposition and Postcomposition

Clear.

Item 2: Interaction With Composition I

Clear.

Item 3: Interaction With Composition II

Clear.

Item 4: Interaction With Composition III

Clear.

Item 5: Interaction With Identities

Clear.

11.2 Examples of Categories

11.2.1 The Empty Category

Example 11.2.1.1.1 ▶ The Empty Category

The **empty category** is the category $\emptyset_{\mathsf{cat}}$ where

- *Objects.* We have
$$\mathrm{Obj}(\emptyset_{\mathsf{cat}}) \stackrel{\text{def}}{=} \emptyset.$$
- *Morphisms.* We have
$$\mathrm{Mor}(\emptyset_{\mathsf{cat}}) \stackrel{\text{def}}{=} \emptyset.$$
- *Identities and Composition.* Having no objects, $\emptyset_{\mathsf{cat}}$ has no unit nor composition maps.

11.2.2 The Punctual Category

Example 11.2.2.1.1 ▶ The Punctual Category

The **punctual category**[1] is the category pt where

- *Objects.* We have
$$\mathrm{Obj}(\mathsf{pt}) \stackrel{\text{def}}{=} \{\star\}.$$
- *Morphisms.* The unique Hom-set of pt is defined by
$$\mathrm{Hom}_{\mathsf{pt}}(\star, \star) \stackrel{\text{def}}{=} \{\mathrm{id}_\star\}.$$
- *Identities.* The unit map
$$\mathbb{1}^{\mathsf{pt}}_\star \colon \mathsf{pt} \longrightarrow \mathrm{Hom}_{\mathsf{pt}}(\star, \star)$$
of pt at $\star$ is defined by
$$\mathrm{id}^{\mathsf{pt}}_\star \stackrel{\text{def}}{=} \mathrm{id}_\star\,.$$

- *Composition*. The composition map

$$\circ^{\mathrm{pt}}_{\star,\star,\star}\colon \mathrm{Hom}_{\mathrm{pt}}(\star,\star)\times\mathrm{Hom}_{\mathrm{pt}}(\star,\star)\longrightarrow\mathrm{Hom}_{\mathrm{pt}}(\star,\star)$$

of pt at $(\star,\star,\star)$ is given by the bijection $\mathrm{pt}\times\mathrm{pt}\cong\mathrm{pt}$.

[1] *Further Terminology*: Also called the **singleton category**.

11.2.3 Monoids as One-Object Categories

Example 11.2.3.1.1 ► Monoids as One-Object Categories

We have an isomorphism of categories[1]

$$\mathrm{Mon}\cong\mathrm{pt}\underset{\mathrm{Sets}}{\times}\mathrm{Cats},\qquad \begin{tikzcd} \mathrm{Mon} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Cats} \arrow[d, "\mathrm{Obj}"] \\ \mathrm{pt} \arrow[r, "{[\mathrm{pt}]}"'] & \mathrm{Sets} \end{tikzcd}$$

via the delooping functor $\mathrm{B}\colon\mathrm{Mon}\longrightarrow\mathrm{Cats}$ of **??** of **??**, exhibiting monoids as exactly those categories having a single object.

[1] This can be enhanced to an isomorphism of 2-categories

$$\mathrm{Mon}_{2\mathrm{disc}}\cong\mathrm{pt}_{\mathrm{bi}}\underset{\mathrm{Sets}_{2\mathrm{disc}}}{\times}\mathrm{Cats}_{2,*},\qquad \begin{tikzcd} \mathrm{Mon}_{2\mathrm{disc}} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Cats}_{2,*} \arrow[d, "\mathrm{Obj}"] \\ \mathrm{pt}_{\mathrm{bi}} \arrow[r, "{[\mathrm{pt}]}"'] & \mathrm{Sets}_{2\mathrm{disc}} \end{tikzcd}$$

between the discrete 2-category $\mathrm{Mon}_{2\mathrm{disc}}$ on Mon and the 2-category of pointed categories with one object.

Proof 11.2.3.1.2 ► Proof of Example 11.2.3.1.1

Omitted. ∎

11.2.4 Ordinal Categories

Example 11.2.4.1.1 ▶ Ordinal Categories

The n**th ordinal category** is the category $\mathbb{n}$ where[1]

- *Objects.* We have

$$\mathrm{Obj}(\mathbb{n}) \stackrel{\text{def}}{=} \{[0], \ldots, [n]\}.$$

- *Morphisms.* For each $[i], [j] \in \mathrm{Obj}(\mathbb{n})$, we have

$$\mathrm{Hom}_{\mathbb{n}}([i], [j]) \stackrel{\text{def}}{=} \begin{cases} \{\mathrm{id}_{[i]}\} & \text{if } [i] = [j], \\ \{[i] \to [j]\} & \text{if } [j] < [i], \\ \emptyset & \text{if } [j] > [i]. \end{cases}$$

- *Identities.* For each $[i] \in \mathrm{Obj}(\mathbb{n})$, the unit map

$$\mathbb{1}^{\mathbb{n}}_{[i]} \colon \mathrm{pt} \to \mathrm{Hom}_{\mathbb{n}}([i], [i])$$

  of $\mathbb{n}$ at $[i]$ is defined by

$$\mathrm{id}^{\mathbb{n}}_{[i]} \stackrel{\text{def}}{=} \mathrm{id}_{[i]}.$$

- *Composition.* For each $[i], [j], [k] \in \mathrm{Obj}(\mathbb{n})$, the composition map

$$\circ^{\mathbb{n}}_{[i],[j],[k]} \colon \mathrm{Hom}_{\mathbb{n}}([j], [k]) \times \mathrm{Hom}_{\mathbb{n}}([i], [j]) \to \mathrm{Hom}_{\mathbb{n}}([i], [k])$$

  of $\mathbb{n}$ at $([i], [j], [k])$ is defined by

$$\mathrm{id}_{[i]} \circ \mathrm{id}_{[i]} = \mathrm{id}_{[i]},$$
$$([j] \to [k]) \circ ([i] \to [j]) = ([i] \to [k]).$$

[1] In other words, $\mathbb{n}$ is the category associated to the poset

$$[0] \to [1] \to \cdots \to [n-1] \to [n].$$

The category $\mathbb{n}$ for $n \geq 2$ may also be defined in terms of $\mathbb{0}$ and joins (Constructions With Categories, **??**): we have isomorphisms of categories

$$\mathbb{1} \cong \mathbb{0} \star \mathbb{0},$$

$$
\begin{aligned}
\mathbb{2} &\cong \mathbb{1} \star \mathbb{0} \\
&\cong (\mathbb{0} \star \mathbb{0}) \star \mathbb{0}, \\
\mathbb{3} &\cong \mathbb{2} \star \mathbb{0} \\
&\cong (\mathbb{1} \star \mathbb{0}) \star \mathbb{0} \\
&\cong ((\mathbb{0} \star \mathbb{0}) \star \mathbb{0}) \star \mathbb{0}, \\
\mathbb{4} &\cong \mathbb{3} \star \mathbb{0} \\
&\cong (\mathbb{2} \star \mathbb{0}) \star \mathbb{0} \\
&\cong ((\mathbb{1} \star \mathbb{0}) \star \mathbb{0}) \star \mathbb{0} \\
&\cong (((\mathbb{0} \star \mathbb{0}) \star \mathbb{0}) \star \mathbb{0}) \star \mathbb{0},
\end{aligned}
$$

and so on.

11.2.5 The Walking Arrow

Definition 11.2.5.1.1 ► The Walking Arrow

The **walking arrow** is the category $\mathbb{1}$ defined as the first ordinal category.

Remark 11.2.5.1.2 ► Unwinding Definition 11.2.5.1.1

In detail, the walking arrow is the category $\mathbb{1}$ where:

- *Objects.* We have $\mathrm{Obj}(\mathbb{1}) = \{0, 1\}$.
- *Morphisms.* We have

$$
\begin{aligned}
\mathrm{Hom}_{\mathbb{1}}(0, 0) &= \{\mathrm{id}_0\}, \\
\mathrm{Hom}_{\mathbb{1}}(1, 1) &= \{\mathrm{id}_1\}, \\
\mathrm{Hom}_{\mathbb{1}}(0, 1) &= \{f_{01}\}, \\
\mathrm{Hom}_{\mathbb{1}}(1, 0) &= \emptyset.
\end{aligned}
$$

- *Identities and Composition.* The identities and composition of $\mathbb{1}$ are completely determined by the unitality and associativity axioms for $\mathbb{1}$.

11.2.6 More Examples of Categories

Example 11.2.6.1.1 ► More Examples of Categories

Here we list some of the other categories appearing throughout this work.

1. The category Sets_* of pointed sets of Pointed Sets, Definition 6.1.3.1.1.
2. The category Rel of sets and relations of Relations, Definition 8.2.2.1.1.
3. The category $\mathsf{Span}(A, B)$ of spans from a set A to a set B of **??**, **??**.
4. The category $\mathsf{ISets}(K)$ of K-indexed sets of Indexed Sets, **??**.
5. The category ISets of indexed sets of Indexed Sets, **??**.
6. The category $\mathsf{FibSets}(K)$ of K-fibred sets of Fibred Sets, **??**.
7. The category $\mathsf{FibSets}$ of fibred sets of Fibred Sets, **??**.
8. Categories of functors $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$ as in Definition 11.10.1.1.1.
9. The category of categories Cats of Definition 11.10.2.1.1.
10. The category of groupoids Grpd of Definition 11.10.4.1.1.

11.2.7 Posetal Categories

Definition 11.2.7.1.1 ► Posetal Categories

Let $(X, \preceq_X)$ be a poset.

1. The **posetal category associated to** $(X, \preceq_X)$ is the category X_{pos} where
   - *Objects.* We have
   $$\mathsf{Obj}(X_{\mathsf{pos}}) \stackrel{\text{def}}{=} X.$$
   - *Morphisms.* For each $a, b \in \mathsf{Obj}(X_{\mathsf{pos}})$, we have
   $$\mathsf{Hom}_{X_{\mathsf{pos}}}(a, b) \stackrel{\text{def}}{=} \begin{cases} \mathsf{pt} & \text{if } a \preceq_X b, \\ \emptyset & \text{otherwise.} \end{cases}$$

- *Identities.* For each $a \in \mathrm{Obj}(X_{\mathrm{pos}})$, the unit map

$$\mathbb{1}_a^{X_{\mathrm{pos}}} \colon \mathrm{pt} \longrightarrow \mathrm{Hom}_{X_{\mathrm{pos}}}(a, a)$$

  of X_{pos} at a is given by the identity map.

- *Composition.* For each $a, b, c \in \mathrm{Obj}(X_{\mathrm{pos}})$, the composition map

$$\circ_{a,b,c}^{X_{\mathrm{pos}}} \colon \mathrm{Hom}_{X_{\mathrm{pos}}}(b, c) \times \mathrm{Hom}_{X_{\mathrm{pos}}}(a, b) \longrightarrow \mathrm{Hom}_{X_{\mathrm{pos}}}(a, c)$$

  of X_{pos} at (a, b, c) is defined as either the inclusion $\emptyset \hookrightarrow \mathrm{pt}$ or the identity map of pt, depending on whether we have $a \preceq_X b$, $b \preceq_X c$, and $a \preceq_X c$.

2. A category $\mathcal{C}$ is **posetal**[1] if $\mathcal{C}$ is equivalent to X_{pos} for some poset $(X, \preceq_X)$.

[1] *Further Terminology:* Also called a **thin** category or a $(0, 1)$**-category**.

Proposition 11.2.7.1.2 ► Properties of Posetal Categories

Let $(X, \preceq_X)$ be a poset and let $\mathcal{C}$ be a category.

1. *Functoriality.* The assignment $(X, \preceq_X) \mapsto X_{\mathrm{pos}}$ defines a functor

$$(-)_{\mathrm{pos}} \colon \mathrm{Pos} \longrightarrow \mathrm{Cats}.$$

2. *Fully Faithfulness.* The functor $(-)_{\mathrm{pos}}$ of Item 1 is fully faithful.

3. *Characterisations.* The following conditions are equivalent:

   (a) The category $\mathcal{C}$ is posetal.

   (b) For each $A, B \in \mathrm{Obj}(\mathcal{C})$ and each $f, g \in \mathrm{Hom}_{\mathcal{C}}(A, B)$, we have $f = g$.

4. *Automatic Commutativity of Diagrams.* Every diagram in a posetal category commutes.

Proof 11.2.7.1.3 ▶ Proof of Proposition 11.2.7.1.2

Item 1: Functoriality

Omitted.

Item 2: Fully Faithfulness

Omitted.

Item 3: Characterisations

Clear.

Item 4: Automatic Commutativity of Diagrams

This follows from the fact that if $\mathcal{C}$ is posetal, then there,s at most one morphism between any two objects. ∎

11.3 The Quadruple Adjunction With Sets

11.3.1 Statement

Let $\mathcal{C}$ be a category.

Proposition 11.3.1.1.1 ▶ The Quadruple Adjunction Between Sets and Cats

We have a quadruple adjunction

$$(\pi_0 \dashv (-)_{\mathsf{disc}} \dashv \mathsf{Obj} \dashv (-)_{\mathsf{indisc}}): \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left=60, "(-)_{\mathsf{disc}}"] \arrow[r, bend right=60, "(-)_{\mathsf{indisc}}"'] \arrow[r, phantom, "\perp" near start] \arrow[r, phantom, "\perp" near end] & \mathsf{Cats}, \arrow[l, bend right=60, "\pi_0"'] \arrow[l, bend left=60, "\mathsf{Obj}"] \arrow[l, phantom, "\perp" near start] \arrow[l, phantom, "\perp" near end] \end{tikzcd}$$

witnessed by bijections of sets

$$\begin{aligned} \mathsf{Hom}_{\mathsf{Sets}}(\pi_0(\mathcal{C}), X) &\cong \mathsf{Hom}_{\mathsf{Cats}}(\mathcal{C}, X_{\mathsf{disc}}), \\ \mathsf{Hom}_{\mathsf{Cats}}(X_{\mathsf{disc}}, \mathcal{C}) &\cong \mathsf{Hom}_{\mathsf{Sets}}(X, \mathsf{Obj}(\mathcal{C})), \\ \mathsf{Hom}_{\mathsf{Sets}}(\mathsf{Obj}(\mathcal{C}), X) &\cong \mathsf{Hom}_{\mathsf{Cats}}(\mathcal{C}, X_{\mathsf{indisc}}), \end{aligned}$$

natural in $\mathcal{C} \in \mathsf{Obj}(\mathsf{Cats})$ and $X \in \mathsf{Obj}(\mathsf{Sets})$, where

- The functor
$$\pi_0 \colon \mathsf{Cats} \to \mathsf{Sets},$$
the **connected components functor**, is the functor sending a category to its set of connected components of Definition 11.3.2.2.1.

- The functor
$$(-)_{\mathsf{disc}} \colon \mathsf{Sets} \to \mathsf{Cats},$$
the **discrete category functor**, is the functor sending a set to its associated discrete category of Item 1.

- The functor
$$\mathrm{Obj} \colon \mathsf{Cats} \to \mathsf{Sets},$$
the **object functor**, is the functor sending a category to its set of objects.

- The functor
$$(-)_{\mathsf{indisc}} \colon \mathsf{Sets} \to \mathsf{Cats},$$
the **indiscrete category functor**, is the functor sending a set to its associated indiscrete category of Item 1.

Proof 11.3.1.1.2 ▶ Proof of Proposition 11.3.1.1.1

Omitted. ∎

11.3.2 Connected Components and Connected Categories

11.3.2.1 Connected Components of Categories

Let $\mathcal{C}$ be a category.

Definition 11.3.2.1.1 ▶ Connected Components of Categories

A **connected component** of $\mathcal{C}$ is a full subcategory $\mathcal{I}$ of $\mathcal{C}$ satisfying the following conditions:[1]

1. *Non-Emptiness.* We have $\mathrm{Obj}(\mathcal{I}) \neq \emptyset$.

2. *Connectedness.* There exists a zigzag of arrows between any two objects of $\mathcal{I}$.

[1] In other words, a **connected component** of $\mathcal{C}$ is an element of the set $\mathrm{Obj}(\mathcal{C})/\sim$ with $\sim$ the equivalence relation generated by the relation $\sim'$ obtained by declaring $A \sim' B$ iff there exists a morphism of $\mathcal{C}$ from A to B.

11.3.2.2 Sets of Connected Components of Categories

Let $\mathcal{C}$ be a category.

Definition 11.3.2.2.1 ▶ Sets of Connected Components of Categories

The **set of connected components of** $\mathcal{C}$ is the set $\pi_0(\mathcal{C})$ whose elements are the connected components of $\mathcal{C}$.

Proposition 11.3.2.2.2 ▶ Properties of Sets of Connected Components

Let $\mathcal{C}$ be a category.

1. *Functoriality.* The assignment $\mathcal{C} \mapsto \pi_0(\mathcal{C})$ defines a functor

$$\pi_0 \colon \mathsf{Cats} \to \mathsf{Sets}.$$

2. *Adjointness.* We have a quadruple adjunction

$$(\pi_0 \dashv (-)_{\mathrm{disc}} \dashv \mathrm{Obj} \dashv (-)_{\mathrm{indisc}}) \colon \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left=60, "(-)_{\mathrm{disc}}"] \arrow[r, bend right=60, "(-)_{\mathrm{indisc}}"'] & \mathsf{Cats}. \arrow[l, bend right=60, "\pi_0"'] \arrow[l, bend left=60, "\mathrm{Obj}"] \end{tikzcd}$$

3. *Interaction With Groupoids.* If $\mathcal{C}$ is a groupoid, then we have an isomorphism of categories

$$\pi_0(\mathcal{C}) \cong \mathrm{K}(\mathcal{C}),$$

where $\mathrm{K}(\mathcal{C})$ is the set of isomorphism classes of $\mathcal{C}$ of **??**.

4. *Preservation of Colimits.* The functor π_0 of Item 1 preserves colimits. In

particular, we have bijections of sets

$$\begin{aligned}
\pi_0(\mathcal{C} \coprod \mathcal{D}) &\cong \pi_0(\mathcal{C}) \coprod \pi_0(\mathcal{D}), \\
\pi_0(\mathcal{C} \coprod_{\mathcal{E}} \mathcal{D}) &\cong \pi_0(\mathcal{C}) \coprod_{\pi_0(\mathcal{E})} \pi_0(\mathcal{D}), \\
\pi_0\left(\mathsf{CoEq}\left(\mathcal{C} \underset{G}{\overset{F}{\rightrightarrows}} \mathcal{D}\right)\right) &\cong \mathsf{CoEq}\left(\pi_0(\mathcal{C}) \underset{\pi_0(G)}{\overset{\pi_0(F)}{\rightrightarrows}} \pi_0(\mathcal{D})\right),
\end{aligned}$$

natural in $\mathcal{C}, \mathcal{D}, \mathcal{E} \in \mathsf{Obj}(\mathsf{Cats})$.

5. *Symmetric Strong Monoidality With Respect to Coproducts.* The connected components functor of Item 1 has a symmetric strong monoidal structure

$$\left(\pi_0, \pi_0^{\coprod}, \pi_{0|\mathbb{1}}^{\coprod}\right) \colon (\mathsf{Cats}, \coprod, \emptyset_{\mathsf{cat}}) \to (\mathsf{Sets}, \coprod, \emptyset),$$

being equipped with isomorphisms

$$\begin{aligned}
\pi_{0|\mathcal{C},\mathcal{D}}^{\coprod} &\colon \pi_0(\mathcal{C}) \coprod \pi_0(\mathcal{D}) \overset{\sim}{\dashrightarrow} \pi_0(\mathcal{C} \coprod \mathcal{D}), \\
\pi_{0|\mathbb{1}}^{\coprod} &\colon \emptyset \overset{\sim}{\dashrightarrow} \pi_0(\emptyset_{\mathsf{cat}}),
\end{aligned}$$

natural in $\mathcal{C}, \mathcal{D} \in \mathsf{Obj}(\mathsf{Cats})$.

6. *Symmetric Strong Monoidality With Respect to Products.* The connected components functor of Item 1 has a symmetric strong monoidal structure

$$\left(\pi_0, \pi_0^{\times}, \pi_{0|\mathbb{1}}^{\times}\right) \colon (\mathsf{Cats}, \times, \mathsf{pt}) \to (\mathsf{Sets}, \times, \mathsf{pt}),$$

being equipped with isomorphisms

$$\begin{aligned}
\pi_{0|\mathcal{C},\mathcal{D}}^{\times} &\colon \pi_0(\mathcal{C}) \times \pi_0(\mathcal{D}) \overset{\sim}{\dashrightarrow} \pi_0(\mathcal{C} \times \mathcal{D}), \\
\pi_{0|\mathbb{1}}^{\times} &\colon \mathsf{pt} \overset{\sim}{\dashrightarrow} \pi_0(\mathsf{pt}),
\end{aligned}$$

natural in $\mathcal{C}, \mathcal{D} \in \mathsf{Obj}(\mathsf{Cats})$.

Proof 11.3.2.2.3 ▶ Proof of Proposition 11.3.2.2.2

Item 1: Functoriality

Clear.

Item 2: Adjointness

This is proved in Proposition 11.3.1.1.1.

Item 3: Interaction With Groupoids

Clear.

Item 4: Preservation of Colimits

This follows from Item 2 and **??** of **??**.

Item 5: Symmetric Strong Monoidality With Respect to Coproducts

Clear.

Item 6: Symmetric Strong Monoidality With Respect to Products

Clear.

11.3.2.3 Connected Categories

Definition 11.3.2.3.1 ▶ Connected Categories

A category C is **connected** if $\pi_0(C) \cong \mathrm{pt}$.[1,2]

[1] *Further Terminology:* A category is **disconnected** if it is not connected.
[2] *Example:* A groupoid is connected iff any two of its objects are isomorphic.

11.3.3 Discrete Categories

Definition 11.3.3.1.1 ▶ Discrete Categories

Let X be a set.

1. The **discrete category on** X is the category X_{disc} where

   - *Objects.* We have

$$\mathrm{Obj}(X_{\mathrm{disc}}) \overset{\mathrm{def}}{=} X.$$

- *Morphisms.* For each $A, B \in \mathrm{Obj}(X_{\mathrm{disc}})$, we have

$$\mathrm{Hom}_{X_{\mathrm{disc}}}(A, B) \overset{\mathrm{def}}{=} \begin{cases} \mathrm{id}_A & \text{if } A = B, \\ \emptyset & \text{if } A \neq B. \end{cases}$$

- *Identities.* For each $A \in \mathrm{Obj}(X_{\mathrm{disc}})$, the unit map

$$\mathbb{1}_A^{X_{\mathrm{disc}}} \colon \mathrm{pt} \longrightarrow \mathrm{Hom}_{X_{\mathrm{disc}}}(A, A)$$

of X_{disc} at A is defined by

$$\mathrm{id}_A^{X_{\mathrm{disc}}} \overset{\mathrm{def}}{=} \mathrm{id}_A .$$

- *Composition.* For each $A, B, C \in \mathrm{Obj}(X_{\mathrm{disc}})$, the composition map

$$\circ_{A,B,C}^{X_{\mathrm{disc}}} \colon \mathrm{Hom}_{X_{\mathrm{disc}}}(B, C) \times \mathrm{Hom}_{X_{\mathrm{disc}}}(A, B) \longrightarrow \mathrm{Hom}_{X_{\mathrm{disc}}}(A, C)$$

of X_{disc} at (A, B, C) is defined by

$$\mathrm{id}_A \circ \mathrm{id}_A \overset{\mathrm{def}}{=} \mathrm{id}_A .$$

2. A category $\mathcal{C}$ is **discrete** if it is equivalent to X_{disc} for some set X.

Proposition 11.3.3.1.2 ► Properties of Discrete Categories on Sets

Let X be a set.

1. *Functoriality.* The assignment $X \mapsto X_{\mathrm{disc}}$ defines a functor

$$(-)_{\mathrm{disc}} \colon \mathsf{Sets} \longrightarrow \mathsf{Cats}.$$

2. *Adjointness.* We have a quadruple adjunction

$$(\pi_0 \dashv (-)_{\mathrm{disc}} \dashv \mathrm{Obj} \dashv (-)_{\mathrm{indisc}}) \colon \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left=15, "(-)_{\mathrm{disc}}"] \arrow[r, bend right=15, "(-)_{\mathrm{indisc}}"'] \arrow[r, phantom, "\perp"] & \mathsf{Cats}. \arrow[l, bend right=15, "\pi_0"'] \arrow[l, bend left=15, "\mathrm{Obj}"] \arrow[l, phantom, "\perp"] \arrow[l, phantom, "\perp"] \end{tikzcd}$$

3. *Symmetric Strong Monoidality With Respect to Coproducts.* The functor of Item 1 has a symmetric strong monoidal structure

$$\left((-)_{\mathsf{disc}}, (-)^{\coprod}_{\mathsf{disc}}, (-)^{\coprod}_{\mathsf{disc}|\mathbb{1}}\right) \colon (\mathsf{Sets}, \coprod, \emptyset) \to (\mathsf{Cats}, \coprod, \emptyset_{\mathsf{cat}}),$$

being equipped with isomorphisms

$$(-)^{\coprod}_{\mathsf{disc}|X,Y} \colon X_{\mathsf{disc}} \coprod Y_{\mathsf{disc}} \overset{\sim}{\dashrightarrow} (X \coprod Y)_{\mathsf{disc}},$$
$$(-)^{\coprod}_{\mathsf{disc}|\mathbb{1}} \colon \emptyset_{\mathsf{cat}} \overset{\sim}{\dashrightarrow} \emptyset_{\mathsf{disc}},$$

natural in $X, Y \in \mathsf{Obj}(\mathsf{Sets})$.

4. *Symmetric Strong Monoidality With Respect to Products.* The functor of Item 1 has a symmetric strong monoidal structure

$$\left((-)_{\mathsf{disc}}, (-)^{\times}_{\mathsf{disc}}, (-)^{\times}_{\mathsf{disc}|\mathbb{1}}\right) \colon (\mathsf{Sets}, \times, \mathsf{pt}) \to (\mathsf{Cats}, \times, \mathsf{pt}),$$

being equipped with isomorphisms

$$(-)^{\times}_{\mathsf{disc}|X,Y} \colon X_{\mathsf{disc}} \times Y_{\mathsf{disc}} \overset{\sim}{\dashrightarrow} (X \times Y)_{\mathsf{disc}},$$
$$(-)^{\times}_{\mathsf{disc}|\mathbb{1}} \colon \mathsf{pt} \overset{\sim}{\dashrightarrow} \mathsf{pt}_{\mathsf{disc}},$$

natural in $X, Y \in \mathsf{Obj}(\mathsf{Sets})$.

Proof 11.3.3.1.3 ▶ Proof of Proposition 11.3.3.1.2

Item 1: Functoriality

Clear.

Item 2: Adjointness

This is proved in Proposition 11.3.1.1.1.

Item 3: Symmetric Strong Monoidality With Respect to Coproducts

Clear.

Item 4: Symmetric Strong Monoidality With Respect to Products

Clear. ∎

11.3.4 Indiscrete Categories

Definition 11.3.4.1.1 ► Indiscrete Categories

Let X be a set.

1. The **indiscrete category on** X[1] is the category X_{indisc} where

   - *Objects.* We have

   $$\text{Obj}(X_{\text{indisc}}) \stackrel{\text{def}}{=} X.$$

   - *Morphisms.* For each $A, B \in \text{Obj}(X_{\text{indisc}})$, we have

   $$\begin{aligned}\text{Hom}_{X_{\text{disc}}}(A, B) &\stackrel{\text{def}}{=} \{[A] \longrightarrow [B]\} \\ &\cong \text{pt}.\end{aligned}$$

   - *Identities.* For each $A \in \text{Obj}(X_{\text{indisc}})$, the unit map

   $$\mathbb{1}_A^{X_{\text{indisc}}} \colon \text{pt} \longrightarrow \text{Hom}_{X_{\text{indisc}}}(A, A)$$

   of X_{indisc} at A is defined by

   $$\text{id}_A^{X_{\text{indisc}}} \stackrel{\text{def}}{=} \{[A] \longrightarrow [A]\}.$$

   - *Composition.* For each $A, B, C \in \text{Obj}(X_{\text{indisc}})$, the composition map

   $$\circ_{A,B,C}^{X_{\text{indisc}}} \colon \text{Hom}_{X_{\text{indisc}}}(B, C) \times \text{Hom}_{X_{\text{indisc}}}(A, B) \longrightarrow \text{Hom}_{X_{\text{indisc}}}(A, C)$$

   of X_{disc} at (A, B, C) is defined by

   $$([B] \longrightarrow [C]) \circ ([A] \longrightarrow [B]) \stackrel{\text{def}}{=} ([A] \longrightarrow [C]).$$

2. A category $\mathcal{C}$ is **indiscrete** if it is equivalent to X_{indisc} for some set X.

[1] *Further Terminology:* Sometimes called the **chaotic category on** X.

Proposition 11.3.4.1.2 ► Properties of Indiscrete Categories on Sets

Let X be a set.

1. *Functoriality.* The assignment $X \mapsto X_{\mathsf{indisc}}$ defines a functor

$$(-)_{\mathsf{indisc}} \colon \mathsf{Sets} \to \mathsf{Cats}.$$

2. *Adjointness.* We have a quadruple adjunction

$$(\pi_0 \dashv (-)_{\mathsf{disc}} \dashv \mathsf{Obj} \dashv (-)_{\mathsf{indisc}}) \colon \quad \begin{tikzcd} \mathsf{Sets} \arrow[r, bend left=60, "(-)_{\mathsf{disc}}"] \arrow[r, bend right=60, "(-)_{\mathsf{indisc}}"'] \arrow[r, phantom, "\perp"] & \mathsf{Cats}. \arrow[l, bend right=60, "\pi_0"'] \arrow[l, bend left=60, "\mathsf{Obj}"] \arrow[l, phantom, "\perp"] \arrow[l, phantom, "\perp"] \end{tikzcd}$$

3. *Symmetric Strong Monoidality With Respect to Products.* The functor of Item 1 has a symmetric strong monoidal structure

$$\left((-)_{\mathsf{indisc}}, (-)^{\times}_{\mathsf{indisc}}, (-)^{\times}_{\mathsf{indisc}|\mathbb{1}}\right) \colon (\mathsf{Sets}, \times, \mathsf{pt}) \to (\mathsf{Cats}, \times, \mathsf{pt}),$$

being equipped with isomorphisms

$$(-)^{\times}_{\mathsf{indisc}|X,Y} \colon X_{\mathsf{indisc}} \times Y_{\mathsf{indisc}} \xrightarrow{\sim} (X \times Y)_{\mathsf{indisc}},$$
$$(-)^{\times}_{\mathsf{indisc}|\mathbb{1}} \colon \mathsf{pt} \xrightarrow{\sim} \mathsf{pt}_{\mathsf{indisc}},$$

natural in $X, Y \in \mathsf{Obj}(\mathsf{Sets})$.

Proof 11.3.4.1.3 ▶ Proof of Proposition 11.3.4.1.2

Item 1: Functoriality

Clear.

Item 2: Adjointness

This is proved in Proposition 11.3.1.1.1.

Item 3: Symmetric Strong Monoidality With Respect to Products

Clear. ∎

11.4 Groupoids

11.4.1 Isomorphisms

Let $\mathcal{C}$ be a category.

Definition 11.4.1.1.1 ▶ Isomorphisms

A morphism $f\colon A \longrightarrow B$ of $\mathcal{C}$ is an **isomorphism** if there exists a morphism $f^{-1}\colon B \longrightarrow A$ of $\mathcal{C}$ such that

$$\begin{aligned} f \circ f^{-1} &= \mathrm{id}_B, \\ f^{-1} \circ f &= \mathrm{id}_A\,. \end{aligned}$$

Notation 11.4.1.1.2 ▶ The Set of Isomorphisms Between Two Objects in a Category

We write $\mathrm{Iso}_{\mathcal{C}}(A, B)$ for the set of all isomorphisms in $\mathcal{C}$ from A to B.

11.4.2 Groupoids

Definition 11.4.2.1.1 ▶ Groupoids

A **groupoid** is a category in which every morphism is an isomorphism.

Example 11.4.2.1.2 ▶ Groups as One-Object Groupoids

The isomorphism of categories of Example 11.2.3.1.1 restricts to an isomorphism

$$\mathrm{Grp} \cong \mathrm{pt} \underset{\mathrm{Sets}}{\times} \mathrm{Grpd}, \qquad \begin{tikzcd} \mathrm{Grp} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Grpd} \arrow[d, "\mathrm{Obj}"] \\ \mathrm{pt} \arrow[r, "{[\mathrm{pt}]}"'] & \mathrm{Sets} \end{tikzcd}$$

where Grpd is the full subcategory of Cats spanned by the groupoids.
In other words, we have an identification

$$\{\text{Groups}\} \cong \{\text{One-object groupoids}\}.$$

11.4.3 The Groupoid Completion of a Category

Let $\mathcal{C}$ be a category.

Definition 11.4.3.1.1 ▶ The Groupoid Completion of a Category

The **groupoid completion of** $\mathcal{C}$[1] is the pair $(\mathrm{K}_0(\mathcal{C}), \iota_{\mathcal{C}})$ consisting of

- A groupoid $\mathrm{K}_0(\mathcal{C})$;
- A functor $\iota_{\mathcal{C}} \colon \mathcal{C} \to \mathrm{K}_0(\mathcal{C})$;

satisfying the following universal property:[2]

(**UP**) Given another such pair $(\mathcal{G}, i)$, there exists a unique functor $\mathrm{K}_0(\mathcal{C}) \xrightarrow{\exists!} \mathcal{G}$ making the diagram

$$\begin{tikzcd} & \mathrm{K}_0(\mathcal{C}) \arrow[d, dashed, "\exists!"] \\ \mathcal{C} \arrow[ur, "\iota_{\mathcal{C}}"] \arrow[r, "i"'] & \mathcal{G} \end{tikzcd}$$

commute.

[1] *Further Terminology:* Also called the **Grothendieck groupoid of** $\mathcal{C}$ or the **Grothendieck groupoid completion of** $\mathcal{C}$.

[2] See Item 3 of Proposition 11.4.3.1.4 for an explicit construction.

Construction 11.4.3.1.2 ▶ Construction of the Groupoid Completion of a Category

Concretely, the groupoid completion of $\mathcal{C}$ is the Gabriel–Zisman localisation $\mathrm{Mor}(\mathcal{C})^{-1}\mathcal{C}$ of $\mathcal{C}$ at the set $\mathrm{Mor}(\mathcal{C})$ of all morphisms of $\mathcal{C}$; see Constructions With Categories, **??**.
(To be expanded upon later on.)

Proof 11.4.3.1.3 ▶ Proof of Construction 11.4.3.1.2

Omitted. ∎

Proposition 11.4.3.1.4 ► Properties of Groupoid Completion

Let $\mathcal{C}$ be a category.

1. *Functoriality.* The assignment $\mathcal{C} \mapsto \mathrm{K}_0(\mathcal{C})$ defines a functor

$$\mathrm{K}_0 \colon \mathsf{Cats} \to \mathsf{Grpd}.$$

2. *2-Functoriality.* The assignment $\mathcal{C} \mapsto \mathrm{K}_0(\mathcal{C})$ defines a 2-functor

$$\mathrm{K}_0 \colon \mathsf{Cats}_2 \to \mathsf{Grpd}_2.$$

3. *Adjointness.* We have an adjunction

$$(\mathrm{K}_0 \dashv \iota)\colon \quad \begin{tikzcd} \mathsf{Cats} \arrow[r, shift left, "\mathrm{K}_0"] \arrow[r, phantom, "\perp"] & \mathsf{Grpd}, \arrow[l, shift left, hook, "\iota"] \end{tikzcd}$$

witnessed by a bijection of sets

$$\mathrm{Hom}_{\mathsf{Grpd}}(\mathrm{K}_0(\mathcal{C}), \mathcal{G}) \cong \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{C}, \mathcal{G}),$$

natural in $\mathcal{C} \in \mathrm{Obj}(\mathsf{Cats})$ and $\mathcal{G} \in \mathrm{Obj}(\mathsf{Grpd})$, forming, together with the functor Core of Item 1 of Proposition 11.4.4.1.5, a triple adjunction

$$(\mathrm{K}_0 \dashv \iota \dashv \mathrm{Core})\colon \quad \begin{tikzcd} \mathsf{Cats} \arrow[r, bend left, "\mathrm{K}_0"] \arrow[r, bend right, "\mathrm{Core}"'] \arrow[r, phantom, "\perp"] \arrow[r, phantom, "\perp"] & \mathsf{Grpd}, \arrow[l, hook, "\iota"'] \end{tikzcd}$$

witnessed by bijections of sets

$$\begin{aligned} \mathrm{Hom}_{\mathsf{Grpd}}(\mathrm{K}_0(\mathcal{C}), \mathcal{G}) &\cong \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{C}, \mathcal{G}), \\ \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{G}, \mathcal{D}) &\cong \mathrm{Hom}_{\mathsf{Grpd}}(\mathcal{G}, \mathrm{Core}(\mathcal{D})), \end{aligned}$$

natural in $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$ and $\mathcal{G} \in \mathrm{Obj}(\mathsf{Grpd})$.

4. *2-Adjointness.* We have a 2-adjunction

$$(\mathrm{K}_0 \dashv \iota)\colon \quad \begin{tikzcd} \mathsf{Cats} \arrow[r, shift left, "\mathrm{K}_0"] \arrow[r, phantom, "\perp_2"] & \mathsf{Grpd}, \arrow[l, shift left, hook, "\iota"] \end{tikzcd}$$

witnessed by an isomorphism of categories

$$\mathrm{Fun}(\mathrm{K}_0(\mathcal{C}), \mathcal{G}) \cong \mathrm{Fun}(\mathcal{C}, \mathcal{G}),$$

natural in $\mathcal{C} \in \mathrm{Obj}(\mathsf{Cats})$ and $\mathcal{G} \in \mathrm{Obj}(\mathsf{Grpd})$, forming, together with the 2-functor Core of Item 2 of Proposition 11.4.4.1.5, a triple 2-adjunction

$$(\mathrm{K}_0 \dashv \iota \dashv \mathrm{Core}): \quad \begin{tikzcd} \mathsf{Cats} \arrow[r, bend left=60, "\mathrm{K}_0"] \arrow[r, bend right=60, "\mathrm{Core}"'] \arrow[r, phantom, "\perp_2"] \arrow[r, phantom, "\perp_2"'] & \mathsf{Grpd} \arrow[l, hook, "\iota"'] \end{tikzcd},$$

witnessed by isomorphisms of categories

$$\begin{aligned} \mathrm{Fun}(\mathrm{K}_0(\mathcal{C}), \mathcal{G}) &\cong \mathrm{Fun}(\mathcal{C}, \mathcal{G}), \\ \mathrm{Fun}(\mathcal{G}, \mathcal{D}) &\cong \mathrm{Fun}(\mathcal{G}, \mathrm{Core}(\mathcal{D})), \end{aligned}$$

natural in $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$ and $\mathcal{G} \in \mathrm{Obj}(\mathsf{Grpd})$.

5. *Interaction With Classifying Spaces.* We have an isomorphism of groupoids

$$\mathrm{K}_0(\mathcal{C}) \cong \Pi_{\leq 1}(|\mathrm{N}_\bullet(\mathcal{C})|),$$

natural in $\mathcal{C} \in \mathrm{Obj}(\mathsf{Cats})$; i.e. the diagram

$$\begin{tikzcd} \mathsf{Cats} \arrow[r, "\mathrm{K}_0"] \arrow[d, "\mathrm{N}_\bullet"'] \arrow[dr, phantom, "\wr"] & \mathsf{Grp} \\ \mathsf{sSets} \arrow[r, "|-|"'] & \mathbb{T} \arrow[u, "\Pi_{\leq 1}"'] \end{tikzcd}$$

commutes up to natural isomorphism.

6. *Symmetric Strong Monoidality With Respect to Coproducts.* The groupoid completion functor of Item 1 has a symmetric strong monoidal structure

$$\left(\mathrm{K}_0, \mathrm{K}_0^{\coprod}, \mathrm{K}_{0|\mathbb{1}}^{\coprod}\right): (\mathsf{Cats}, \textstyle\coprod, \emptyset_{\mathsf{cat}}) \to (\mathsf{Grpd}, \textstyle\coprod, \emptyset_{\mathsf{cat}})$$

being equipped with isomorphisms

$$\mathrm{K}^{\coprod}_{0|\mathcal{C},\mathcal{D}} \colon \mathrm{K}_0(\mathcal{C}) \coprod \mathrm{K}_0(\mathcal{D}) \overset{\sim}{\dashrightarrow} \mathrm{K}_0(\mathcal{C} \coprod \mathcal{D}),$$
$$\mathrm{K}^{\coprod}_{0|\mathbb{1}} \colon \emptyset_{\mathsf{cat}} \overset{\sim}{\dashrightarrow} \mathrm{K}_0(\emptyset_{\mathsf{cat}}),$$

natural in $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$.

7. *Symmetric Strong Monoidality With Respect to Products.* The groupoid completion functor of Item 1 has a symmetric strong monoidal structure

$$\left(\mathrm{K}_0, \mathrm{K}^{\times}_0, \mathrm{K}^{\times}_{0|\mathbb{1}}\right) \colon (\mathsf{Cats}, \times, \mathrm{pt}) \longrightarrow (\mathsf{Grpd}, \times, \mathrm{pt})$$

being equipped with isomorphisms

$$\mathrm{K}^{\times}_{0|\mathcal{C},\mathcal{D}} \colon \mathrm{K}_0(\mathcal{C}) \times \mathrm{K}_0(\mathcal{D}) \overset{\sim}{\dashrightarrow} \mathrm{K}_0(\mathcal{C} \times \mathcal{D}),$$
$$\mathrm{K}^{\times}_{0|\mathbb{1}} \colon \mathrm{pt} \overset{\sim}{\dashrightarrow} \mathrm{K}_0(\mathrm{pt}),$$

natural in $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$.

Proof 11.4.3.1.5 ► Proof of Proposition 11.4.3.1.4

Item 1: Functoriality

Omitted.

Item 2: 2-Functoriality

Omitted.

Item 3: Adjointness

Omitted.

Item 4: 2-Adjointness

Omitted.

Item 5: Interaction With Classifying Spaces

See Corollary 18.33 of https://web.ma.utexas.edu/users/dafr/M392C-2012/Notes/lecture18.pdf.

Item 6: Symmetric Strong Monoidality With Respect to Coproducts

Omitted.

Item 7: Symmetric Strong Monoidality With Respect to Products

Omitted.

11.4.4 The Core of a Category

Let $\mathcal{C}$ be a category.

Definition 11.4.4.1.1 ▶ The Core of a Category

The **core** of $\mathcal{C}$ is the pair $(\mathrm{Core}(\mathcal{C}), \iota_{\mathcal{C}})$ consisting of

- A groupoid $\mathrm{Core}(\mathcal{C})$;
- A functor $\iota_{\mathcal{C}} \colon \mathrm{Core}(\mathcal{C}) \hookrightarrow \mathcal{C}$;

satisfying the following universal property:

(**UP**) Given another such pair $(\mathcal{G}, i)$, there exists a unique functor $\mathcal{G} \xrightarrow{\exists!} \mathrm{Core}(\mathcal{C})$ making the diagram

$$\begin{tikzcd} & \mathrm{Core}(\mathcal{C}) \arrow[d, hook, "\iota_{\mathcal{C}}"] \\ \mathcal{G} \arrow[ur, dashed, "\exists!"] \arrow[r, "i"'] & \mathcal{C} \end{tikzcd}$$

commute.

Notation 11.4.4.1.2 ▶ Alternative Notation for the Core of a Category

We also write $\mathcal{C}^{\simeq}$ for $\mathrm{Core}(\mathcal{C})$.

Construction 11.4.4.1.3 ▶ Construction of the Core of a Category

The core of $\mathcal{C}$ is the wide subcategory of $\mathcal{C}$ spanned by the isomorphisms of $\mathcal{C}$, i.e. the category $\mathrm{Core}(\mathcal{C})$ where[1]

1. *Objects.* We have

$$\mathrm{Obj}(\mathrm{Core}(\mathcal{C})) \overset{\text{def}}{=} \mathrm{Obj}(\mathcal{C}).$$

2. *Morphisms.* The morphisms of $\mathrm{Core}(\mathcal{C})$ are the isomorphisms of $\mathcal{C}$.

[1] *Slogan:* The groupoid $\mathrm{Core}(\mathcal{C})$ is the maximal subgroupoid of $\mathcal{C}$.

Proof 11.4.4.1.4 ▶ Proof of Construction 11.4.4.1.3

This follows from the fact that functors preserve isomorphisms (Item 1 of Proposition 11.5.1.1.8). ∎

Proposition 11.4.4.1.5 ▶ Properties of the Core of a Category

Let $\mathcal{C}$ be a category.

1. *Functoriality.* The assignment $\mathcal{C} \mapsto \mathrm{Core}(\mathcal{C})$ defines a functor

$$\mathrm{Core}\colon \mathrm{Cats} \to \mathrm{Grpd}.$$

2. *2-Functoriality.* The assignment $\mathcal{C} \mapsto \mathrm{Core}(\mathcal{C})$ defines a 2-functor

$$\mathrm{Core}\colon \mathrm{Cats}_2 \to \mathrm{Grpd}_2.$$

3. *Adjointness.* We have an adjunction

$$(\iota \dashv \mathrm{Core})\colon \quad \begin{tikzcd} \mathrm{Grpd} \arrow[r, bend left, "\iota"] \arrow[r, phantom, "\perp"] & \mathrm{Cats}, \arrow[l, bend left, "\mathrm{Core}"] \end{tikzcd}$$

witnessed by a bijection of sets

$$\mathrm{Hom}_{\mathrm{Cats}}(\mathcal{G}, \mathcal{D}) \cong \mathrm{Hom}_{\mathrm{Grpd}}(\mathcal{G}, \mathrm{Core}(\mathcal{D})),$$

natural in $\mathcal{G} \in \mathrm{Obj}(\mathsf{Grpd})$ and $\mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$, forming, together with the functor K_0 of Item 1 of Proposition 11.4.3.1.4, a triple adjunction

$$(\mathrm{K}_0 \dashv \iota \dashv \mathrm{Core}): \quad \begin{tikzcd} \mathsf{Cats} \arrow[r, bend left=60, "\mathrm{K}_0"] \arrow[r, bend right=60, "\mathrm{Core}"'] \arrow[r, phantom, "\perp" near start] \arrow[r, phantom, "\perp" near end] & \mathsf{Grpd} \arrow[l, hook, "\iota"'] \end{tikzcd},$$

witnessed by bijections of sets

$$\begin{aligned} \mathrm{Hom}_{\mathsf{Grpd}}(\mathrm{K}_0(\mathcal{C}), \mathcal{G}) &\cong \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{C}, \mathcal{G}), \\ \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{G}, \mathcal{D}) &\cong \mathrm{Hom}_{\mathsf{Grpd}}(\mathcal{G}, \mathrm{Core}(\mathcal{D})), \end{aligned}$$

natural in $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$ and $\mathcal{G} \in \mathrm{Obj}(\mathsf{Grpd})$.

4. *2-Adjointness.* We have an adjunction

$$(\iota \dashv \mathrm{Core}): \quad \begin{tikzcd} \mathsf{Grpd} \arrow[r, hook, bend left, "\iota"] \arrow[r, phantom, "\perp_2"] & \mathsf{Cats} \arrow[l, bend left, "\mathrm{Core}"] \end{tikzcd},$$

witnessed by an isomorphism of categories

$$\mathrm{Fun}(\mathcal{G}, \mathcal{D}) \cong \mathrm{Fun}(\mathcal{G}, \mathrm{Core}(\mathcal{D})),$$

natural in $\mathcal{G} \in \mathrm{Obj}(\mathsf{Grpd})$ and $\mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$, forming, together with the 2-functor K_0 of Item 2 of Proposition 11.4.3.1.4, a triple 2-adjunction

$$(\mathrm{K}_0 \dashv \iota \dashv \mathrm{Core}): \quad \begin{tikzcd} \mathsf{Cats} \arrow[r, bend left=60, "\mathrm{K}_0"] \arrow[r, bend right=60, "\mathrm{Core}"'] \arrow[r, phantom, "\perp_2" near start] \arrow[r, phantom, "\perp_2" near end] & \mathsf{Grpd} \arrow[l, hook, "\iota"'] \end{tikzcd},$$

witnessed by isomorphisms of categories

$$\begin{aligned} \mathrm{Fun}(\mathrm{K}_0(\mathcal{C}), \mathcal{G}) &\cong \mathrm{Fun}(\mathcal{C}, \mathcal{G}), \\ \mathrm{Fun}(\mathcal{G}, \mathcal{D}) &\cong \mathrm{Fun}(\mathcal{G}, \mathrm{Core}(\mathcal{D})), \end{aligned}$$

natural in $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$ and $\mathcal{G} \in \mathrm{Obj}(\mathsf{Grpd})$.

5. *Symmetric Strong Monoidality With Respect to Products.* The core functor of Item 1 has a symmetric strong monoidal structure

$$\left(\mathrm{Core}, \mathrm{Core}^{\times}, \mathrm{Core}^{\times}_{\mathbb{1}}\right) \colon (\mathsf{Cats}, \times, \mathrm{pt}) \to (\mathsf{Grpd}, \times, \mathrm{pt})$$

being equipped with isomorphisms

$$\mathrm{Core}^{\times}_{\mathcal{C},\mathcal{D}} \colon \mathrm{Core}(\mathcal{C}) \times \mathrm{Core}(\mathcal{D}) \overset{\sim}{\dashrightarrow} \mathrm{Core}(\mathcal{C} \times \mathcal{D}),$$
$$\mathrm{Core}^{\times}_{\mathbb{1}} \colon \mathrm{pt} \overset{\sim}{\dashrightarrow} \mathrm{Core}(\mathrm{pt}),$$

natural in $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$.

6. *Symmetric Strong Monoidality With Respect to Coproducts.* The core functor of Item 1 has a symmetric strong monoidal structure

$$\left(\mathrm{Core}, \mathrm{Core}^{\coprod}, \mathrm{Core}^{\coprod}_{\mathbb{1}}\right) \colon (\mathsf{Cats}, \coprod, \emptyset_{\mathsf{cat}}) \to (\mathsf{Grpd}, \coprod, \emptyset_{\mathsf{cat}})$$

being equipped with isomorphisms

$$\mathrm{Core}^{\coprod}_{\mathcal{C},\mathcal{D}} \colon \mathrm{Core}(\mathcal{C}) \coprod \mathrm{Core}(\mathcal{D}) \overset{\sim}{\dashrightarrow} \mathrm{Core}(\mathcal{C} \coprod \mathcal{D}),$$
$$\mathrm{Core}^{\coprod}_{\mathbb{1}} \colon \emptyset_{\mathsf{cat}} \overset{\sim}{\dashrightarrow} \mathrm{Core}(\emptyset_{\mathsf{cat}}),$$

natural in $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$.

Proof 11.4.4.1.6 ► Proof of Proposition 11.4.4.1.5

Item 1: Functoriality

Omitted.

Item 2: 2-Functoriality

Omitted.

Item 3: Adjointness

Omitted.

Item 4: 2-Adjointness

Omitted.

Item 5: Symmetric Strong Monoidality With Respect to Products

Omitted.

Item 6: Symmetric Strong Monoidality With Respect to Coproducts

Omitted. ∎

11.5 Functors

11.5.1 Foundations

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.5.1.1.1 ► Functors

A **functor** $F\colon \mathcal{C} \to \mathcal{D}$ **from** $\mathcal{C}$ **to** $\mathcal{D}$[1] consists of:

1. *Action on Objects.* A map of sets

$$F\colon \operatorname{Obj}(\mathcal{C}) \to \operatorname{Obj}(\mathcal{D}),$$

called the **action on objects of** F.

2. *Action on Morphisms.* For each $A, B \in \operatorname{Obj}(\mathcal{C})$, a map

$$F_{A,B}\colon \operatorname{Hom}_{\mathcal{C}}(A, B) \to \operatorname{Hom}_{\mathcal{D}}(F(A), F(B)),$$

called the **action on morphisms of** F **at** (A, B)[2].

satisfying the following conditions:

1. *Preservation of Identities.* For each $A \in \operatorname{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
\mathrm{pt} \arrow[d, "\mathbb{1}^{\mathcal{C}}_{A}"'] \arrow[dr, "\mathbb{1}^{\mathcal{D}}_{F(A)}"] & \\
\operatorname{Hom}_{\mathcal{C}}(A, A) \arrow[r, "F_{A,A}"'] & \operatorname{Hom}_{\mathcal{D}}(F(A), F(A))
\end{tikzcd}$$

commutes, i.e. we have

$$F(\mathrm{id}_A) = \mathrm{id}_{F(A)}\,.$$

2. *Preservation of Composition.* For each $A, B, C \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\mathcal{C}}(B,C) \times \mathrm{Hom}_{\mathcal{C}}(A,B) \arrow[r, "\circ^{\mathcal{C}}_{A,B,C}"] \arrow[d, "F_{B,C} \times F_{A,B}"'] & \mathrm{Hom}_{\mathcal{C}}(A,C) \arrow[d, "F_{A,C}"] \\
\mathrm{Hom}_{\mathcal{D}}(F(B),F(C)) \times \mathrm{Hom}_{\mathcal{D}}(F(A),F(B)) \arrow[r, "\circ^{\mathcal{D}}_{F(A),F(B),F(C)}"'] & \mathrm{Hom}_{\mathcal{D}}(F(A),F(C))
\end{tikzcd}$$

commutes, i.e. for each composable pair (g, f) of morphisms of $\mathcal{C}$, we have

$$F(g \circ f) = F(g) \circ F(f).$$

[1] *Further Terminology:* Also called a **covariant functor**.
[2] *Further Terminology:* Also called **action on Hom-sets of** F **at** (A, B).

Notation 11.5.1.1.2 ► Subscript and Superscript Notation for Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and write $\mathcal{C}^{\mathrm{op}}$ for the opposite category of $\mathcal{C}$ of Constructions With Categories, **??**.

1. Given a functor
$$F\colon \mathcal{C} \to \mathcal{D},$$
we also write F_A for $F(A)$.

2. Given a functor
$$F\colon \mathcal{C}^{\mathrm{op}} \to \mathcal{D},$$
we also write F^A for $F(A)$.

3. Given a functor
$$F\colon \mathcal{C} \times \mathcal{C} \to \mathcal{D},$$
we also write $F_{A,B}$ for $F(A, B)$.

4. Given a functor
$$F\colon \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D},$$
we also write F^A_B for $F(A, B)$.

We employ a similar notation for morphisms, writing e.g. F_f for $F(f)$ given a functor $F\colon \mathcal{C} \to \mathcal{D}$.

Following the notation $[\![x \mapsto f(x)]\!]$ for a function $f\colon X \to Y$ introduced in Sets, Notation 3.1.1.1.2, we will sometimes denote a functor $F\colon \mathcal{C} \to \mathcal{D}$ by

$$F \stackrel{\text{def}}{=} [\![A \mapsto F(A)]\!],$$

specially when the action on morphisms of F is clear from its action on objects.

Example 11.5.1.1.4 ▶ Identity Functors

The **identity functor** of a category $\mathcal{C}$ is the functor $\mathrm{id}_{\mathcal{C}}\colon \mathcal{C} \to \mathcal{C}$ where

1. *Action on Objects.* For each $A \in \mathrm{Obj}(\mathcal{C})$, we have

$$\mathrm{id}_{\mathcal{C}}(A) \stackrel{\text{def}}{=} A.$$

2. *Action on Morphisms.* For each $A, B \in \mathrm{Obj}(\mathcal{C})$, the action on morphisms

$$(\mathrm{id}_{\mathcal{C}})_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \underbrace{\mathrm{Hom}_{\mathcal{C}}(\mathrm{id}_{\mathcal{C}}(A), \mathrm{id}_{\mathcal{C}}(B))}_{\stackrel{\text{def}}{=}\mathrm{Hom}_{\mathcal{C}}(A,B)}$$

of $\mathrm{id}_{\mathcal{C}}$ at (A, B) is defined by

$$(\mathrm{id}_{\mathcal{C}})_{A,B} \stackrel{\text{def}}{=} \mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(A,B)}.$$

Proof 11.5.1.1.5 ▶ Proof of Example 11.5.1.1.4

Preservation of Identities

We have $\mathrm{id}_{\mathcal{C}}(\mathrm{id}_A) \stackrel{\text{def}}{=} \mathrm{id}_A$ for each $A \in \mathrm{Obj}(\mathcal{C})$ by definition.

Preservation of Compositions

For each composable pair $A \xrightarrow{f} B \xrightarrow{g} B$ of morphisms of $\mathcal{C}$, we have

$$\begin{aligned}\mathrm{id}_{\mathcal{C}}(g \circ f) &\stackrel{\text{def}}{=} g \circ f \\ &\stackrel{\text{def}}{=} \mathrm{id}_{\mathcal{C}}(g) \circ \mathrm{id}_{\mathcal{C}}(f).\end{aligned}$$

This finishes the proof. ∎

Definition 11.5.1.1.6 ▶ Composition of Functors

The **composition** of two functors $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$ is the functor $G \circ F$ where

- *Action on Objects.* For each $A \in \mathrm{Obj}(\mathcal{C})$, we have

$$[G \circ F](A) \overset{\text{def}}{=} G(F(A)).$$

- *Action on Morphisms.* For each $A, B \in \mathrm{Obj}(\mathcal{C})$, the action on morphisms

$$(G \circ F)_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{E}}\left(G_{F_A}, G_{F_B}\right)$$

  of $G \circ F$ at (A, B) is defined by

$$[G \circ F](f) \overset{\text{def}}{=} G(F(f)).$$

Proof 11.5.1.1.7 ▶ Proof of Definition 11.5.1.1.6

Preservation of Identities

For each $A \in \mathrm{Obj}(\mathcal{C})$, we have

$$\begin{aligned} G_{F_{\mathrm{id}_A}} &= G_{\mathrm{id}_{F_A}} && \text{(functoriality of } F\text{)} \\ &= \mathrm{id}_{G_{F_A}}. && \text{(functoriality of } G\text{)} \end{aligned}$$

Preservation of Composition

For each composable pair (g, f) of morphisms of $\mathcal{C}$, we have

$$\begin{aligned} G_{F_{g \circ f}} &= G_{F_g \circ F_f} && \text{(functoriality of } F\text{)} \\ &= G_{F_g} \circ G_{F_f}. && \text{(functoriality of } G\text{)} \end{aligned}$$

This finishes the proof. ∎

Proposition 11.5.1.1.8 ▶ Elementary Properties of Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Preservation of Isomorphisms.* If f is an isomorphism in $\mathcal{C}$, then $F(f)$ is

an isomorphism in $\mathcal{D}$.[1]

[1]When the converse holds, we call F *conservative*, see Definition 11.6.4.1.1.

Proof 11.5.1.1.9 ▶ Proof of Proposition 11.5.1.1.8

Item 1: Preservation of Isomorphisms

Indeed, we have

$$\begin{aligned} F(f)^{-1} \circ F(f) &= F\left(f^{-1} \circ f\right) \\ &= F(\mathrm{id}_A) \\ &= \mathrm{id}_{F(A)} \end{aligned}$$

and

$$\begin{aligned} F(f) \circ F(f)^{-1} &= F\left(f \circ f^{-1}\right) \\ &= F(\mathrm{id}_B) \\ &= \mathrm{id}_{F(B)}, \end{aligned}$$

showing $F(f)$ to be an isomorphism. ∎

11.5.2 Contravariant Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and let $\mathcal{C}^{\mathrm{op}}$ denote the opposite category of $\mathcal{C}$ of Constructions With Categories, **??**.

Definition 11.5.2.1.1 ▶ Contravariant Functors

A **contravariant functor** from $\mathcal{C}$ to $\mathcal{D}$ is a functor from $\mathcal{C}^{\mathrm{op}}$ to $\mathcal{D}$.

Remark 11.5.2.1.2 ▶ Unwinding Definition 11.5.2.1.1

In detail, a **contravariant functor** from $\mathcal{C}$ to $\mathcal{D}$ consists of:

1. *Action on Objects.* A map of sets

$$F\colon \mathrm{Obj}(\mathcal{C}) \to \mathrm{Obj}(\mathcal{D}),$$

called the **action on objects of** F.

2. *Action on Morphisms.* For each $A, B \in \mathrm{Obj}(\mathcal{C})$, a map

$$F_{A,B}\colon\ \mathrm{Hom}_{\mathcal{C}}(A,B) \longrightarrow \mathrm{Hom}_{\mathcal{D}}(F(B),F(A)),$$

called the **action on morphisms of** F **at** (A,B).

satisfying the following conditions:

1. *Preservation of Identities.* For each $A \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
\mathrm{pt} \arrow[d, "\mathbb{1}^{\mathcal{C}}_{A}"'] \arrow[dr, "\mathbb{1}^{\mathcal{D}}_{F(A)}"] & \\
\mathrm{Hom}_{\mathcal{C}}(A,A) \arrow[r, "F_{A,A}"'] & \mathrm{Hom}_{\mathcal{D}}(F(A),F(A))
\end{tikzcd}$$

commutes, i.e. we have

$$F(\mathrm{id}_A) = \mathrm{id}_{F(A)}\,.$$

2. *Preservation of Composition.* For each $A, B, C \in \mathrm{Obj}(\mathcal{C})$, the diagram

$$\begin{tikzcd}
 & \mathrm{Hom}_{\mathcal{D}}(F(C),F(B)) \times \mathrm{Hom}_{\mathcal{D}}(F(B),F(A)) \arrow[dr, dashed, "\sim", "\sigma^{\mathsf{Sets}}_{\mathrm{Hom}_{\mathcal{D}}(F(C),F(B)),\mathrm{Hom}_{\mathcal{D}}(F(B),F(A))}"'] & \\
\mathrm{Hom}_{\mathcal{C}}(B,C) \times \mathrm{Hom}_{\mathcal{C}}(A,B) \arrow[ur, "F_{B,C}\times F_{A,B}"] \arrow[dd, "\circ^{\mathcal{C}}_{A,B,C}"'] & & \mathrm{Hom}_{\mathcal{D}}(F(B),F(A)) \times \mathrm{Hom}_{\mathcal{D}}(F(C),F(B)) \arrow[dd, "\circ^{\mathcal{D}}_{F(C),F(B),F(A)}"] \\
 & & \\
\mathrm{Hom}_{\mathcal{C}}(A,C) \arrow[r, "F_{A,C}"'] & & \mathrm{Hom}_{\mathcal{D}}(F(C),F(A))
\end{tikzcd}$$

commutes, i.e. for each composable pair (g,f) of morphisms of $\mathcal{C}$, we have

$$F(g \circ f) = F(f) \circ F(g).$$

Remark 11.5.2.1.3 ► On the Term Contravariant Functor

Throughout this work we will not use the term "contravariant' functor, speaking instead simply of functors $F\colon \mathcal{C}^{\mathsf{op}} \longrightarrow \mathcal{D}$. We will usually, however, write

$$F_{A,B}\colon \operatorname{Hom}_{\mathcal{C}}(A,B) \longrightarrow \operatorname{Hom}_{\mathcal{D}}(F(B),F(A))$$

for the action on morphisms

$$F_{A,B}\colon \operatorname{Hom}_{\mathcal{C}^{\mathsf{op}}}(A,B) \longrightarrow \operatorname{Hom}_{\mathcal{D}}(F(A),F(B))$$

of F, as well as write $F(g \circ f) = F(f) \circ F(g)$.

11.5.3 Forgetful Functors

Definition 11.5.3.1.1 ► Forgetful Functors

There isn,t a precise definition of a **forgetful functor**.

Remark 11.5.3.1.2 ► Unwinding Definition 11.5.3.1.1

Despite there not being a formal or precise definition of a forgetful functor, the term is often very useful in practice, similarly to the word "canonical'. The idea is that a "forgetful functor' is a functor that forgets structure or properties, and is best explained through examples, such as the ones below (see Examples 11.5.3.1.3 and 11.5.3.1.4).

Example 11.5.3.1.3 ► Forgetful Functors That Forget Structure

Examples of forgetful functors that forget structure include:

1. *Forgetting Group Structures.* The functor $\mathsf{Grp} \longrightarrow \mathsf{Sets}$ sending a group (G, μ_G, η_G) to its underlying set G, forgetting the multiplication and unit maps μ_G and η_G of G.

2. *Forgetting Topologies.* The functor $\mathsf{Top} \longrightarrow \mathsf{Sets}$ sending a topological space $(X, \mathcal{T}_X)$ to its underlying set X, forgetting the topology $\mathcal{T}_X$.

3. *Forgetting Fibrations.* The functor $\mathsf{FibSets}(K) \longrightarrow \mathsf{Sets}$ sending a K-fibred set $\phi_X\colon X \longrightarrow K$ to the set X, forgetting the map ϕ_X and the base

set K.

Example 11.5.3.1.4 ► Forgetful Functors That Forget Properties

Examples of forgetful functors that forget properties include:

1. *Forgetting Commutativity.* The inclusion functor $\iota\colon \mathrm{CMon} \hookrightarrow \mathrm{Mon}$ which forgets the property of being commutative.

2. *Forgetting Inverses.* The inclusion functor $\iota\colon \mathrm{Grp} \hookrightarrow \mathrm{Mon}$ which forgets the property of having inverses.

Notation 11.5.3.1.5 ► Notation For Forgetful Functors That Forget Structure

Throughout this work, we will denote forgetful functors that forget structure by 忘, e.g. as in

$$忘\colon \mathrm{Grp} \to \mathrm{Sets}.$$

The symbol 忘, pronounced *wasureru* (see Item 1 of Remark 11.5.3.1.6 below), means *to forget*, and is a kanji found in the following words in Japanese and Chinese:

1. 忘れる, transcribed as *wasureru*, meaning *to forget*.

2. 忘却関手, transcribed as *boukyaku kanshu*, meaning *forgetful functor*.

3. 忘记 or 忘記, transcribed as *wàngjì*, meaning *to forget*.

4. 遗忘函子 or 遺忘函子, transcribed as *yíwàng hánzǐ*, meaning *forgetful functor*.

Remark 11.5.3.1.6 ► Pronunciation of the Words in Notation 11.5.3.1.5

Here we collect the pronunciation of the words in Notation 11.5.3.1.5 for accuracy and completeness.

1. Pronunciation of 忘れる:

   - See here.
   - IPA broad transcription: [wäsɯɾɛɾɯ].
   - IPA narrow transcription: [ɯ̟ˀäsiˀɾɛɾɯ̟ˀ].

2. Pronunciation of 忘却関手: Pronunciation:
   - See here.
   - IPA broad transcription: [bo̞ːkʲäkɯ kɐ̃ũɻɕɯ].
   - IPA narrow transcription: [bo̞ːkʲäkɯ̟ˀ kɐ̃ũɻɕɯˀ].
3. Pronunciation of 忘记:
   - See here.
   - Broad IPA transcription: [wɑŋtɕi].
   - Sinological IPA transcription: [wɑŋ$^{51-53}$t͡ɕi^{51}].
4. Pronunciation of 遗忘函子:
   - See here.
   - Broad IPA transcription: [iwɑŋ xänt͡ʂʐɨ].
   - Sinological IPA transcription: [i^{35}wɑŋ^{51} xän^{35}t͡ʂʐ$^{214-21(4)}$].

11.5.4 The Natural Transformation Associated to a Functor

Definition 11.5.4.1.1 ▶ The Natural Transformation Associated to a Functor

Every functor $F\colon \mathcal{C} \to \mathcal{D}$ defines a natural transformation[1]

$$F^{\dagger}\colon \mathrm{Hom}_{\mathcal{C}} \Longrightarrow \mathrm{Hom}_{\mathcal{D}} \circ (F^{\mathrm{op}} \times F),$$

$$\begin{tikzcd}
\mathcal{C}^{\mathrm{op}} \times \mathcal{C} \arrow[rr, "F^{\mathrm{op}} \times F"] \arrow[dr, "\mathrm{Hom}_{\mathcal{C}}"'] & \arrow[d, phantom, "\overset{F^{\dagger}}{\Longrightarrow}"] & \mathcal{D}^{\mathrm{op}} \times \mathcal{D} \arrow[dl, "\mathrm{Hom}_{\mathcal{D}}"] \\
& \mathsf{Sets}, &
\end{tikzcd}$$

called the **natural transformation associated to** F, consisting of the collection

$$\left\{F^{\dagger}_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A,B) \to \mathrm{Hom}_{\mathcal{D}}(F_A, F_B)\right\}_{(A,B)\in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}}\times\mathcal{C})}$$

with

$$F^{\dagger}_{A,B} \overset{\text{def}}{=} F_{A,B}.$$

[1]This is the 1-categorical version of Constructions With Sets, **??** of **??**.

Proof 11.5.4.1.2 ▶ Proof of Definition 11.5.4.1.1

The naturality condition for $F^{\dagger}$ is the requirement that for each morphism

$$(\phi, \psi)\colon (X, Y) \to (A, B)$$

of $\mathcal{C}^{\mathrm{op}} \times \mathcal{C}$, the diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\mathcal{C}}(X,Y) \arrow[r, "\phi^* \circ \psi_* = \psi_* \circ \phi^*"] \arrow[d, "F_{X,Y}"'] & \mathrm{Hom}_{\mathcal{C}}(A,B) \arrow[d, "F_{A,B}"] \\
\mathrm{Hom}_{\mathcal{D}}(F_X, F_Y) \arrow[r, "F(\phi)^* \circ F(\psi)_* = F(\psi)_* \circ F(\phi)^*"'] & \mathrm{Hom}_{\mathcal{D}}(F_A, F_B),
\end{tikzcd}$$

acting on elements as

$$\begin{tikzcd}
f \arrow[r, mapsto] \arrow[d, mapsto] & \psi \circ f \circ \phi \arrow[d, mapsto] \\
F(f) \arrow[r, mapsto] & F(\psi) \circ F(f) \circ F(\psi) = F(\psi \circ f \circ \phi)
\end{tikzcd}$$

commutes, which follows from the functoriality of F. ∎

Proposition 11.5.4.1.3 ► Properties of Natural Transformations Associated to Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$ be functors.

1. *Interaction With Natural Isomorphisms.* The following conditions are equivalent:

   (a) The natural transformation $F^\dagger\colon \mathrm{Hom}_{\mathcal{C}} \Longrightarrow \mathrm{Hom}_{\mathcal{D}} \circ (F^{\mathrm{op}} \times F)$ associated to F is a natural isomorphism.

   (b) The functor F is fully faithful.

2. *Interaction With Composition.* We have an equality of pasting diagrams

$$\begin{tikzcd}
\mathcal{C}^{\mathrm{op}} \times \mathcal{C} \arrow[r, "F^{\mathrm{op}}\times F"] \arrow[dr, "\mathrm{Hom}_{\mathcal{C}}"'] & \mathcal{D}^{\mathrm{op}} \times \mathcal{D} \arrow[r, "G^{\mathrm{op}}\times G"] \arrow[d, "\mathrm{Hom}_{\mathcal{D}}"] \arrow[dl, phantom, "\overset{F^\dagger}{\Nearrow}"] & \mathcal{E}^{\mathrm{op}} \times \mathcal{E} \arrow[dl, "\mathrm{Hom}_{\mathcal{E}}"] \arrow[dl, phantom, "\overset{G^\dagger}{\Nearrow}"] \\
& \mathrm{Sets} &
\end{tikzcd}
=
\begin{tikzcd}
\mathcal{C}^{\mathrm{op}} \times \mathcal{C} \arrow[r, "(G\circ F)^{\mathrm{op}}\times(G\circ F)"] \arrow[dr, "\mathrm{Hom}_{\mathcal{C}}"'] & \mathcal{E}^{\mathrm{op}} \times \mathcal{E}, \arrow[d, "\mathrm{Hom}_{\mathcal{E}}"] \arrow[dl, phantom, "\overset{(G\circ F)^\dagger}{\Nearrow}"] \\
& \mathrm{Sets}
\end{tikzcd}$$

   in Cats_2, i.e. we have

$$(G \circ F)^\dagger = \left(G^\dagger \star \mathrm{id}_{F^{\mathrm{op}}\times F}\right) \circ F^\dagger.$$

3. *Interaction With Identities.* We have

$$\mathrm{id}_{\mathcal{C}}^\dagger = \mathrm{id}_{\mathrm{Hom}_{\mathcal{C}}(-_1,-_2)},$$

   i.e. the natural transformation associated to $\mathrm{id}_{\mathcal{C}}$ is the identity natural transformation of the functor $\mathrm{Hom}_{\mathcal{C}}(-_1, -_2)$.

Proof 11.5.4.1.4 ▶ Proof of Proposition 11.5.4.1.3

Item 1: Interaction With Natural Isomorphisms

Clear.

Item 2: Interaction With Composition

Clear.

Item 3: Interaction With Identities

Clear. ∎

11.6 Conditions on Functors

11.6.1 Faithful Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.6.1.1.1 ▶ Faithful Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **faithful** if, for each $A, B \in \mathrm{Obj}(\mathcal{C})$, the action on morphisms

$$F_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F_A, F_B)$$

of F at (A, B) is injective.

Proposition 11.6.1.1.2 ▶ Properties of Faithful Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$ be functors.

1. *Interaction With Composition.* If F and G are faithful, then so is $G \circ F$.
2. *Interaction With Postcomposition.* The following conditions are equivalent:
   (a) The functor $F\colon \mathcal{C} \to \mathcal{D}$ is faithful.
   (b) For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

   $$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

   is faithful.

(c) The functor $F\colon \mathcal{C} \to \mathcal{D}$ is a representably faithful morphism in Cats_2 in the sense of Types of Morphisms in Bicategories, Definition 13.1.1.1.1.

3. *Interaction With Precomposition I.* Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

   (a) If F is faithful, then the precomposition functor

   $$F^*\colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

   can fail to be faithful.

   (b) Conversely, if the precomposition functor

   $$F^*\colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

   is faithful, then F ***can fail*** to be faithful.

4. *Interaction With Precomposition II.* If F is essentially surjective, then the precomposition functor

   $$F^*\colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

   is faithful.

5. *Interaction With Precomposition III.* The following conditions are equivalent:

   (a) For each $\mathcal{X} \in \mathsf{Obj}(\mathsf{Cats})$, the precomposition functor

   $$F^*\colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

   is faithful.

   (b) For each $\mathcal{X} \in \mathsf{Obj}(\mathsf{Cats})$, the precomposition functor

   $$F^*\colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

   is conservative.

   (c) For each $\mathcal{X} \in \mathsf{Obj}(\mathsf{Cats})$, the precomposition functor

   $$F^*\colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

   is monadic.

(d) The functor $F\colon \mathcal{C} \to \mathcal{D}$ is a corepresentably faithful morphism in Cats_2 in the sense of Types of Morphisms in Bicategories, Definition 13.2.1.1.1.

(e) The components

$$\eta_G\colon G \Longrightarrow \mathsf{Ran}_F(G \circ F)$$

of the unit

$$\eta\colon \mathsf{id}_{\mathsf{Fun}(\mathcal{D},\mathcal{X})} \Longrightarrow \mathsf{Ran}_F \circ F^*$$

of the adjunction $F^* \dashv \mathsf{Ran}_F$ are all monomorphisms.

(f) The components

$$\epsilon_G\colon \mathsf{Lan}_F(G \circ F) \Longrightarrow G$$

of the counit

$$\epsilon\colon \mathsf{Lan}_F \circ F^* \Longrightarrow \mathsf{id}_{\mathsf{Fun}(\mathcal{D},\mathcal{X})}$$

of the adjunction $\mathsf{Lan}_F \dashv F^*$ are all epimorphisms.

(g) The functor F is dominant (Definition 11.7.1.1.1), i.e. every object of $\mathcal{D}$ is a retract of some object in $\mathsf{Im}(F)$:

($\star$) For each $B \in \mathsf{Obj}(\mathcal{D})$, there exist:
- An object A of $\mathcal{C}$;
- A morphism $s\colon B \to F(A)$ of $\mathcal{D}$;
- A morphism $r\colon F(A) \to B$ of $\mathcal{D}$;

such that $r \circ s = \mathsf{id}_B$.

Proof 11.6.1.1.3 ► Proof of Proposition 11.6.1.1.2

Item 1: Interaction With Composition

Since the map

$$(G \circ F)_{A,B} \colon \operatorname{Hom}_{\mathcal{C}}(A, B) \to \operatorname{Hom}_{\mathcal{D}}\left(G_{F_A}, G_{F_B}\right),$$

defined as the composition

$$\operatorname{Hom}_{\mathcal{C}}(A, B) \xrightarrow{F_{A,B}} \operatorname{Hom}_{\mathcal{D}}(F_A, F_B) \xrightarrow{G_{F(A),F(B)}} \operatorname{Hom}_{\mathcal{D}}\left(G_{F_A}, G_{F_B}\right),$$

is a composition of injective functions, it follows from **??** that it is also injective. Therefore $G \circ F$ is faithful.

Item 2: Interaction With Postcomposition

Omitted.

Item 3: Interaction With Precomposition I

See [MSE 733163] for Item 3a. Item 3b follows from Item 4 and the fact that there are essentially surjective functors that are not faithful.

Item 4: Interaction With Precomposition II

Omitted, but see `https://unimath.github.io/doc/UniMath/d4de26f//UniMath.CategoryTheory.precomp_fully_faithful.html` for a formalised proof.

Item 5: Interaction With Precomposition III

We claim Items 5a to 5g are equivalent:

- *Items 5a and 5d Are Equivalent:* This is true by the definition of corepresentably faithful morphism; see Types of Morphisms in Bicategories, Definition 13.2.1.1.1.
- *Items 5a to 5c and 5g Are Equivalent:* See [Adá–01, Proposition 4.1] or alternatively [Fre09, Lemmas 3.1 and 3.2] for the equivalence between Items 5a and 5g.
- *Items 5a, 5e and 5f Are Equivalent:* See **??**, **??** of **??**.

This finishes the proof. ∎

11.6.2 Full Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.6.2.1.1 ▶ Full Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **full** if, for each $A, B \in \mathrm{Obj}(\mathcal{C})$, the action on morphisms

$$F_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F_A, F_B)$$

of F at (A, B) is surjective.

Proposition 11.6.2.1.2 ▶ Properties of Full Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$ be functors.

1. *Interaction With Composition.* If F and G are full, then so is $G \circ F$.
2. *Interaction With Postcomposition I.* If F is full, then the postcomposition functor
$$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$
can fail to be full.
3. *Interaction With Postcomposition II.* If, for each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor
$$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$
is full, then F is also full.
4. *Interaction With Precomposition I.* If F is full, then the precomposition functor
$$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$
can fail to be full.
5. *Interaction With Precomposition II.* If, for each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor
$$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$
is full, then F ***can fail*** to be full.
6. *Interaction With Precomposition III.* If F is essentially surjective and full, then the precomposition functor
$$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$
is full (and also faithful by Item 4 of Proposition 11.6.1.1.2).

7. *Interaction With Precomposition IV.* The following conditions are equivalent:

   (a) For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

   $$F^*\colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

   is full.

   (b) The functor $F\colon \mathcal{C} \to \mathcal{D}$ is a corepresentably full morphism in Cats_2 in the sense of Types of Morphisms in Bicategories, Definition 13.2.1.1.1.

   (c) The components

   $$\eta_G\colon G \Longrightarrow \mathsf{Ran}_F(G \circ F)$$

   of the unit

   $$\eta\colon \mathrm{id}_{\mathsf{Fun}(\mathcal{D},\mathcal{X})} \Longrightarrow \mathsf{Ran}_F \circ F^*$$

   of the adjunction $F^* \dashv \mathsf{Ran}_F$ are all retractions/split epimorphisms.

   (d) The components

   $$\epsilon_G\colon \mathsf{Lan}_F(G \circ F) \Longrightarrow G$$

   of the counit

   $$\epsilon\colon \mathsf{Lan}_F \circ F^* \Longrightarrow \mathrm{id}_{\mathsf{Fun}(\mathcal{D},\mathcal{X})}$$

   of the adjunction $\mathsf{Lan}_F \dashv F^*$ are all sections/split monomorphisms.

   (e) For each $B \in \mathrm{Obj}(\mathcal{D})$, there exist:

   - An object A_B of $\mathcal{C}$;
   - A morphism $s_B\colon B \to F(A_B)$ of $\mathcal{D}$;
   - A morphism $r_B\colon F(A_B) \to B$ of $\mathcal{D}$;

   satisfying the following condition:

($\star$) For each $A \in \mathrm{Obj}(\mathcal{C})$ and each pair of morphisms

$$\begin{aligned} r &\colon F(A) \to B, \\ s &\colon B \to F(A) \end{aligned}$$

of $\mathcal{D}$, we have

$$[(A_B, s_B, r_B)] = [(A, s, r \circ s_B \circ r_B)]$$

in $\int^{A \in \mathcal{C}} h^{B'}_{F_A} \times h^{F_A}_B$.

Proof 11.6.2.1.3 ▶ Proof of Proposition 11.6.2.1.2

Item 1: Interaction With Composition

Since the map

$$(G \circ F)_{A,B} \colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}\left(G_{F_A}, G_{F_B}\right),$$

defined as the composition

$$\mathrm{Hom}_{\mathcal{C}}(A, B) \xrightarrow{F_{A,B}} \mathrm{Hom}_{\mathcal{D}}(F_A, F_B) \xrightarrow{G_{F(A),F(B)}} \mathrm{Hom}_{\mathcal{D}}\left(G_{F_A}, G_{F_B}\right),$$

is a composition of surjective functions, it follows from **??** that it is also surjective. Therefore $G \circ F$ is full.

Item 2: Interaction With Postcomposition I

We follow the proof (completely formalised in cubical Agda!) given by Naïm Camille Favier in [**favier:postcompose-not-full**]. Let $\mathcal{C}$ be the category where:

- *Objects.* We have $\mathrm{Obj}(\mathcal{C}) = \{A, B\}$.
- *Morphisms.* We have

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}}(A, A) &= \{e_A, \mathrm{id}_A\}, \\ \mathrm{Hom}_{\mathcal{C}}(B, B) &= \{e_B, \mathrm{id}_B\}, \\ \mathrm{Hom}_{\mathcal{C}}(A, B) &= \{f, g\}, \\ \mathrm{Hom}_{\mathcal{C}}(B, A) &= \emptyset. \end{aligned}$$

- *Composition*. The nontrivial compositions in $\mathcal{C}$ are the following:

$$\begin{aligned} e_A \circ e_A &= \mathrm{id}_A, \quad f \circ e_A = g, \quad e_B \circ f = f, \\ e_B \circ e_B &= \mathrm{id}_B, \quad g \circ e_A = f, \quad e_B \circ g = g. \end{aligned}$$

We may picture $\mathcal{C}$ as follows:

$$\begin{tikzcd} A \arrow[loop left, "e_A"] \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \arrow[loop right, "e_B"] \end{tikzcd}.$$

Next, let $\mathcal{D}$ be the walking arrow category $\mathbb{1}$ of Definition 11.2.5.1.1 and let $F\colon \mathcal{C} \longrightarrow \mathbb{1}$ be the functor given on objects by

$$\begin{aligned} F(A) &= 0, \\ F(B) &= 1 \end{aligned}$$

and on non-identity morphisms by

$$\begin{aligned} F(f) &= f_{01}, \quad F(e_A) = \mathrm{id}_0, \\ F(g) &= f_{01}, \quad F(e_B) = \mathrm{id}_1 . \end{aligned}$$

Finally, let $\mathcal{X} = \mathrm{B}\mathbb{Z}_{/2}$ be the walking involution and let $\iota_A, \iota_B\colon \mathrm{B}\mathbb{Z}_{/2} \rightrightarrows \mathcal{C}$ be the inclusion functors from $\mathrm{B}\mathbb{Z}_{/2}$ to $\mathcal{C}$ with

$$\begin{aligned} \iota_A(\bullet) &= A, \\ \iota_B(\bullet) &= B. \end{aligned}$$

Since every morphism in $\mathbb{1}$ has a preimage in $\mathcal{C}$ by F, the functor F is full. Now, for F_* to be full, the map

$$\begin{aligned} F_{*|\iota_A,\iota_B}\colon \ \mathrm{Nat}(\iota_A, \iota_B) &\longrightarrow \mathrm{Nat}(F \circ \iota_A, F \circ \iota_B) \\ \alpha &\longmapsto \mathrm{id}_F \star \alpha \end{aligned}$$

would need to be surjective. However, as we will show next, we have

$$\begin{aligned} \mathrm{Nat}(\iota_A, \iota_B) &= \emptyset, \\ \mathrm{Nat}(F \circ \iota_A, F \circ \iota_B) &\cong \mathrm{pt}, \end{aligned}$$

so this is impossible:

- *Proof of* $\mathrm{Nat}(\iota_A, \iota_B) = \emptyset$: A natural transformation $\alpha\colon \iota_A \Rightarrow \iota_B$ consists of a morphism

$$\alpha\colon \underbrace{\iota_A(\bullet)}_{=A} \to \underbrace{\iota_B(\bullet)}_{=B}$$

in $\mathcal{C}$ making the diagram

$$\begin{tikzcd} \iota_A(\bullet) \arrow[r, "\iota_A(e)"] \arrow[d, "\alpha"'] & \iota_A(\bullet) \arrow[d, "\alpha"] \\ \iota_B(\bullet) \arrow[r, "\iota_B(e)"'] & \iota_B(\bullet) \end{tikzcd}$$

commute for each $e \in \mathrm{Hom}_{\mathsf{B}\mathbb{Z}_{/2}}(\bullet, \bullet) \cong \mathbb{Z}_{/2}$. We have two cases:

1. If $\alpha = f$, the naturality diagram for the unique nonidentity element of $\mathbb{Z}_{/2}$ is given by

$$\begin{tikzcd} A \arrow[r, "e_A"] \arrow[d, "f"'] & A \arrow[d, "f"] \\ B \arrow[r, "e_B"'] & B. \end{tikzcd}$$

However, $e_B \circ f = f$ and $f \circ e_A = g$, so this diagram does not commute.

2. If $\alpha = g$, the naturality diagram for the unique nonidentity element of $\mathbb{Z}_{/2}$ is given by

$$\begin{tikzcd} A \arrow[r, "e_A"] \arrow[d, "g"'] & A \arrow[d, "g"] \\ B \arrow[r, "e_B"'] & B. \end{tikzcd}$$

However, $e_B \circ g = g$ and $g \circ e_A = f$, so this diagram does not commute.

As a result, there are no natural transformations from ι_A to ι_B.

- *Proof of* $\mathrm{Nat}(F \circ \iota_A, F \circ \iota_B) \cong pt$: A natural transformation

$$\beta \colon F \circ \iota_A \Rightarrow F \circ \iota_B$$

consists of a morphism

$$\beta \colon \underbrace{[F \circ \iota_A](\bullet)}_{=0} \to \underbrace{[F \circ \iota_B](\bullet)}_{=1}$$

in $\mathbb{1}$ making the diagram

$$\begin{tikzcd}
{[F \circ \iota_A](\bullet)} \arrow[r, "{[F \circ \iota_A](e)}"] \arrow[d, "\beta"'] & {[F \circ \iota_A](\bullet)} \arrow[d, "\beta"] \\
{[F \circ \iota_B](\bullet)} \arrow[r, "{[F \circ \iota_B](e)}"'] & {[F \circ \iota_B](\bullet)}
\end{tikzcd}$$

commute for each $e \in \mathrm{Hom}_{\mathrm{B}\mathbb{Z}_{/2}}(\bullet, \bullet) \cong \mathbb{Z}_{/2}$. Since the only morphism from 0 to 1 in $\mathbb{1}$ is f_{01}, we must have $\beta = f_{01}$ if such a transformation were to exist, and in fact it indeed does, as in this case the naturality diagram above becomes

$$\begin{tikzcd}
0 \arrow[r, "\mathrm{id}_0"] \arrow[d, "f_{01}"'] & 0 \arrow[d, "f_{01}"] \\
1 \arrow[r, "\mathrm{id}_1"'] & 1
\end{tikzcd}$$

for each $e \in \mathbb{Z}_{/2}$, and this diagram indeed commutes, making β into a natural transformation.

This finishes the proof.

Item 3: Interaction With Postcomposition II

Taking $\mathcal{X} = \mathrm{pt}$, it follows by assumption that the functor

$$F_* \colon \mathrm{Fun}(\mathrm{pt}, \mathcal{C}) \to \mathrm{Fun}(\mathrm{pt}, \mathcal{D})$$

is full. However, by Item 5 of Proposition 11.10.1.1.2, we have isomorphisms of categories

$$\begin{aligned} \mathrm{Fun}(\mathrm{pt}, \mathcal{C}) &\cong \mathcal{C}, \\ \mathrm{Fun}(\mathrm{pt}, \mathcal{D}) &\cong \mathcal{D} \end{aligned}$$

and the diagram

$$\begin{tikzcd} \mathrm{Fun}(\mathrm{pt}, \mathcal{C}) \arrow[r, "F_*"] \arrow[d, dashed, "\wr"] & \mathrm{Fun}(\mathrm{pt}, \mathcal{D}) \arrow[d, dashed, "\wr"'] \\ \mathcal{C} \arrow[r, "F"'] & \mathcal{D} \end{tikzcd}$$

commutes. It then follows from Item 1 that F is full.

Item 4: Interaction With Precomposition I

Omitted.

Item 5: Interaction With Precomposition II

See [BS10, p. 47].

Item 6: Interaction With Precomposition III

Omitted, but see `https://unimath.github.io/doc/UniMath/d4de26f//UniMath.CategoryTheory.precomp_fully_faithful.html` for a formalised proof.

Item 7: Interaction With Precomposition IV

We claim Items 7a to 7e are equivalent:

- *Items 7a and 7b Are Equivalent:* This is true by the definition of corepresentably full morphism; see Types of Morphisms in Bicategories, Definition 13.2.2.1.1.
- *Items 7a, 7c and 7d Are Equivalent:* See **??**, **??** of **??**.

- *Items 7a and 7e Are Equivalent:* See [Adá–01, Item (b) of Remark 4.3].

This finishes the proof.

Question 11.6.2.1.4 ► Better Characterisations of Functors With Full Precomposition

Item 7 of Proposition 11.6.2.1.2 gives a characterisation of the functors F for which F^* is full, but the characterisations given there are really messy. Are there better ones?
This question also appears as [MO 468121b].

11.6.3 Fully Faithful Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.6.3.1.1 ► Fully Faithful Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **fully faithful** if F is full and faithful, i.e. if, for each $A, B \in \mathrm{Obj}(\mathcal{C})$, the action on morphisms

$$F_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F_A, F_B)$$

of F at (A, B) is bijective.

Proposition 11.6.3.1.2 ► Properties of Fully Faithful Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$ be functors.

1. *Characterisations.* The following conditions are equivalent:

   (a) The functor F is fully faithful.

   (b) We have a pullback square

$$\mathrm{Arr}(\mathcal{C}) \cong (\mathcal{C} \times \mathcal{C}) \times_{\mathcal{D}\times\mathcal{D}} \mathrm{Arr}(\mathcal{D}),$$

$$\begin{tikzcd} \mathrm{Arr}(\mathcal{C}) \arrow[r, "\mathrm{Arr}(F)"] \arrow[d, "\mathrm{src}\times\mathrm{tgt}"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Arr}(\mathcal{D}) \arrow[d, "\mathrm{src}\times\mathrm{tgt}"] \\ \mathcal{C}\times\mathcal{C} \arrow[r, "F\times F"'] & \mathcal{D}\times\mathcal{D} \end{tikzcd}$$

in Cats.

2. *Interaction With Composition.* If F and G are fully faithful, then so is $G \circ F$.

3. *Conservativity.* If F is fully faithful, then F is conservative.

4. *Essential Injectivity.* If F is fully faithful, then F is essentially injective.

5. *Interaction With Co/Limits.* If F is fully faithful, then F reflects co/limits.

6. *Interaction With Postcomposition.* The following conditions are equivalent:

   (a) The functor $F\colon \mathcal{C} \to \mathcal{D}$ is fully faithful.

   (b) For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

   $$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

   is fully faithful.

   (c) The functor $F\colon \mathcal{C} \to \mathcal{D}$ is a representably fully faithful morphism in Cats_2 in the sense of Types of Morphisms in Bicategories, Definition 13.1.3.1.1.

7. *Interaction With Precomposition I.* If F is fully faithful, then the precomposition functor

   $$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

   can fail to be fully faithful.

8. *Interaction With Precomposition II.* If the precomposition functor

   $$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

   is fully faithful, then F ***can fail*** to be fully faithful (and in fact it can also fail to be either full or faithful).

9. *Interaction With Precomposition III.* If F is essentially surjective and full, then the precomposition functor

   $$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

   is fully faithful.

10. *Interaction With Precomposition IV.* The following conditions are equivalent:

    (a) For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

    $$F^* \colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

    is fully faithful.

    (b) The precomposition functor

    $$F^* \colon \mathsf{Fun}(\mathcal{D}, \mathsf{Sets}) \to \mathsf{Fun}(\mathcal{C}, \mathsf{Sets})$$

    is fully faithful.

    (c) The functor

    $$\mathsf{Lan}_F \colon \mathsf{Fun}(\mathcal{C}, \mathsf{Sets}) \to \mathsf{Fun}(\mathcal{D}, \mathsf{Sets})$$

    is fully faithful.

    (d) The functor F is a corepresentably fully faithful morphism in Cats_2 in the sense of Types of Morphisms in Bicategories, Definition 13.2.3.1.1.

    (e) The functor F is absolutely dense.

    (f) The components

    $$\eta_G \colon G \Longrightarrow \mathsf{Ran}_F(G \circ F)$$

    of the unit

    $$\eta \colon \mathrm{id}_{\mathsf{Fun}(\mathcal{D},\mathcal{X})} \Longrightarrow \mathsf{Ran}_F \circ F^*$$

    of the adjunction $F^* \dashv \mathsf{Ran}_F$ are all isomorphisms.

    (g) The components

    $$\epsilon_G \colon \mathsf{Lan}_F(G \circ F) \Longrightarrow G$$

    of the counit

    $$\epsilon \colon \mathsf{Lan}_F \circ F^* \Longrightarrow \mathrm{id}_{\mathsf{Fun}(\mathcal{D},\mathcal{X})}$$

    of the adjunction $\mathsf{Lan}_F \dashv F^*$ are all isomorphisms.

(h) The natural transformation

$$\alpha\colon \operatorname{Lan}_{h_F}\left(h^F\right) \Longrightarrow h$$

with components

$$\alpha_{B',B}\colon \int^{A\in\mathcal{C}} h^{B'}_{F_A} \times h^{F_A}_{B} \longrightarrow h^{B'}_{B}$$

given by

$$\alpha_{B',B}([(\phi,\psi)]) = \psi \circ \phi$$

is a natural isomorphism.

(i) For each $B \in \operatorname{Obj}(\mathcal{D})$, there exist:

- An object A_B of $\mathcal{C}$;
- A morphism $s_B\colon B \longrightarrow F(A_B)$ of $\mathcal{D}$;
- A morphism $r_B\colon F(A_B) \longrightarrow B$ of $\mathcal{D}$;

satisfying the following conditions:

i. The triple $(F(A_B), r_B, s_B)$ is a retract of B, i.e. we have $r_B \circ s_B = \mathrm{id}_B$.

ii. For each morphism $f\colon B' \longrightarrow B$ of $\mathcal{D}$, we have

$$[(A_B, s_{B'}, f \circ r_{B'})] = [(A_B, s_B \circ f, r_B)]$$

in $\int^{A\in\mathcal{C}} h^{B'}_{F_A} \times h^{F_A}_{B}$.

Proof 11.6.3.1.3 ▶ Proof of Proposition 11.6.3.1.2

Item 1: Characterisations

Omitted.

Item 2: Interaction With Composition

Since the map

$$(G \circ F)_{A,B}\colon \operatorname{Hom}_{\mathcal{C}}(A,B) \longrightarrow \operatorname{Hom}_{\mathcal{D}}\left(G_{F_A}, G_{F_B}\right),$$

defined as the composition

$$\mathrm{Hom}_{\mathcal{C}}(A, B) \xrightarrow{F_{A,B}} \mathrm{Hom}_{\mathcal{D}}(F_A, F_B) \xrightarrow{G_{F(A),F(B)}} \mathrm{Hom}_{\mathcal{D}}\left(G_{F_A}, G_{F_B}\right),$$

is a composition of bijective functions, it follows from **??** that it is also bijective. Therefore $G \circ F$ is fully faithful.

Item 3: Conservativity

This is a repetition of Item 2 of Proposition 11.6.4.1.2, and is proved there.

Item 4: Essential Injectivity

Omitted.

Item 5: Interaction With Co/Limits

Omitted.

Item 6: Interaction With Postcomposition

This follows from Item 2 of Proposition 11.6.1.1.2 and **??** of Proposition 11.6.2.1.2.

Item 7: Interaction With Precomposition I

See [MSE 733161] for an example of a fully faithful functor whose precomposition with which fails to be full.

Item 8: Interaction With Precomposition II

See [MSE 749304, Item 3].

Item 9: Interaction With Precomposition III

Omitted, but see `https://unimath.github.io/doc/UniMath/d4de26f//UniMath.CategoryTheory.precomp_fully_faithful.html` for a formalised proof.

Item 10: Interaction With Precomposition IV

We claim Items 10a to 10i are equivalent:

- *Items 10a and 10d Are Equivalent:* This is true by the definition of corepresentably fully faithful morphism; see Types of Morphisms in Bicategories, Definition 13.2.3.1.1.
- *Items 10a, 10f and 10g Are Equivalent:* See **??**, **??** of **??**.

- *Items 10a to 10c Are Equivalent:* This follows from [Low15, Proposition A.1.5].
- *Items 10a, 10e, 10h and 10i Are Equivalent:* See [Fre09, Theorem 4.1] and [Adá–01, Theorem 1.1].

This finishes the proof. ∎

11.6.4 Conservative Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.6.4.1.1 ▶ Conservative Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **conservative** if it satisfies the following condition:[1]

($\star$) For each $f \in \mathrm{Mor}(\mathcal{C})$, if $F(f)$ is an isomorphism in $\mathcal{D}$, then f is an isomorphism in $\mathcal{C}$.

[1] *Slogan:* A functor F is **conservative** if it reflects isomorphisms.

Proposition 11.6.4.1.2 ▶ Properties of Conservative Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Characterisations.* The following conditions are equivalent:
   (a) The functor F is conservative.
   (b) For each $f \in \mathrm{Mor}(\mathcal{C})$, the morphism $F(f)$ is an isomorphism in $\mathcal{D}$ iff f is an isomorphism in $\mathcal{C}$.
2. *Interaction With Fully Faithfulness.* Every fully faithful functor is conservative.
3. *Interaction With Precomposition.* The following conditions are equivalent:
   (a) For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor
   $$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$
   is conservative.
   (b) The equivalent conditions of Item 5 of Proposition 11.6.1.1.2 are satisfied.

Proof 11.6.4.1.3 ▶ Proof of Proposition 11.6.4.1.2

Item 1: Characterisations

This follows from Item 1 of Proposition 11.5.1.1.8.

Item 2: Interaction With Fully Faithfulness

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a fully faithful functor, let $f\colon A \to B$ be a morphism of $\mathcal{C}$, and suppose that F_f is an isomorphism. We have

$$\begin{aligned} F(\mathrm{id}_B) &= \mathrm{id}_{F(B)} \\ &= F(f) \circ F(f)^{-1} \\ &= F\left(f \circ f^{-1}\right). \end{aligned}$$

Similarly, $F(\mathrm{id}_A) = F\left(f^{-1} \circ f\right)$. But since F is fully faithful, we must have

$$\begin{aligned} f \circ f^{-1} &= \mathrm{id}_B, \\ f^{-1} \circ f &= \mathrm{id}_A, \end{aligned}$$

showing f to be an isomorphism. Thus F is conservative. ∎

Question 11.6.4.1.4 ▶ Characterisations of Functors With Conservative Pre/Post-composition

Is there a characterisation of functors $F\colon \mathcal{C} \to \mathcal{D}$ satisfying the following condition:

($\star$) For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

$$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

is conservative?

This question also appears as [MO 468121a].

11.6.5 Essentially Injective Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.6.5.1.1 ▶ Essentially Injective Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **essentially injective** if it satisfies the following condition:

($\star$) For each $A, B \in \mathrm{Obj}(\mathcal{C})$, if $F(A) \cong F(B)$, then $A \cong B$.

Question 11.6.5.1.2 ▶ Characterisations of Functors With Essentially Injective Pre/Postcomposition

Is there a characterisation of functors $F\colon \mathcal{C} \to \mathcal{D}$ such that:

1. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

$$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

is essentially injective, i.e. if $\phi \circ F \cong \psi \circ F$, then $\phi \cong \psi$ for all functors ϕ and ψ?

2. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

$$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

is essentially injective, i.e. if $F \circ \phi \cong F \circ \psi$, then $\phi \cong \psi$?

This question also appears as [MO 468121a].

11.6.6 Essentially Surjective Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.6.6.1.1 ▶ Essentially Surjective Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **essentially surjective**[1] if it satisfies the following condition:

($\star$) For each $D \in \mathrm{Obj}(\mathcal{D})$, there exists some object A of $\mathcal{C}$ such that $F(A) \cong D$.

[1] *Further Terminology:* Also called an **eso** functor, meaning *essentially surjective on objects*.

Question 11.6.6.1.2 ► Characterisations of Functors With Essentially Surjective Pre/Postcomposition

Is there a characterisation of functors $F\colon \mathcal{C} \to \mathcal{D}$ such that:

1. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

$$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

   is essentially surjective?

2. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

$$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

   is essentially surjective?

This question also appears as [MO 468121a].

11.6.7 Equivalences of Categories

Definition 11.6.7.1.1 ► Equivalences of Categories

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

1. An **equivalence of categories** between $\mathcal{C}$ and $\mathcal{D}$ consists of a pair of functors

$$\begin{aligned} F &\colon \mathcal{C} \to \mathcal{D}, \\ G &\colon \mathcal{D} \to \mathcal{C} \end{aligned}$$

   together with natural isomorphisms

$$\begin{aligned} \eta &\colon \mathrm{id}_{\mathcal{C}} \overset{\sim}{\Longrightarrow} G \circ F, \\ \epsilon &\colon F \circ G \overset{\sim}{\Longrightarrow} \mathrm{id}_{\mathcal{D}}\,. \end{aligned}$$

2. An **adjoint equivalence of categories** between $\mathcal{C}$ and $\mathcal{D}$ is an equivalence (F, G, η, ϵ) between $\mathcal{C}$ and $\mathcal{D}$ which is also an adjunction.

Proposition 11.6.7.1.2 ► Properties of Equivalences of Categories

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Characterisations.* If $\mathcal{C}$ and $\mathcal{D}$ are small[1], then the following conditions are equivalent:[2]

   (a) The functor F is an equivalence of categories.

   (b) The functor F is fully faithful and essentially surjective.

   (c) The induced functor

   $$\restriction F\mathsf{Sk}(\mathcal{C})\colon \mathsf{Sk}(\mathcal{C}) \to \mathsf{Sk}(\mathcal{D})$$

   is an *isomorphism* of categories.

   (d) For each $X \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

   $$F^*\colon \mathsf{Fun}(\mathcal{D}, \mathcal{X}) \to \mathsf{Fun}(\mathcal{C}, \mathcal{X})$$

   is an equivalence of categories.

   (e) For each $X \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

   $$F_*\colon \mathsf{Fun}(\mathcal{X}, \mathcal{C}) \to \mathsf{Fun}(\mathcal{X}, \mathcal{D})$$

   is an equivalence of categories.

2. *Two-Out-of-Three.* Let

   $$\begin{tikzcd} \mathcal{C} \arrow[rr, "G\circ F"] \arrow[dr, "F"'] & & \mathcal{E} \\ & \mathcal{D} \arrow[ur, "G"'] & \end{tikzcd}$$

   be a diagram in Cats. If two out of the three functors among F, G, and $G \circ F$ are equivalences of categories, then so is the third.

3. *Stability Under Composition.* Let

   $$\begin{tikzcd} \mathcal{C} \arrow[r, shift left, "F"] & \mathcal{D} \arrow[l, shift left, "G"] \arrow[r, shift left, "F'"] & \mathcal{E} \arrow[l, shift left, "G'"] \end{tikzcd}$$

   be a diagram in Cats. If (F, G) and (F', G') are equivalences of categories, then so is their composite $(F' \circ F, G' \circ G)$.

4. *Equivalences vs.Adjoint Equivalences.* Every equivalence of categories can be promoted to an adjoint equivalence.[3]

5. *Interaction With Groupoids.* If $\mathcal{C}$ and $\mathcal{D}$ are groupoids, then the following conditions are equivalent:

   (a) The functor F is an equivalence of groupoids.

   (b) The following conditions are satisfied:

      i. The functor F induces a bijection

      $$\pi_0(F)\colon \pi_0(\mathcal{C}) \to \pi_0(\mathcal{D})$$

      of sets.

      ii. For each $A \in \mathrm{Obj}(\mathcal{C})$, the induced map

      $$F_{x,x}\colon \mathrm{Aut}_{\mathcal{C}}(A) \to \mathrm{Aut}_{\mathcal{D}}(F_A)$$

      is an isomorphism of groups.

[1]Otherwise there will be size issues. One can also work with large categories and universes, or require F to be *constructively* essentially surjective; see [MSE 1465107].

[2]In ZFC, the equivalence between Item 1a and Item 1b is equivalent to the axiom of choice; see [MO 119454].

In Univalent Foundations, this is true without requiring neither the axiom of choice nor the law of excluded middle.

[3]More precisely, we can promote an equivalence of categories (F, G, η, ϵ) to adjoint equivalences (F, G, η', ϵ) and (F, G, η, ϵ').

Proof 11.6.7.1.3 ▶ Proof of Proposition 11.6.7.1.2

Item 1: Characterisations

We claim that Items 1a to 1e are indeed equivalent:

1. *Item 1a* $\Longrightarrow$ *Item 1b*: Clear.

2. *Item 1b* $\Longrightarrow$ *Item 1a*: Since F is essentially surjective and $\mathcal{C}$ and $\mathcal{D}$ are small, we can choose, using the axiom of choice, for each $B \in \mathrm{Obj}(\mathcal{D})$, an object j_B of $\mathcal{C}$ and an isomorphism $i_B\colon B \to F_{j_B}$ of $\mathcal{D}$.

Since F is fully faithful, we can extend the assignment $B \mapsto j_B$ to a *unique* functor $j\colon \mathcal{D} \to \mathcal{C}$ such that the isomorphisms $i_B\colon B \to F_{j_B}$ assemble into a natural isomorphism $\eta\colon \mathrm{id}_{\mathcal{D}} \overset{\sim}{\Longrightarrow} F \circ j$, with a similar natural isomorphism $\epsilon\colon \mathrm{id}_{\mathcal{C}} \overset{\sim}{\Longrightarrow} j \circ F$. Hence F is an equivalence.

3. *Item 1a* $\Longrightarrow$ *Item 1c*: This follows from Item 4 of Proposition 11.1.3.1.3.

4. *Item 1c* $\Longrightarrow$ *Item 1a*: Omitted.

5. *Items 1a, 1d and 1e Are Equivalent:* This follows from **??**.

This finishes the proof of Item 1.

Item 2: Two-Out-of-Three

Omitted.

Item 3: Stability Under Composition

Clear.

Item 4: Equivalences vs.Adjoint Equivalences

See [Rie16, Proposition 4.4.5].

Item 5: Interaction With Groupoids

See [nLa25, Proposition 4.4].

11.6.8 Isomorphisms of Categories

Definition 11.6.8.1.1 ▶ Isomorphisms of Categories

An **isomorphism of categories** is a pair of functors

$$F\colon \mathcal{C} \to \mathcal{D},$$
$$G\colon \mathcal{D} \to \mathcal{C}$$

such that we have

$$G \circ F = \mathrm{id}_{\mathcal{C}},$$

$$F \circ G = \mathrm{id}_{\mathcal{D}}\,.$$

Example 11.6.8.1.2 ► Equivalent But Non-Isomorphic Categories

Categories can be equivalent but non-isomorphic. For example, the category consisting of two isomorphic objects is equivalent to pt, but not isomorphic to it.

Proposition 11.6.8.1.3 ► Properties of Isomorphisms of Categories

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Characterisations.* If $\mathcal{C}$ and $\mathcal{D}$ are small, then the following conditions are equivalent:

   (a) The functor F is an isomorphism of categories.

   (b) The functor F is fully faithful and bijective on objects.

   (c) For each $X \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

   $$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

   is an isomorphism of categories.

   (d) For each $X \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

   $$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

   is an isomorphism of categories.

Proof 11.6.8.1.4 ► Proof of Proposition 11.6.8.1.3

Item 1: Characterisations

We claim that Items 1a to 1d are indeed equivalent:

1. *Items 1a and 1b Are Equivalent:* Omitted, but similar to Item 1 of Proposition 11.6.7.1.2.

2. *Items 1a, 1c and 1d Are Equivalent:* This follows from **??**.

This finishes the proof. ∎

11.7 More Conditions on Functors

11.7.1 Dominant Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.7.1.1.1 ▶ Dominant Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **dominant** if every object of $\mathcal{D}$ is a retract of some object in $\mathrm{Im}(F)$, i.e.:

($\star$) For each $B \in \mathrm{Obj}(\mathcal{D})$, there exist:

- An object A of $\mathcal{C}$;
- A morphism $r\colon F(A) \to B$ of $\mathcal{D}$;
- A morphism $s\colon B \to F(A)$ of $\mathcal{D}$;

such that we have

$$r \circ s = \mathrm{id}_B, \qquad \begin{tikzcd} B \arrow[r, "s"] \arrow[dr, "\mathrm{id}_B"'] & F(A) \arrow[d, "r"] \\ & B. \end{tikzcd}$$

Proposition 11.7.1.1.2 ▶ Properties of Dominant Functors

Let $F, G\colon \mathcal{C} \rightrightarrows \mathcal{D}$ be functors and let $I\colon \mathcal{X} \to \mathcal{C}$ be a functor.

1. *Interaction With Right Whiskering.* If I is full and dominant, then the map

$$- \star \mathrm{id}_I\colon \mathrm{Nat}(F, G) \to \mathrm{Nat}(F \circ I, G \circ I)$$

is a bijection.

2. *Interaction With Adjunctions.* Let $(F, G)\colon \mathcal{C} \rightleftarrows \mathcal{D}$ be an adjunction.

   (a) If F is dominant, then G is faithful.

   (b) The following conditions are equivalent:

      i. The functor G is full.

ii. The restriction

$$\restriction G\mathrm{Im}_F \colon \mathrm{Im}(F) \to \mathcal{C}$$

of G to $\mathrm{Im}(F)$ is full.

PROOF 11.7.1.1.3 ▶ PROOF OF PROPOSITION 11.7.1.1.2

Item 1: Interaction With Right Whiskering

See [DFH75, Proposition 1.4].

Item 2: Interaction With Adjunctions

See [DFH75, Proposition 1.7].

QUESTION 11.7.1.1.4 ▶ CHARACTERISATIONS OF FUNCTORS WITH DOMINANT PRE/POST-COMPOSITION

Is there a characterisation of functors $F \colon \mathcal{C} \to \mathcal{D}$ such that:

1. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

$$F^* \colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

is dominant?

2. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

$$F_* \colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

is dominant?

This question also appears as [MO 468121a].

11.7.2 Monomorphisms of Categories

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.7.2.1.1 ► Monomorphisms of Categories

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is a **monomorphism of categories** if it is a monomorphism in Cats (see **??**, **??**).

Proposition 11.7.2.1.2 ► Properties of Monomorphisms of Categories

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Characterisations.* The following conditions are equivalent:

   (a) The functor F is a monomorphism of categories.

   (b) The functor F is injective on objects and morphisms, i.e. F is injective on objects and the map

   $$F\colon \operatorname{Mor}(\mathcal{C}) \to \operatorname{Mor}(\mathcal{D})$$

   is injective.

Proof 11.7.2.1.3 ► Proof of Proposition 11.7.2.1.2

Item 1: Characterisations

Omitted. ∎

Question 11.7.2.1.4 ► Characterisations of Functors With Monic Pre/Postcomposition

Is there a characterisation of functors $F\colon \mathcal{C} \to \mathcal{D}$ such that:

1. For each $\mathcal{X} \in \operatorname{Obj}(\mathsf{Cats})$, the precomposition functor

   $$F^*\colon \operatorname{Fun}(\mathcal{D}, \mathcal{X}) \to \operatorname{Fun}(\mathcal{C}, \mathcal{X})$$

   is a monomorphism of categories?

2. For each $\mathcal{X} \in \operatorname{Obj}(\mathsf{Cats})$, the postcomposition functor

   $$F_*\colon \operatorname{Fun}(\mathcal{X}, \mathcal{C}) \to \operatorname{Fun}(\mathcal{X}, \mathcal{D})$$

   is a monomorphism of categories?

This question also appears as [MO 468121a].

11.7.3 Epimorphisms of Categories

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.7.3.1.1 ► Epimorphisms of Categories

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is a **epimorphism of categories** if it is a epimorphism in Cats (see **??**, **??**).

Proposition 11.7.3.1.2 ► Properties of Epimorphisms of Categories

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Characterisations.* The following conditions are equivalent:[1]

   (a) The functor F is a epimorphism of categories.

   (b) For each morphism $f\colon A \to B$ of $\mathcal{D}$, we have a diagram

$$\begin{tikzcd}
 & & & A \arrow[dlll, "\phi_1"'] \arrow[dl, "\phi_2"'] \arrow[d, "\phi_3"] \arrow[dr, "\phi_m"] \arrow[drrr, dashed, "\mathrm{id}_A"] & & & \\
X_1 \arrow[dr, dashed, "\alpha_1"] \arrow[ddrrr, dashed, bend right, "\alpha_0"'] & & X_2 \arrow[dl, dashed, "\alpha_2"'] \arrow[dr, dashed, "\alpha_3"] & X_3 \arrow[dl, dashed, "\alpha_4"'] \arrow[dr, dashed, "\alpha_6"] & \cdots \arrow[dl, dashed, "\alpha_{2m-2}"'] & X_m \arrow[dr, dashed, "\alpha_{2m-1}"] & & A \arrow[dl, dashed, "\alpha_{2m}"'] \\
 & Y_1 \arrow[dr, "\psi_1"] & & Y_2 \arrow[d, "\psi_2"] & \cdots \arrow[d, "\psi_3"] & & Y_m \arrow[dl, "\psi_m"'] & \\
 & & & B & & & &
\end{tikzcd}$$

   in $\mathcal{D}$ satisfying the following conditions:

   i. We have $f = \alpha_0 \circ \phi_1$.

   ii. We have $f = \psi_m \circ \alpha_{2m}$.

   iii. For each $0 \leq i \leq 2m$, we have $\alpha_i \in \mathrm{Mor}(\mathrm{Im}(F))$.

2. *Surjectivity on Objects.* If F is an epimorphism of categories, then F is surjective on objects.

[1] *Further Terminology:* This statement is known as **Isbell's zigzag theorem**.

Proof 11.7.3.1.3 ▶ Proof of Proposition 11.7.3.1.2

Item 1: Characterisations

See [Isb68].

Item 2: Surjectivity on Objects

Omitted.

Question 11.7.3.1.4 ▶ Characterisations of Functors With Epic Pre/Postcomposition

Is there a characterisation of functors $F\colon \mathcal{C} \to \mathcal{D}$ such that:

1. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

$$F^*\colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

   is an epimorphism of categories?

2. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

$$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

   is an epimorphism of categories?

This question also appears as [MO 468121a].

11.7.4 Pseudomonic Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.7.4.1.1 ▶ Pseudomonic Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **pseudomonic** if it satisfies the following conditions:

1. For all diagrams of the form

$$\begin{tikzcd} \mathcal{X} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\Downarrow\alpha \quad \Downarrow\beta"] & \mathcal{C} \arrow[r, "F"] & \mathcal{D}, \end{tikzcd}$$

if we have

$$\mathrm{id}_F \star \alpha = \mathrm{id}_F \star \beta,$$

then $\alpha = \beta$.

2. For each $\mathcal{X} \in \mathrm{Obj}(\mathrm{Cats})$ and each natural isomorphism

$$\beta\colon F \circ \phi \overset{\sim}{\Longrightarrow} F \circ \psi, \qquad \begin{tikzcd} \mathcal{X} \arrow[r, bend left, "F\circ\phi"] \arrow[r, bend right, "F\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & \mathcal{D}, \end{tikzcd}$$

there exists a natural isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} \mathcal{X} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{C} \end{tikzcd}$$

such that we have an equality

$$\begin{tikzcd} \mathcal{X} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{C} \arrow[r, "F"] & \mathcal{D} \end{tikzcd} = \begin{tikzcd} \mathcal{X} \arrow[r, bend left, "F\circ\phi"] \arrow[r, bend right, "F\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & \mathcal{D} \end{tikzcd}$$

of pasting diagrams, i.e. such that we have

$$\beta = \mathrm{id}_F \star \alpha.$$

Proposition 11.7.4.1.2 ► Properties of Pseudomonic Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Characterisations.* The following conditions are equivalent:
   (a) The functor F is pseudomonic.
   (b) The functor F satisfies the following conditions:
      i. The functor F is faithful, i.e. for each $A, B \in \mathrm{Obj}(\mathcal{C})$, the action on morphisms

$$F_{A,B}\colon \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F_A, F_B)$$

of F at (A, B) is injective.

ii. For each $A, B \in \mathrm{Obj}(\mathcal{C})$, the restriction

$$F^{\mathrm{iso}}_{A,B} \colon \mathrm{Iso}_{\mathcal{C}}(A, B) \to \mathrm{Iso}_{\mathcal{D}}(F_A, F_B)$$

of the action on morphisms of F at (A, B) to isomorphisms is surjective.

(c) We have an isocomma square of the form

$$\mathcal{C} \overset{\text{eq.}}{\cong} \mathcal{C} \overset{\leftrightarrow}{\times}_{\mathcal{D}} \mathcal{C}, \qquad \begin{tikzcd} \mathcal{C} \arrow[r, "\mathrm{id}_{\mathcal{C}}"] \arrow[d, "\mathrm{id}_{\mathcal{C}}"'] & \mathcal{C} \arrow[d, "F"] \\ \mathcal{C} \arrow[r, "F"'] \arrow[ur, dashed, leftrightarrow] & \mathcal{D} \end{tikzcd}$$

in Cats_2 up to equivalence.

(d) We have an isocomma square of the form

$$\mathcal{C} \overset{\text{eq.}}{\cong} \mathcal{C} \overset{\leftrightarrow}{\times}_{\mathrm{Arr}(\mathcal{D})} \mathcal{D}, \qquad \begin{tikzcd} \mathcal{C} \arrow[r, hook] \arrow[d, "F"'] & \mathrm{Arr}(\mathcal{C}) \arrow[d, "\mathrm{Arr}(F)"] \\ \mathcal{D} \arrow[r, hook] \arrow[ur, dashed, leftrightarrow] & \mathrm{Arr}(\mathcal{D}) \end{tikzcd}$$

in Cats_2 up to equivalence.

(e) For each $\mathcal{X} \in \mathrm{Obj}(\mathrm{Cats})$, the postcomposition[1] functor

$$F_* \colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

is pseudomonic.

2. *Conservativity.* If F is pseudomonic, then F is conservative.

3. *Essential Injectivity.* If F is pseudomonic, then F is essentially injective.

[1] Asking the precomposition functors

$$F^* \colon \mathrm{Fun}(\mathcal{D}, \mathcal{X}) \to \mathrm{Fun}(\mathcal{C}, \mathcal{X})$$

to be pseudomonic leads to pseudoepic functors; see Item 1b of Item 1 of Proposition 11.7.5.1.2.

Proof 11.7.4.1.3 ► Proof of Proposition 11.7.4.1.2

Item 1: Characterisations

Omitted.

Item 2: Conservativity

Omitted.

Item 3: Essential Injectivity

Omitted. ∎

11.7.5 Pseudoepic Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.7.5.1.1 ► Pseudoepic Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **pseudoepic** if it satisfies the following conditions:

1. For all diagrams of the form

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] & \mathcal{D} \arrow[r, bend left, "\phi"{name=U}] \arrow[r, bend right, "\psi"'{name=D}] & \mathcal{X}, \arrow[from=U, to=D, Rightarrow, "\alpha"'] \arrow[from=U, to=D, Rightarrow, "\beta"] \end{tikzcd}$$

   if we have

$$\alpha \star \mathrm{id}_F = \beta \star \mathrm{id}_F,$$

   then $\alpha = \beta$.

2. For each $X \in \mathrm{Obj}(\mathcal{C})$ and each 2-isomorphism

$$\beta\colon \phi \circ F \overset{\sim}{\Longrightarrow} \psi \circ F, \qquad \begin{tikzcd} \mathcal{C} \arrow[r, bend left, "\phi\circ F"{name=U}] \arrow[r, bend right, "\psi\circ F"'{name=D}] & \mathcal{X} \arrow[from=U, to=D, Rightarrow, "\beta"'] \end{tikzcd}$$

   of $\mathcal{C}$, there exists a 2-isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} \mathcal{D} \arrow[r, bend left, "\phi"{name=U}] \arrow[r, bend right, "\psi"'{name=D}] & \mathcal{X} \arrow[from=U, to=D, Rightarrow, "\alpha"'] \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] & \mathcal{D} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{X} \end{tikzcd} = \begin{tikzcd} \mathcal{C} \arrow[r, bend left, "\phi\circ F"] \arrow[r, bend right, "\psi\circ F"'] \arrow[r, phantom, "\beta\Downarrow"] & \mathcal{X} \end{tikzcd}$$

of pasting diagrams in $\mathcal{C}$, i.e. such that we have

$$\beta = \alpha \star \mathrm{id}_F\,.$$

Proposition 11.7.5.1.2 ▶ Properties of Pseudoepic Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Characterisations.* The following conditions are equivalent:

   (a) The functor F is pseudoepic.

   (b) For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the functor

   $$F^*\colon \mathrm{Fun}(\mathcal{D},\mathcal{X}) \to \mathrm{Fun}(\mathcal{C},\mathcal{X})$$

   given by precomposition by F is pseudomonic.

   (c) We have an isococomma square of the form

   $$\mathcal{D} \overset{\text{eq.}}{\cong} \mathcal{D} \overset{\leftrightarrow}{\coprod}_{\mathcal{C}} \mathcal{D}, \qquad \begin{tikzcd} \mathcal{D} & \mathcal{D} \arrow[l, "\mathrm{id}_{\mathcal{D}}"'] \\ \mathcal{D} \arrow[u, "\mathrm{id}_{\mathcal{D}}"] & \mathcal{C} \arrow[l, "F"] \arrow[u, "F"'] \arrow[ul, dashed] \end{tikzcd}$$

   in Cats_2 up to equivalence.

2. *Dominance.* If F is pseudoepic, then F is dominant (Definition 11.7.1.1.1).

Proof 11.7.5.1.3 ▶ Proof of Proposition 11.7.5.1.2

Item 1: Characterisations

Omitted.

Item 2: Dominance

If F is pseudoepic, then

$$F^*\colon \mathrm{Fun}(\mathcal{D},\mathcal{X}) \to \mathrm{Fun}(\mathcal{C},\mathcal{X})$$

is pseudomonic for all $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, and thus in particular faithful. By Item 5g of Item 5 of Proposition 11.6.1.1.2, this is equivalent to requiring F to be dominant. ∎

Question 11.7.5.1.4 ▶ Characterisations of Pseudoepic Functors

Is there a nice characterisation of the pseudoepic functors, similarly to the characterisaiton of pseudomonic functors given in Item 1b of Item 1 of Proposition 11.7.4.1.2?
This question also appears as [MO 321971].

Question 11.7.5.1.5 ▶ Must a Pseudomonic and Pseudoepic Functor Be an Equivalence of Categories

A pseudomonic and pseudoepic functor is dominant, faithful, essentially injective, and full on isomorphisms. Is it necessarily an equivalence of categories? If not, how bad can this fail, i.e. how far can a pseudomonic and pseudoepic functor be from an equivalence of categories?
This question also appears as [MO 468334].

Question 11.7.5.1.6 ▶ Characterisations of Functors With Pseudoepic Pre/Post-composition

Is there a characterisation of functors $F\colon \mathcal{C} \to \mathcal{D}$ such that:

1. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the precomposition functor

$$F^*\colon \mathrm{Fun}(\mathcal{D},\mathcal{X}) \to \mathrm{Fun}(\mathcal{C},\mathcal{X})$$

is pseudoepic?

2. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition functor

$$F_*\colon \mathrm{Fun}(\mathcal{X}, \mathcal{C}) \to \mathrm{Fun}(\mathcal{X}, \mathcal{D})$$

is pseudoepic?

This question also appears as [MO 468121a].

11.8 Even More Conditions on Functors

11.8.1 Injective on Objects Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.1.1.1 ▶ Injective on Objects Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **injective on objects** if the action on objects

$$F\colon \mathrm{Obj}(\mathcal{C}) \to \mathrm{Obj}(\mathcal{D})$$

of F is injective.

Proposition 11.8.1.1.2 ▶ Properties of Injective on Objects Functors

Let $F\colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Characterisations.* The following conditions are equivalent:
   (a) The functor F is injective on objects.
   (b) The functor F is an isocofibration in Cats_2.

Proof 11.8.1.1.3 ▶ Proof of Proposition 11.8.1.1.2

Item 1: Characterisations

Omitted. ∎

11.8.2 Surjective on Objects Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.2.1.1 ▶ Surjective on Objects Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **surjective on objects** if the action on objects

$$F\colon \operatorname{Obj}(\mathcal{C}) \to \operatorname{Obj}(\mathcal{D})$$

of F is surjective.

11.8.3 Bijective on Objects Functors

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.3.1.1 ▶ Bijective on Objects Functors

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **bijective on objects**[1] if the action on objects

$$F\colon \operatorname{Obj}(\mathcal{C}) \to \operatorname{Obj}(\mathcal{D})$$

of F is a bijection.

[1] *Further Terminology:* Also called a **bo** functor.

11.8.4 Functors Representably Faithful on Cores

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.4.1.1 ▶ Functors Representably Faithful on Cores

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **representably faithful on cores** if, for each $X \in \operatorname{Obj}(\mathsf{Cats})$, the postcomposition by F functor

$$F_*\colon \operatorname{Core}(\operatorname{Fun}(\mathcal{X}, \mathcal{C})) \to \operatorname{Core}(\operatorname{Fun}(\mathcal{X}, \mathcal{D}))$$

is faithful.

Remark 11.8.4.1.2 ▶ Unwinding Definition 11.8.4.1.1

In detail, a functor $F\colon \mathcal{C} \to \mathcal{D}$ is **representably faithful on cores** if, given a diagram of the form

$$\begin{tikzcd} \mathcal{X} \arrow[r, bend left, "\phi"{name=U}] \arrow[r, bend right, "\psi"'{name=D}] & \mathcal{C} \arrow[r, "F"] & \mathcal{D}, \arrow[Rightarrow, from=U, to=D, "\alpha"', "\beta"] \end{tikzcd}$$

if α and β are natural isomorphisms and we have

$$\mathrm{id}_F \star \alpha = \mathrm{id}_F \star \beta,$$

then $\alpha = \beta$.

Question 11.8.4.1.3 ▶ Characterisation of Functors Representably Faithful on Cores

Is there a characterisation of functors representably faithful on cores?

11.8.5 Functors Representably Full on Cores

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.5.1.1 ▶ Functors Representably Full on Cores

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **representably full on cores** if, for each $X \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition by F functor

$$F_*\colon \mathrm{Core}(\mathrm{Fun}(\mathcal{X}, \mathcal{C})) \to \mathrm{Core}(\mathrm{Fun}(\mathcal{X}, \mathcal{D}))$$

is full.

Remark 11.8.5.1.2 ▶ Unwinding Definition 11.8.5.1.1

In detail, a functor $F\colon \mathcal{C} \to \mathcal{D}$ is **representably full on cores** if, for each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$ and each natural isomorphism

$$\beta\colon F \circ \phi \overset{\sim}{\Longrightarrow} F \circ \psi, \qquad \begin{tikzcd} \mathcal{X} \arrow[r, bend left, "F\circ\phi"{name=U}] \arrow[r, bend right, "F\circ\psi"'{name=D}] & \mathcal{D}, \arrow[Rightarrow, from=U, to=D, "\beta"'] \end{tikzcd}$$

there exists a natural isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} \mathcal{X} \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & \mathcal{C} \arrow[Rightarrow, from=U, to=D, "\alpha"] \end{tikzcd}$$

such that we have an equality

$$\begin{tikzcd} \mathcal{X} \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & \mathcal{C} \arrow[Rightarrow, from=U, to=D, "\alpha"] \arrow[r, "F"] & \mathcal{D} \end{tikzcd} = \begin{tikzcd} \mathcal{X} \arrow[r, bend left=60, "F\circ\phi"{name=U}] \arrow[r, bend right=60, "F\circ\psi"'{name=D}] & \mathcal{D} \arrow[Rightarrow, from=U, to=D, "\beta"] \end{tikzcd}$$

of pasting diagrams in Cats_2, i.e. such that we have

$$\beta = \mathrm{id}_F \star \alpha.$$

Question 11.8.5.1.3 ▶ Characterisation of Functors Representably Full on Cores

Is there a characterisation of functors representably full on cores?
This question also appears as [MO 468121a].

11.8.6 Functors Representably Fully Faithful on Cores

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.6.1.1 ▶ Functors Representably Fully Faithful on Cores

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **representably fully faithful on cores** if, for each $X \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition by F functor

$$F_*\colon \mathrm{Core}(\mathrm{Fun}(\mathcal{X}, \mathcal{C})) \to \mathrm{Core}(\mathrm{Fun}(\mathcal{X}, \mathcal{D}))$$

is fully faithful.

Remark 11.8.6.1.2 ▶ Unwinding Definition 11.8.6.1.1

In detail, a functor $F\colon \mathcal{C} \to \mathcal{D}$ is **representably fully faithful on cores** if it satisfies the conditions in Remarks 11.8.4.1.2 and 11.8.5.1.2, i.e.:

1. For all diagrams of the form

$$\begin{tikzcd} \mathcal{X} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\Downarrow\beta"] & \mathcal{C} \arrow[r, "F"] & \mathcal{D}, \end{tikzcd}$$

with α and β natural isomorphisms, if we have $\mathrm{id}_F \star \alpha = \mathrm{id}_F \star \beta$, then $\alpha = \beta$.

2. For each $\mathcal{X} \in \mathrm{Obj}(\mathsf{Cats})$ and each natural isomorphism

$$\beta \colon F \circ \phi \overset{\sim}{\Longrightarrow} F \circ \psi, \qquad \begin{tikzcd} \mathcal{X} \arrow[r, bend left, "F\circ\phi"] \arrow[r, bend right, "F\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & \mathcal{D} \end{tikzcd}$$

of $\mathcal{C}$, there exists a natural isomorphism

$$\alpha \colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} \mathcal{X} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{C} \end{tikzcd}$$

of $\mathcal{C}$ such that we have an equality

$$\begin{tikzcd} \mathcal{X} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{C} \arrow[r, "F"] & \mathcal{D} \end{tikzcd} = \begin{tikzcd} \mathcal{X} \arrow[r, bend left, "F\circ\phi"] \arrow[r, bend right, "F\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & \mathcal{D} \end{tikzcd}$$

of pasting diagrams in Cats_2, i.e. such that we have

$$\beta = \mathrm{id}_F \star \alpha.$$

Question 11.8.6.1.3 ▶ Characterisation of Functors Representably Fully Faithful on Cores

Is there a characterisation of functors representably fully faithful on cores?

11.8.7 Functors Corepresentably Faithful on Cores

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.7.1.1 ► Functors Corepresentably Faithful on Cores

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **corepresentably faithful on cores** if, for each $X \in \mathrm{Obj}(\mathsf{Cats})$, the postcomposition by F functor

$$F_*\colon \mathrm{Core}(\mathrm{Fun}(\mathcal{X}, \mathcal{C})) \to \mathrm{Core}(\mathrm{Fun}(\mathcal{X}, \mathcal{D}))$$

is faithful.

Remark 11.8.7.1.2 ► Unwinding Definition 11.8.7.1.1

In detail, a functor $F\colon \mathcal{C} \to \mathcal{D}$ is **corepresentably faithful on cores** if, given a diagram of the form

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] & \mathcal{D} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\beta"] & \mathcal{X}, \end{tikzcd}$$

if α and β are natural isomorphisms and we have

$$\alpha \star \mathrm{id}_F = \beta \star \mathrm{id}_F,$$

then $\alpha = \beta$.

Question 11.8.7.1.3 ► Characterisation of Functors Corepresentably Faithful on Cores

Is there a characterisation of functors corepresentably faithful on cores?

11.8.8 Functors Corepresentably Full on Cores

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.8.1.1 ▶ Functors Corepresentably Full on Cores

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **corepresentably full on cores** if, for each $X \in \mathrm{Obj}(\mathrm{Cats})$, the postcomposition by F functor

$$F_*\colon \mathrm{Core}(\mathrm{Fun}(\mathcal{X},\mathcal{C})) \to \mathrm{Core}(\mathrm{Fun}(\mathcal{X},\mathcal{D}))$$

is full.

Remark 11.8.8.1.2 ▶ Unwinding Definition 11.8.8.1.1

In detail, a functor $F\colon \mathcal{C} \to \mathcal{D}$ is **corepresentably full on cores** if, for each $\mathcal{X} \in \mathrm{Obj}(\mathrm{Cats})$ and each natural isomorphism

$$\beta\colon \phi \circ F \overset{\sim}{\Longrightarrow} \psi \circ F, \qquad \begin{tikzcd} \mathcal{C} \arrow[r, bend left=60, "\phi\circ F"{name=U}] \arrow[r, bend right=60, "\psi\circ F"'{name=D}] & \mathcal{X}, \arrow[from=U, to=D, Rightarrow, "\beta"] \end{tikzcd}$$

there exists a natural isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} \mathcal{D} \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & \mathcal{X} \arrow[from=U, to=D, Rightarrow, "\alpha"] \end{tikzcd}$$

such that we have an equality

$$\begin{tikzcd} \mathcal{X} \arrow[r, bend left=60, "\phi"{name=U}] \arrow[r, bend right=60, "\psi"'{name=D}] & \mathcal{C} \arrow[r, "F"] & \mathcal{D} \arrow[from=U, to=D, Rightarrow, "\alpha"] \end{tikzcd} = \begin{tikzcd} \mathcal{X} \arrow[r, bend left=60, "F\circ\phi"{name=U}] \arrow[r, bend right=60, "F\circ\psi"'{name=D}] & \mathcal{D} \arrow[from=U, to=D, Rightarrow, "\beta"] \end{tikzcd}$$

of pasting diagrams in Cats_2, i.e. such that we have

$$\beta = \alpha \star \mathrm{id}_F\,.$$

Question 11.8.8.1.3 ▶ Characterisation of Functors Corepresentably Full on Cores

Is there a characterisation of functors corepresentably full on cores?
This question also appears as [MO 468121a].

11.8.9 Functors Corepresentably Fully Faithful on Cores

Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

Definition 11.8.9.1.1 ▶ Functors Corepresentably Fully Faithful on Cores

A functor $F\colon \mathcal{C} \to \mathcal{D}$ is **corepresentably fully faithful on cores** if, for each $X \in \mathrm{Obj}(\mathrm{Cats})$, the postcomposition by F functor

$$F_*\colon \mathrm{Core}(\mathrm{Fun}(\mathcal{X}, \mathcal{C})) \to \mathrm{Core}(\mathrm{Fun}(\mathcal{X}, \mathcal{D}))$$

is fully faithful.

Remark 11.8.9.1.2 ▶ Unwinding Definition 11.8.9.1.1

In detail, a functor $F\colon \mathcal{C} \to \mathcal{D}$ is **corepresentably fully faithful on cores** if it satisfies the conditions in Remarks 11.8.7.1.2 and 11.8.8.1.2, i.e.:

1. For all diagrams of the form

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] & \mathcal{D} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow\Downarrow\Downarrow\beta"] & \mathcal{X}, \end{tikzcd}$$

   if α and β are natural isomorphisms and we have

$$\alpha \star \mathrm{id}_F = \beta \star \mathrm{id}_F,$$

   then $\alpha = \beta$.

2. For each $\mathcal{X} \in \mathrm{Obj}(\mathrm{Cats})$ and each natural isomorphism

$$\beta\colon \phi \circ F \overset{\sim}{\Longrightarrow} \psi \circ F, \qquad \begin{tikzcd} \mathcal{C} \arrow[r, bend left, "\phi\circ F"] \arrow[r, bend right, "\psi\circ F"'] \arrow[r, phantom, "\beta\Downarrow"] & \mathcal{X}, \end{tikzcd}$$

   there exists a natural isomorphism

$$\alpha\colon \phi \overset{\sim}{\Longrightarrow} \psi, \qquad \begin{tikzcd} \mathcal{D} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{X} \end{tikzcd}$$

such that we have an equality

$$\begin{tikzcd} \mathcal{X} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{C} \arrow[r, "F"] & \mathcal{D} \end{tikzcd} = \begin{tikzcd} \mathcal{X} \arrow[r, bend left, "F\circ\phi"] \arrow[r, bend right, "F\circ\psi"'] \arrow[r, phantom, "\beta\Downarrow"] & \mathcal{D} \end{tikzcd}$$

of pasting diagrams in Cats_2, i.e. such that we have

$$\beta = \alpha \star \mathrm{id}_F\,.$$

Question 11.8.9.1.3 ▶ Characterisation of Functors Corepresentably Fully Faithful on Cores

Is there a characterisation of functors corepresentably fully faithful on cores?

11.9 Natural Transformations

11.9.1 Transformations

Let $\mathcal{C}$ and $\mathcal{D}$ be categories and let $F, G\colon \mathcal{C} \rightrightarrows \mathcal{D}$ be functors.

Definition 11.9.1.1.1 ▶ Transformations

A **transformation**[1] $\alpha\colon F \Rightarrow G$ **from** F **to** G is a collection

$$\{\alpha_A\colon F(A) \to G(A)\}_{A\in\mathrm{Obj}(\mathcal{C})}$$

of morphisms of $\mathcal{D}$.

[1] *Further Terminology:* Also called an **unnatural transformation** for emphasis.

Notation 11.9.1.1.2 ▶ The Set of Transformations Between Two Functors

We write $\mathrm{Trans}(F, G)$ for the set of transformations from F to G.

Remark 11.9.1.1.3 ▶ The Set of Transformations as a Product

We have an isomorphism

$$\mathrm{Trans}(F, G) \cong \prod_{A \in \mathcal{C}} \mathrm{Hom}_{\mathcal{D}}(F_A, G_A).$$

Proof 11.9.1.1.4 ▶ Proof of Remark 11.9.1.1.3

Clear. ∎

11.9.2 Natural Transformations

Let $\mathcal{C}$ and $\mathcal{D}$ be categories and $F, G \colon \mathcal{C} \rightrightarrows \mathcal{D}$ be functors.

Definition 11.9.2.1.1 ▶ Natural Transformations

A **natural transformation** $\alpha \colon F \Rightarrow G$ **from** F **to** G is a transformation

$$\{\alpha_A \colon F(A) \to G(A)\}_{A \in \mathrm{Obj}(\mathcal{C})}$$

from F to G such that, for each morphism $f \colon A \to B$ of $\mathcal{C}$, the diagram

$$\begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[d, "\alpha_A"'] & F(B) \arrow[d, "\alpha_B"] \\ G(A) \arrow[r, "G(f)"'] & G(B) \end{tikzcd}$$

commutes.

Remark 11.9.2.1.2 ▶ Further Terminology and Notation for Natural Transformations

Let $\alpha \colon F \Rightarrow G$ be a natural transformation.

1. For each $A \in \mathrm{Obj}(\mathcal{C})$, the morphism $\alpha_A \colon F_A \to G_A$ is called the **component of** α **at** A.

2. We denote natural transformations such as α in diagrams as

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "G"'{name=D}] \arrow[from=U, to=D, Rightarrow, "\alpha"] & \mathcal{D}. \end{tikzcd}$$

NOTATION 11.9.2.1.3 ► THE SET OF NATURAL TRANSFORMATIONS BETWEEN TWO FUNCTORS

We write $\mathrm{Nat}(F, G)$ for the set of natural transformations from F to G.

DEFINITION 11.9.2.1.4 ► EQUALITY OF NATURAL TRANSFORMATIONS

Two natural transformations $\alpha, \beta \colon F \Rightarrow G$ are **equal** if we have

$$\alpha_A = \beta_A$$

for each $A \in \mathrm{Obj}(\mathcal{C})$.

11.9.3 Examples of Natural Transformations

EXAMPLE 11.9.3.1.1 ► IDENTITY NATURAL TRANSFORMATIONS

The **identity natural transformation** $\mathrm{id}_F \colon F \Rightarrow F$ **of** F is the natural transformation consisting of the collection

$$\{(\mathrm{id}_F)_A \colon F(A) \to F(A)\}_{A \in \mathrm{Obj}(\mathcal{C})}$$

defined by

$$(\mathrm{id}_F)_A \overset{\text{def}}{=} \mathrm{id}_{F(A)}$$

for each $A \in \mathrm{Obj}(\mathcal{C})$.

Proof 11.9.3.1.2 ▶ Proof of Example 11.9.3.1.1

The naturality condition for id_F is the requirement that, for each morphism $f\colon A \to B$ of $\mathcal{C}$, the diagram

$$\begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[d, "\mathrm{id}_{F(A)}"'] & F(B) \arrow[d, "\mathrm{id}_{F(B)}"] \\ F(A) \arrow[r, "F(f)"'] & F(B) \end{tikzcd}$$

commutes. This follows from unitality of the composition of $\mathcal{D}$, as we have

$$\begin{aligned} F(f) \circ \mathrm{id}_{F(A)} &= F(f) \\ &= \mathrm{id}_{F(B)} \circ F(f), \end{aligned}$$

where we have applied unitality twice. ∎

Example 11.9.3.1.3 ▶ Natural Transformations Between Morphisms of Monoids

Let A and B be monoids and let $f, g\colon A \rightrightarrows B$ be morphisms of monoids. Applying the delooping construction of **??**, we obtain functors $\mathrm{B}f, \mathrm{B}g\colon \mathrm{B}A \rightrightarrows \mathrm{B}B$. We then have

$$\mathrm{Nat}(\mathrm{B}f, \mathrm{B}g) \cong \left\{ b \in B \;\middle|\; \begin{array}{l} \text{for each } a \in A, \text{ we} \\ \text{have } bf(a) = g(a)b \end{array} \right\}.$$

Proof 11.9.3.1.4 ▶ Proof of Example 11.9.3.1.3

Unwinding the definitions in this case, we see that a transformation α from $\mathrm{B}f$ to $\mathrm{B}g$ consists of a collection

$$\{\alpha_\bullet\colon \bullet \to \bullet\}_{\bullet \in \mathrm{Obj}(\mathrm{B}A)}$$

of morphisms of $\mathrm{B}B$ indexed by $\mathrm{Obj}(\mathrm{B}A)$. Since $\mathrm{Obj}(\mathrm{B}A) = \mathrm{pt}$ and the morphisms of $\mathrm{B}B$ are precisely the elements of B, it follows that α corresponds pre-

cisely to the data of an element $b \in B$. Now, a transformation $[b] \colon \mathsf{B}f \Rightarrow \mathsf{B}g$ is natural precisely if, for each $a \in \mathrm{Hom}_{\mathsf{B}A}(\bullet, \bullet) \overset{\text{def}}{=} A$, the diagram

$$\begin{tikzcd}
\mathsf{B}f(\bullet) \arrow[r, "\mathsf{B}f(a)"] \arrow[d, "{[b]_\bullet}"'] & \mathsf{B}f(\bullet) \arrow[d, "{[b]_\bullet}"] \\
\mathsf{B}g(\bullet) \arrow[r, "\mathsf{B}g(a)"'] & \mathsf{B}g(\bullet)
\end{tikzcd}$$

commutes. Unwinding the definitions, we see that this diagram is given by

$$\begin{tikzcd}
\bullet \arrow[r, "f(a)"] \arrow[d, "b"'] & \bullet \arrow[d, "b"] \\
\bullet \arrow[r, "g(a)"'] & \bullet,
\end{tikzcd}$$

and hence corresponds precisely to the condition $g(a)b = bf(a)$.

11.9.4 Vertical Composition of Natural Transformations

Definition 11.9.4.1.1 ▶ Vertical Composition of Natural Transformations

The **vertical composition** of two natural transformations $\alpha\colon F \Longrightarrow G$ and $\beta\colon G \Longrightarrow H$ as in the diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=60, "F"{name=U}] \arrow[r, "G"{name=M}] \arrow[r, bend right=60, "H"'{name=D}] & \mathcal{D} \arrow[from=U, to=M, Rightarrow, "\alpha"] \arrow[from=M, to=D, Rightarrow, "\beta"] \end{tikzcd}$$

is the natural transformation $\beta \circ \alpha\colon F \Longrightarrow H$ consisting of the collection

$$\left\{(\beta \circ \alpha)_A\colon F(A) \to H(A)\right\}_{A \in \mathrm{Obj}(\mathcal{C})}$$

with

$$(\beta \circ \alpha)_A \overset{\text{def}}{=} \beta_A \circ \alpha_A$$

for each $A \in \mathrm{Obj}(\mathcal{C})$.

Proof 11.9.4.1.2 ▶ Proof of Definition 11.9.4.1.1

The naturality condition for $\beta \circ \alpha$ is the requirement that the boundary of the diagram

$$\begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[d, "\alpha_A"'] \arrow[dr, phantom, "(1)"] & F(B) \arrow[d, "\alpha_B"] \\ G(A) \arrow[r, "G(f)"] \arrow[d, "\beta_A"'] \arrow[dr, phantom, "(2)"] & G(B) \arrow[d, "\beta_B"] \\ H(A) \arrow[r, "H(f)"'] & H(B) \end{tikzcd}$$

commutes. Since

- Subdiagram (1) commutes by the naturality of α.
- Subdiagram (2) commutes by the naturality of β.

so does the boundary diagram. Hence $\beta \circ \alpha$ is a natural transformation. ∎

Proposition 11.9.4.1.3 ► Properties of Vertical Composition of Natural Transformations

Let $\mathcal{C}$, $\mathcal{D}$, and $\mathcal{E}$ be categories.

1. *Functionality.* The assignment $(\beta, \alpha) \mapsto \beta \circ \alpha$ defines a function

$$\circ_{F,G,H} \colon \mathrm{Nat}(G,H) \times \mathrm{Nat}(F,G) \longrightarrow \mathrm{Nat}(F,H).$$

2. *Associativity.* Let $F, G, H, K \colon \mathcal{C} \overset{\rightrightarrows}{\rightrightarrows} \mathcal{D}$ be functors. The diagram

$$\begin{tikzcd}
 & \mathrm{Nat}(H,K) \times (\mathrm{Nat}(G,H) \times \mathrm{Nat}(F,G)) \arrow[dr, "\mathrm{id}_{\mathrm{Nat}(H,K)} \times \circ_{F,G,H}"] & \\
(\mathrm{Nat}(H,K) \times \mathrm{Nat}(G,H)) \times \mathrm{Nat}(F,G) \arrow[ur, dashed, "\alpha^{\mathsf{Sets}}_{\mathrm{Nat}(H,K),\mathrm{Nat}(G,H),\mathrm{Nat}(F,G)}", "\sim"'] \arrow[d, "\circ_{G,H,K} \times \mathrm{id}_{\mathrm{Nat}(F,G)}"'] & & \mathrm{Nat}(H,K) \times \mathrm{Nat}(F,H) \arrow[d, "\circ_{F,H,K}"] \\
\mathrm{Nat}(G,K) \times \mathrm{Nat}(F,G) \arrow[rr, "\circ_{F,G,K}"'] & & \mathrm{Nat}(F,K)
\end{tikzcd}$$

commutes, i.e. given natural transformations

$$F \overset{\alpha}{\Longrightarrow} G \overset{\beta}{\Longrightarrow} H \overset{\gamma}{\Longrightarrow} K,$$

we have

$$(\gamma \circ \beta) \circ \alpha = \gamma \circ (\beta \circ \alpha).$$

3. *Unitality.* Let $F, G \colon \mathcal{C} \rightrightarrows \mathcal{D}$ be functors.

   (a) *Left Unitality.* The diagram

$$\begin{tikzcd}
\mathrm{pt} \times \mathrm{Nat}(F,G) \arrow[d, "{[\mathrm{id}_G] \times \mathrm{id}_{\mathrm{Nat}(F,G)}}"'] \arrow[dr, dashed, "\lambda^{\mathsf{Sets}}_{\mathrm{Nat}(F,G)}", "\sim"'] & \\
\mathrm{Nat}(G,G) \times \mathrm{Nat}(F,G) \arrow[r, "\circ_{F,G,G}"'] & \mathrm{Nat}(F,G)
\end{tikzcd}$$

commutes, i.e. given a natural transformation $\alpha\colon F \Longrightarrow G$, we have

$$\mathrm{id}_G \circ \alpha = \alpha.$$

(b) *Right Unitality.* The diagram

$$\begin{tikzcd}
\mathrm{Nat}(F,G) \times \mathrm{pt} \arrow[d, "\mathrm{id}_{\mathrm{Nat}(F,G)} \times [\mathrm{id}_F]"'] \arrow[dr, dashed, "\rho^{\mathsf{Sets}}_{\mathrm{Nat}(F,G)}", "\sim"'] & \\
\mathrm{Nat}(F,G) \times \mathrm{Nat}(F,F) \arrow[r, "\circ^{\mathcal{C}}_{F,F,G}"'] & \mathrm{Nat}(F,G)
\end{tikzcd}$$

commutes, i.e. given a natural transformation $\alpha\colon F \Longrightarrow G$, we have

$$\alpha \circ \mathrm{id}_F = \alpha.$$

4. *Middle Four Exchange.* Let $F_1, F_2, F_3\colon \mathcal{C} \to \mathcal{D}$ and $G_1, G_2, G_3\colon \mathcal{D} \to \mathcal{E}$ be functors. The diagram

$$\begin{tikzcd}
(\mathrm{Nat}(G_2,G_3) \times \mathrm{Nat}(G_1,G_2)) \times (\mathrm{Nat}(F_2,F_3) \times \mathrm{Nat}(F_1,F_2)) \arrow[rr, leftrightarrow, dashed, "\mu_4", "\sim"'] \arrow[d, "\circ_{G_1,G_2,G_3} \times \circ_{F_1,F_2,F_3}"'] & & (\mathrm{Nat}(G_2,G_3) \times \mathrm{Nat}(F_2,F_3)) \times (\mathrm{Nat}(G_1,G_2) \times \mathrm{Nat}(F_1,F_2)) \arrow[d, "\star_{F_2,F_3,G_2,G_3} \times \star_{F_1,F_2,G_1,G_2}"] \\
\mathrm{Nat}(G_1,G_3) \times \mathrm{Nat}(F_1,F_3) \arrow[dr, "\star_{F_1,F_3,G_1,G_3}"'] & & \mathrm{Nat}(G_2 \circ F_2, G_3 \circ F_3) \times \mathrm{Nat}(G_1 \circ F_1, G_2 \circ F_2) \arrow[dl, "\circ_{G_1 \circ F_1, G_2 \circ F_2, G_3 \circ F_3}"] \\
& \mathrm{Nat}(G_1 \circ F_1, G_3 \circ F_3) &
\end{tikzcd}$$

commutes, i.e. given a diagram

$$\begin{tikzcd}
\mathcal{C} \arrow[r, bend left=60, "F_1"] \arrow[r, "F_2"] \arrow[r, bend right=60, "F_3"'] \arrow[r, phantom, shift left=3, "\alpha\Downarrow"] \arrow[r, phantom, shift right=3, "\alpha'\Downarrow"] & \mathcal{D} \arrow[r, bend left=60, "G_1"] \arrow[r, "G_2"] \arrow[r, bend right=60, "G_3"'] \arrow[r, phantom, shift left=3, "\beta\Downarrow"] \arrow[r, phantom, shift right=3, "\beta'\Downarrow"] & \mathcal{E}
\end{tikzcd}$$

in Cats_2, we have

$$(\beta' \star \alpha') \circ (\beta \star \alpha) = (\beta' \circ \beta) \star (\alpha' \circ \alpha).$$

Proof 11.9.4.1.4 ▶ Proof of Proposition 11.9.4.1.3

Item 1: Functionality

Clear.

Item 2: Associativity

Indeed, we have

$$\begin{aligned}((\gamma \circ \beta) \circ \alpha)_A &\overset{\text{def}}{=} (\gamma \circ \beta)_A \circ \alpha_A \\ &\overset{\text{def}}{=} (\gamma_A \circ \beta_A) \circ \alpha_A \\ &= \gamma_A \circ (\beta_A \circ \alpha_A) \\ &\overset{\text{def}}{=} \gamma_A \circ (\beta \circ \alpha)_A \\ &\overset{\text{def}}{=} (\gamma \circ (\beta \circ \alpha))_A\end{aligned}$$

for each $A \in \mathrm{Obj}(C)$, showing the desired equality.

Item 3: Unitality

We have

$$\begin{aligned}(\mathrm{id}_G \circ \alpha)_A &= \mathrm{id}_G \circ \alpha_A \\ &= \alpha_A, \\ (\alpha \circ \mathrm{id}_F)_A &= \alpha_A \circ \mathrm{id}_F \\ &= \alpha_A\end{aligned}$$

for each $A \in \mathrm{Obj}(C)$, showing the desired equality.

Item 4: Middle Four Exchange

This is proved in Item 4 of Proposition 11.9.5.1.4.

11.9.5 Horizontal Composition of Natural Transformations

Definition 11.9.5.1.1 ▶ Horizontal Composition of Natural Transformations

The **horizontal composition**[1,2] of two natural transformations $\alpha\colon F \Longrightarrow G$ and $\beta\colon H \Longrightarrow K$ as in the diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "G"'{name=D}] & \mathcal{D} \arrow[r, bend left, "H"{name=U'}] \arrow[r, bend right, "K"'{name=D'}] & \mathcal{E} \arrow[from=U, to=D, Rightarrow, "\alpha"] \arrow[from=U', to=D', Rightarrow, "\beta"] \end{tikzcd}$$

of α and β is the natural transformation

$$\beta \star \alpha\colon (H \circ F) \Longrightarrow (K \circ G),$$

as in the diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "H\circ F"{name=U}] \arrow[r, bend right, "K\circ G"'{name=D}] & \mathcal{E}, \arrow[from=U, to=D, Rightarrow, "\beta\star\alpha"] \end{tikzcd}$$

consisting of the collection

$$\left\{(\beta \star \alpha)_A\colon H(F(A)) \longrightarrow K(G(A))\right\}_{A\in\mathrm{Obj}(\mathcal{C})},$$

of morphisms of $\mathcal{E}$ with

$$\begin{aligned} (\beta \star \alpha)_A &\overset{\text{def}}{=} \beta_{G(A)} \circ H(\alpha_A) \\ &= K(\alpha_A) \circ \beta_{F(A)}, \end{aligned} \qquad \begin{tikzcd} H(F(A)) \arrow[r, "H(\alpha_A)"] \arrow[d, "\beta_{F(A)}"'] & H(G(A)) \arrow[d, "\beta_{G(A)}"] \\ K(F(A)) \arrow[r, "K(\alpha_A)"'] & K(G(A)). \end{tikzcd}$$

---

[1] *Further Terminology:* Also called the **Godement product** of α and β.

[2] Horizontal composition forms a map

$$\star_{(F,H),(G,K)}\colon \mathrm{Nat}(H,K) \times \mathrm{Nat}(F,G) \longrightarrow \mathrm{Nat}(H\circ F, K\circ G).$$

Proof 11.9.5.1.2 ▶ Proof of Definition 11.9.5.1.1

First, we claim that we indeed have

$$\beta_{G(A)} \circ H(\alpha_A) = K(\alpha_A) \circ \beta_{F(A)}, \qquad \begin{tikzcd} H(F(A)) \arrow[r, "H(\alpha_A)"] \arrow[d, "\beta_{F(A)}"'] & H(G(A)) \arrow[d, "\beta_{G(A)}"] \\ K(F(A)) \arrow[r, "K(\alpha_A)"'] & K(G(A)). \end{tikzcd}$$

This is, however, simply the naturality square for β applied to the morphism $\alpha_A \colon F(A) \longrightarrow G(A)$. Next, we check the naturality condition for $\beta \star \alpha$, which is the requirement that the boundary of the diagram

$$\begin{tikzcd} H(F(A)) \arrow[r, "H(F(f))"] \arrow[d, "H(\alpha_A)"'] \arrow[dr, phantom, "(1)"] & H(F(B)) \arrow[d, "H(\alpha_B)"] \\ H(G(A)) \arrow[r, "H(G(f))"] \arrow[d, "\beta_{G(A)}"'] \arrow[dr, phantom, "(2)"] & H(G(B)) \arrow[d, "\beta_{G(B)}"] \\ K(G(A)) \arrow[r, "K(G(f))"'] & K(G(B)) \end{tikzcd}$$

commutes. Since

- Subdiagram (1) commutes by the naturality of α.
- Subdiagram (2) commutes by the naturality of β.

so does the boundary diagram. Hence $\beta \circ \alpha$ is a natural transformation.[1]

[1] *Reference:* [Bor94, Proposition 1.3.4].

Definition 11.9.5.1.3 ► Whiskering of Functors With Natural Transformations

Let

$$\begin{tikzcd} \mathcal{X} \arrow[r, "F"] & \mathcal{C} \arrow[r, bend left, "\phi"] \arrow[r, bend right, "\psi"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{D} \arrow[r, "G"] & \mathcal{Y} \end{tikzcd}$$

be a diagram in Cats_2.

1. The **left whiskering of α with G** is the natural transformation[1]

$$\mathrm{id}_G \star \alpha \colon G \circ \phi \Longrightarrow G \circ \psi.$$

2. The **right whiskering of α with F** is the natural transformation[2]

$$\alpha \star \mathrm{id}_F \colon \phi \circ F \Longrightarrow \psi \circ F.$$

[1] *Further Notation*: Also written $G\alpha$ or $G \star \alpha$, although we won,t use either of these notations in this work.

[2] *Further Notation*: Also written αF or $\alpha \star F$, although we won,t use either of these notations in this work.

Proposition 11.9.5.1.4 ► Properties of Horizontal Composition of Natural Transformations

Let $\mathcal{C}$, $\mathcal{D}$, and $\mathcal{E}$ be categories.

1. *Functionality.* The assignment $(\beta, \alpha) \mapsto \beta \star \alpha$ defines a function

$$\star_{(F,G),(H,K)} \colon \mathrm{Nat}(H, K) \times \mathrm{Nat}(F, G) \to \mathrm{Nat}(H \circ F, K \circ G).$$

2. *Associativity.* Let

$$\begin{tikzcd} \mathcal{C} \arrow[r, shift left, "F_1"] \arrow[r, shift right, "G_1"'] & \mathcal{D} \arrow[r, shift left, "F_2"] \arrow[r, shift right, "G_2"'] & \mathcal{E} \arrow[r, shift left, "F_3"] \arrow[r, shift right, "G_3"'] & \mathcal{F} \end{tikzcd}$$

be a diagram in Cats_2. The diagram

$$\begin{tikzcd} \mathrm{Nat}(F_3, G_3) \times \mathrm{Nat}(F_2, G_2) \times \mathrm{Nat}(F_1, G_1) \arrow[r, "\star_{(F_2,G_2),(F_3,G_3)} \times \mathrm{id}"] \arrow[d, "\mathrm{id} \times \star_{(F_1,G_1),(F_2,G_2)}"'] & \mathrm{Nat}(F_3 \circ F_2, G_3 \circ G_2) \times \mathrm{Nat}(F_1, G_1) \arrow[d, "\star_{(F_3 \circ F_2),(G_3 \circ G_2,F_1,G_1)}"] \\ \mathrm{Nat}(F_3, G_3) \times \mathrm{Nat}(F_2 \circ F_1, G_2 \circ G_1) \arrow[r, "\star_{(F_2 \circ F_1),(G_2 \circ G_1,F_3,G_3)}"'] & \mathrm{Nat}(F_3 \circ F_2 \circ F_1, G_3 \circ G_2 \circ G_1) \end{tikzcd}$$

commutes, i.e. given natural transformations

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=60, "F_1", "G_1"'] \arrow[r, phantom, "\alpha\Downarrow"] & \mathcal{D} \arrow[r, bend left=60, "F_2", "G_2"'] \arrow[r, phantom, "\beta\Downarrow"] & \mathcal{E} \arrow[r, bend left=60, "F_3", "G_3"'] \arrow[r, phantom, "\gamma\Downarrow"] & \mathcal{F}, \end{tikzcd}$$

we have

$$(\gamma \star \beta) \star \alpha = \gamma \star (\beta \star \alpha).$$

3. *Interaction With Identities.* Let $F\colon \mathcal{C} \to \mathcal{D}$ and $G\colon \mathcal{D} \to \mathcal{E}$ be functors. The diagram

$$\begin{tikzcd} \mathrm{pt} \times \mathrm{pt} \arrow[r, "{[\mathrm{id}_G]\times[\mathrm{id}_F]}"] & \mathrm{Nat}(G,G) \times \mathrm{Nat}(F,F) \arrow[d, "\star_{(F,F),(G,G)}"] \\ \mathrm{pt} \arrow[u, dashed, "\wr"] \arrow[r, "{[\mathrm{id}_{G\circ F}]}"'] & \mathrm{Nat}(G\circ F, G\circ F) \end{tikzcd}$$

commutes, i.e. we have

$$\mathrm{id}_G \star \mathrm{id}_F = \mathrm{id}_{G\circ F}\,.$$

4. *Middle Four Exchange.* Let $F_1, F_2, F_3\colon \mathcal{C} \to \mathcal{D}$ and $G_1, G_2, G_3\colon \mathcal{D} \to \mathcal{E}$ be functors. The diagram

$$\begin{tikzcd} (\mathrm{Nat}(G_2,G_3) \times \mathrm{Nat}(G_1,G_2)) \times (\mathrm{Nat}(F_2,F_3) \times \mathrm{Nat}(F_1,F_2)) \arrow[rr, leftrightarrow, "\mu_4", "\sim"'] \arrow[d, "{\circ_{G_1,G_2,G_3}\times\circ_{F_1,F_2,F_3}}"'] & & (\mathrm{Nat}(G_2,G_3) \times \mathrm{Nat}(F_2,F_3)) \times (\mathrm{Nat}(G_1,G_2) \times \mathrm{Nat}(F_1,F_2)) \arrow[d, "{\star_{F_2,F_3,G_2,G_3}\times\star_{F_1,F_2,G_1,G_2}}"] \\ \mathrm{Nat}(G_1,G_3) \times \mathrm{Nat}(F_1,F_3) \arrow[dr, "{\star_{F_1,F_3,G_1,G_3}}"'] & & \mathrm{Nat}(G_2\circ F_2, G_3\circ F_3) \times \mathrm{Nat}(G_1\circ F_1, G_2\circ F_2) \arrow[dl, "{\circ_{G_1\circ F_1,G_2\circ F_2,G_3\circ F_3}}"] \\ & \mathrm{Nat}(G_1\circ F_1, G_3\circ F_3) & \end{tikzcd}$$

commutes, i.e. given a diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=60, "F_1"] \arrow[r, phantom, "\alpha\Downarrow"] \arrow[r, bend right=60, "F_3"'] \arrow[r, phantom, "\alpha'\Downarrow"] \arrow[r, "F_2"] & \mathcal{D} \arrow[r, bend left=60, "G_1"] \arrow[r, phantom, "\beta\Downarrow"] \arrow[r, bend right=60, "G_3"'] \arrow[r, phantom, "\beta'\Downarrow"] \arrow[r, "G_2"] & \mathcal{E} \end{tikzcd}$$

in Cats_2, we have

$$(\beta' \star \alpha') \circ (\beta \star \alpha) = (\beta' \circ \beta) \star (\alpha' \circ \alpha).$$

PROOF 11.9.5.1.5 ► PROOF OF PROPOSITION 11.9.5.1.4

Item 1: Functionality

Clear.

Item 2: Associativity

Omitted.

Item 3: Interaction With Identities

We have

$$\begin{aligned}
(\mathrm{id}_G \star \mathrm{id}_F)_A &\overset{\text{def}}{=} (\mathrm{id}_G)_{F_A} \circ G_{(\mathrm{id}_F)_A} \\
&\overset{\text{def}}{=} \mathrm{id}_{G_{F_A}} \circ G_{\mathrm{id}_{F_A}} \\
&= \mathrm{id}_{G_{F_A}} \circ \mathrm{id}_{G_{F_A}} \\
&= \mathrm{id}_{G_{F_A}} \\
&\overset{\text{def}}{=} (\mathrm{id}_{G \circ F})_A
\end{aligned}$$

for each $A \in \mathrm{Obj}(\mathcal{C})$, showing the desired equality.

Item 4: Middle Four Exchange

Let $A \in \mathrm{Obj}(\mathcal{C})$ and consider the diagram

$$\begin{tikzcd}
 & & G_1(F_3(A)) \arrow[dr, "\beta_{F_3(A)}"] & & \\
G_1(F_1(A)) \arrow[r, "G_1(\alpha_A)"] & G_1(F_2(A)) \arrow[ur, "G_1(\alpha'_A)"] \arrow[dr, "\beta_{F_2(A)}"'] \arrow[r, phantom, "(1)"] & & G_2(F_3(A)) \arrow[r, "\beta'_{F_3(A)}"] & G_3(F_3(A)). \\
 & & G_2(F_2(A)) \arrow[ur, "G_2(\alpha'_A)"'] & &
\end{tikzcd}$$

The top composition

$$\begin{tikzcd}
 & & G_1(F_3(A)) \arrow[dr, "\beta_{F_3(A)}"] & & \\
G_1(F_1(A)) \arrow[r, "G_1(\alpha_A)"] & G_1(F_2(A)) \arrow[ur, "G_1(\alpha'_A)"] \arrow[dr, "\beta_{F_2(A)}"] \arrow[r, phantom, "(1)"] & & G_2(F_3(A)) \arrow[r, "\beta'_{F_3(A)}"] & G_3(F_3(A)). \\
 & & G_2(F_2(A)) \arrow[ur, "G_2(\alpha'_A)"] & &
\end{tikzcd}$$

is given by $((\beta' \circ \beta) \star (\alpha' \circ \alpha))_A$, while the bottom composition

$$\begin{tikzcd}
 & & G_1(F_3(A)) \arrow[dr, "\beta_{F_3(A)}"] & & \\
G_1(F_1(A)) \arrow[r, "G_1(\alpha_A)"] & G_1(F_2(A)) \arrow[ur, "G_1(\alpha'_A)"] \arrow[dr, "\beta_{F_2(A)}"] \arrow[r, phantom, "(1)"] & & G_2(F_3(A)) \arrow[r, "\beta'_{F_3(A)}"] & G_3(F_3(A)). \\
 & & G_2(F_2(A)) \arrow[ur, "G_2(\alpha'_A)"] & &
\end{tikzcd}$$

is given by $((\beta' \star \alpha') \circ (\beta \star \alpha))_A$. Now, Subdiagram (1) corresponds to the naturality condition

$$G_2(\alpha'_A) \circ \beta_{F_2(A)} = \beta_{F_3}(A) \circ G_1(\alpha'_A), \qquad \begin{tikzcd}
G_1(F_2(A)) \arrow[r, "G_1(\alpha'_A)"] \arrow[d, "\beta_{F_2(A)}"'] & G_1(F_3(A)) \arrow[d, "\beta_{F_3(A)}"] \\
G_2(F_2(A)) \arrow[r, "G_2(\alpha'_A)"'] & G_2(F_3(A))
\end{tikzcd}$$

for $\beta\colon G_1 \Longrightarrow G_2$ at $\alpha'_A\colon F_2(A) \to F_3(A)$, and thus commutes. Thus we have

$$((\beta' \circ \beta) \star (\alpha' \circ \alpha))_A = ((\beta' \star \alpha') \circ (\beta \star \alpha))_A$$

for each $A \in \mathrm{Obj}(\mathcal{C})$ and therefore

$$(\beta' \star \alpha') \circ (\beta \star \alpha) = (\beta' \circ \beta) \star (\alpha' \circ \alpha).$$

This finishes the proof. ∎

11.9.6 Properties of Natural Transformations

Proposition 11.9.6.1.1 ► Natural Transformations as Categorical Homotopies

Let $F, G \colon \mathcal{C} \rightrightarrows \mathcal{D}$ be functors. The following data are equivalent:[1]

1. A natural transformation $\alpha \colon F \Longrightarrow G$.
2. A functor $[\alpha] \colon \mathcal{C} \to \mathcal{D}^{\mathbb{1}}$ filling the diagram

$$\begin{tikzcd}
 & \mathcal{D} \\
\mathcal{C} \arrow[ur, "F"] \arrow[r, "{[\alpha]}"] \arrow[dr, "G"'] & \mathcal{D}^{\mathbb{1}}. \arrow[u, two heads, "\mathrm{ev}_0"'] \arrow[d, two heads, "\mathrm{ev}_1"] \\
 & \mathcal{D}
\end{tikzcd}$$

3. A functor $[\alpha] \colon \mathcal{C} \times \mathbb{1} \to \mathcal{D}$ filling the diagram

$$\begin{tikzcd}
\mathcal{C} \arrow[dr, "F"] & \\
\mathcal{C} \times \mathbb{1} \arrow[u, two heads, "\mathrm{ev}_0"] \arrow[r, "{[\alpha]}"] \arrow[d, two heads, "\mathrm{ev}_1"'] & \mathcal{D}. \\
\mathcal{C} \arrow[ur, "G"'] &
\end{tikzcd}$$

[1] Taken from [MO 64365].

Proof 11.9.6.1.2 ▶ Proof of Proposition 11.9.6.1.1

From Item 1 to Item 2 and Back

We may identify $\mathcal{D}^{\mathbb{1}}$ with $\mathrm{Arr}(\mathcal{D})$. Given a natural transformation $\alpha\colon F \Longrightarrow G$, we have a functor

$$\begin{aligned}
[\alpha]\colon \mathcal{C} &\longrightarrow \mathcal{D}^{\mathbb{1}} \\
A &\longmapsto \alpha_A \\
(f\colon A \to B) &\longmapsto \left(\begin{tikzcd} F_A \arrow[r, "F_f"] \arrow[d, "\alpha_A"'] & F_B \arrow[d, "\alpha_B"] \\ G_A \arrow[r, "G_f"'] & G_B \end{tikzcd}\right)
\end{aligned}$$

making the diagram in Item 2 commute. Conversely, every such functor gives rise to a natural transformation from F to G, and these constructions are inverse to each other.

From Item 2 to Item 3 and Back

This follows from Item 3 of Proposition 11.10.1.1.2.

11.9.7 Natural Isomorphisms

Let $\mathcal{C}$ and $\mathcal{D}$ be categories and let $F, G\colon \mathcal{C} \rightrightarrows \mathcal{D}$ be functors.

Definition 11.9.7.1.1 ▶ Natural Isomorphisms

A natural transformation $\alpha\colon F \Longrightarrow G$ is a **natural isomorphism** if there exists a natural transformation $\alpha^{-1}\colon G \Longrightarrow F$ such that

$$\begin{aligned}
\alpha^{-1} \circ \alpha &= \mathrm{id}_F, \\
\alpha \circ \alpha^{-1} &= \mathrm{id}_G\,.
\end{aligned}$$

PROPOSITION 11.9.7.1.2 ► PROPERTIES OF NATURAL ISOMORPHISMS

Let $\alpha\colon F \Longrightarrow G$ be a natural transformation.

1. *Characterisations.* The following conditions are equivalent:

   (a) The natural transformation α is a natural isomorphism.

   (b) For each $A \in \mathrm{Obj}(C)$, the morphism $\alpha_A\colon F_A \longrightarrow G_A$ is an isomorphism.

2. *Componentwise Inverses of Natural Transformations Assemble Into Natural Transformations.* Let $\alpha^{-1}\colon G \Longrightarrow F$ be a transformation such that, for each $A \in \mathrm{Obj}(C)$, we have

$$\begin{aligned}\alpha_A^{-1} \circ \alpha_A &= \mathrm{id}_{F(A)},\\ \alpha_A \circ \alpha_A^{-1} &= \mathrm{id}_{G(A)}\,.\end{aligned}$$

   Then α^{-1} is a natural transformation.

PROOF 11.9.7.1.3 ► PROOF OF PROPOSITION 11.9.7.1.2

Item 1: Characterisations

The implication Item 1a $\Longrightarrow$ Item 1b is clear, whereas the implication Item 1b $\Longrightarrow$ Item 1a follows from Item 2.

Item 2: Componentwise Inverses of Natural Transformations Assemble Into

The naturality condition for α^{-1} corresponds to the commutativity of the diagram

$$\begin{tikzcd} G(A) \arrow[r, "G(f)"] \arrow[d, "\alpha_A^{-1}"'] & G(B) \arrow[d, "\alpha_B^{-1}"] \\ F(A) \arrow[r, "F(f)"'] & F(B) \end{tikzcd}$$

for each $A, B \in \mathrm{Obj}(\mathcal{C})$ and each $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Considering the diagram

$$\begin{tikzcd}
G(A) \arrow[r, "G(f)"] \arrow[d, "\alpha_A^{-1}"'] \arrow[dr, phantom, "(1)"] & G(B) \arrow[d, "\alpha_B^{-1}"] \\
F(A) \arrow[r, "F(f)"] \arrow[d, "\alpha_A"'] \arrow[dr, phantom, "(2)"] & F(B) \arrow[d, "\alpha_B"] \\
G(A) \arrow[r, "G(f)"'] & G(B),
\end{tikzcd}$$

where the boundary diagram as well as Subdiagram (2) commute, we have

$$\begin{aligned}
G(f) &= G(f) \circ \mathrm{id}_{G(A)} \\
&= G(f) \circ \alpha_A \circ \alpha_A^{-1} \\
&= \alpha_B \circ F(f) \circ \alpha_A^{-1}.
\end{aligned}$$

Postcomposing both sides with α_B^{-1}, we get

$$\begin{aligned}
\alpha_B^{-1} \circ G(f) &= \alpha_B^{-1} \circ \alpha_B \circ F(f) \circ \alpha_A^{-1} \\
&= \mathrm{id}_{F(B)} \circ F(f) \circ \alpha_A^{-1} \\
&= F(f) \circ \alpha_A^{-1},
\end{aligned}$$

which is the naturality condition we wanted to show. Thus α^{-1} is a natural transformation.

11.10 Categories of Categories

11.10.1 Functor Categories

Let $\mathcal{C}$ be a category and $\mathcal{D}$ be a small category.

Definition 11.10.1.1.1 ▶ Functor Categories

The **category of functors from $\mathcal{C}$ to $\mathcal{D}$**[1] is the category $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$[2] where

- *Objects.* The objects of $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$ are functors from $\mathcal{C}$ to $\mathcal{D}$.
- *Morphisms.* For each $F, G \in \mathsf{Obj}(\mathsf{Fun}(\mathcal{C}, \mathcal{D}))$, we have

$$\mathsf{Hom}_{\mathsf{Fun}(\mathcal{C},\mathcal{D})}(F, G) \stackrel{\text{def}}{=} \mathsf{Nat}(F, G).$$

- *Identities.* For each $F \in \mathsf{Obj}(\mathsf{Fun}(\mathcal{C}, \mathcal{D}))$, the unit map

$$\mathbb{1}_F^{\mathsf{Fun}(\mathcal{C},\mathcal{D})} \colon \mathsf{pt} \to \mathsf{Nat}(F, F)$$

  of $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$ at F is given by

$$\mathsf{id}_F^{\mathsf{Fun}(\mathcal{C},\mathcal{D})} \stackrel{\text{def}}{=} \mathsf{id}_F,$$

  where $\mathsf{id}_F \colon F \Longrightarrow F$ is the identity natural transformation of F of Example 11.9.3.1.1.

- *Composition.* For each $F, G, H \in \mathsf{Obj}(\mathsf{Fun}(\mathcal{C}, \mathcal{D}))$, the composition map

$$\circ_{F,G,H}^{\mathsf{Fun}(\mathcal{C},\mathcal{D})} \colon \mathsf{Nat}(G, H) \times \mathsf{Nat}(F, G) \to \mathsf{Nat}(F, H)$$

  of $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$ at (F, G, H) is given by

$$\beta \circ_{F,G,H}^{\mathsf{Fun}(\mathcal{C},\mathcal{D})} \alpha \stackrel{\text{def}}{=} \beta \circ \alpha,$$

  where $\beta \circ \alpha$ is the vertical composition of α and β of Item 1 of Proposition 11.9.4.1.3.

[1] *Further Terminology:* Also called the **functor category** $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$.
[2] *Further Notation:* Also written $\mathcal{D}^{\mathcal{C}}$ and $[\mathcal{C}, \mathcal{D}]$.

Proposition 11.10.1.1.2 ▶ Properties of Functor Categories

Let $\mathcal{C}$ and $\mathcal{D}$ be categories and let $F \colon \mathcal{C} \to \mathcal{D}$ be a functor.

1. *Functoriality.* The assignments $\mathcal{C}, \mathcal{D}, (\mathcal{C}, \mathcal{D}) \mapsto \mathsf{Fun}(\mathcal{C}, \mathcal{D})$ define

functors

$$\begin{aligned}
\mathrm{Fun}(\mathcal{C},-)\colon &\quad \mathrm{Cats} &&\to \mathrm{Cats},\\
\mathrm{Fun}(-,\mathcal{D})\colon &\quad \mathrm{Cats}^{\mathrm{op}} &&\to \mathrm{Cats},\\
\mathrm{Fun}(-_1,-_2)\colon &\quad \mathrm{Cats}^{\mathrm{op}}\times\mathrm{Cats} &&\to \mathrm{Cats}.
\end{aligned}$$

2. *2-Functoriality.* The assignments $\mathcal{C},\mathcal{D},(\mathcal{C},\mathcal{D})\mapsto \mathrm{Fun}(\mathcal{C},\mathcal{D})$ define 2-functors

$$\begin{aligned}
\mathrm{Fun}(\mathcal{C},-)\colon &\quad \mathrm{Cats}_2 &&\to \mathrm{Cats}_2,\\
\mathrm{Fun}(-,\mathcal{D})\colon &\quad \mathrm{Cats}_2^{\mathrm{op}} &&\to \mathrm{Cats}_2,\\
\mathrm{Fun}(-_1,-_2)\colon &\quad \mathrm{Cats}_2^{\mathrm{op}}\times\mathrm{Cats}_2 &&\to \mathrm{Cats}_2.
\end{aligned}$$

3. *Adjointness.* We have adjunctions

$$(\mathcal{C}\times - \dashv \mathrm{Fun}(\mathcal{C},-))\colon \quad \begin{tikzcd} \mathrm{Cats} \arrow[r, bend left, "\mathcal{C}\times-"] \arrow[r, phantom, "\perp"] & \mathrm{Cats}, \arrow[l, bend left, "\mathrm{Fun}(\mathcal{C},-)"] \end{tikzcd}$$

$$(-\times\mathcal{D} \dashv \mathrm{Fun}(\mathcal{D},-))\colon \quad \begin{tikzcd} \mathrm{Cats} \arrow[r, bend left, "-\times\mathcal{D}"] \arrow[r, phantom, "\perp"] & \mathrm{Cats}, \arrow[l, bend left, "\mathrm{Fun}(\mathcal{D},-)"] \end{tikzcd}$$

witnessed by bijections of sets

$$\begin{aligned}
\mathrm{Hom}_{\mathrm{Cats}}(\mathcal{C}\times\mathcal{D},\mathcal{E}) &\cong \mathrm{Hom}_{\mathrm{Cats}}(\mathcal{D},\mathrm{Fun}(\mathcal{C},\mathcal{E})),\\
\mathrm{Hom}_{\mathrm{Cats}}(\mathcal{C}\times\mathcal{D},\mathcal{E}) &\cong \mathrm{Hom}_{\mathrm{Cats}}(\mathcal{C},\mathrm{Fun}(\mathcal{D},\mathcal{E})),
\end{aligned}$$

natural in $\mathcal{C},\mathcal{D},\mathcal{E}\in\mathrm{Obj}(\mathrm{Cats})$.

4. *2-Adjointness.* We have 2-adjunctions

$$(\mathcal{C}\times - \dashv \mathrm{Fun}(\mathcal{C},-))\colon \quad \begin{tikzcd} \mathrm{Cats}_2 \arrow[r, bend left, "\mathcal{C}\times-"] \arrow[r, phantom, "\perp_2"] & \mathrm{Cats}_2, \arrow[l, bend left, "\mathrm{Fun}(\mathcal{C},-)"] \end{tikzcd}$$

$$(-\times\mathcal{D} \dashv \mathrm{Fun}(\mathcal{D},-))\colon \quad \begin{tikzcd} \mathrm{Cats}_2 \arrow[r, bend left, "-\times\mathcal{D}"] \arrow[r, phantom, "\perp_2"] & \mathrm{Cats}_2, \arrow[l, bend left, "\mathrm{Fun}(\mathcal{D},-)"] \end{tikzcd}$$

witnessed by isomorphisms of categories

$$\begin{aligned}\mathrm{Fun}(\mathcal{C}\times\mathcal{D},\mathcal{E}) &\cong \mathrm{Fun}(\mathcal{D},\mathrm{Fun}(\mathcal{C},\mathcal{E})),\\ \mathrm{Fun}(\mathcal{C}\times\mathcal{D},\mathcal{E}) &\cong \mathrm{Fun}(\mathcal{C},\mathrm{Fun}(\mathcal{D},\mathcal{E})),\end{aligned}$$

natural in $\mathcal{C},\mathcal{D},\mathcal{E}\in\mathrm{Obj}(\mathsf{Cats}_2)$.

5. *Interaction With Punctual Categories.* We have a canonical isomorphism of categories
$$\mathrm{Fun}(\mathrm{pt},\mathcal{C})\cong\mathcal{C},$$
natural in $\mathcal{C}\in\mathrm{Obj}(\mathsf{Cats})$.

6. *Objectwise Computation of Co/Limits.* Let
$$D\colon\mathcal{I}\to\mathrm{Fun}(\mathcal{C},\mathcal{D})$$
be a diagram in $\mathrm{Fun}(\mathcal{C},\mathcal{D})$. We have isomorphisms
$$\begin{aligned}\lim(D)_A &\cong \lim_{i\in\mathcal{I}}(D_i(A)),\\ \mathrm{colim}(D)_A &\cong \mathrm{colim}_{i\in\mathcal{I}}(D_i(A)),\end{aligned}$$
naturally in $A\in\mathrm{Obj}(\mathcal{C})$.

7. *Interaction With Co/Completeness.* If $\mathcal{E}$ is co/complete, then so is $\mathrm{Fun}(\mathcal{C},\mathcal{E})$.

8. *Monomorphisms and Epimorphisms.* Let $\alpha\colon F\Longrightarrow G$ be a morphism of $\mathrm{Fun}(\mathcal{C},\mathcal{D})$. The following conditions are equivalent:

   (a) The natural transformation
   $$\alpha\colon F\Longrightarrow G$$
   is a monomorphism (resp. epimorphism) in $\mathrm{Fun}(\mathcal{C},\mathcal{D})$.

   (b) For each $A\in\mathrm{Obj}(\mathcal{C})$, the morphism
   $$\alpha_A\colon F_A\to G_A$$
   is a monomorphism (resp. epimorphism) in $\mathcal{D}$.

Proof 11.10.1.1.3 ▶ Proof of Proposition 11.10.1.1.2

Item 1: Functoriality

Omitted.

Item 2: 2-Functoriality

Omitted.

Item 3: Adjointness

Omitted.

Item 4: 2-Adjointness

Omitted.

Item 5: Interaction With Punctual Categories

Omitted.

Item 6: Objectwise Computation of Co/Limits

Omitted.

Item 7: Interaction With Co/Completeness

This follows from **??**.

Item 8: Monomorphisms and Epimorphisms

Omitted. ∎

11.10.2 The Category of Categories and Functors

Definition 11.10.2.1.1 ▶ The Category of Categories and Functors

The **category of** (**small**) **categories and functors** is the category Cats where

- *Objects.* The objects of Cats are small categories.
- *Morphisms.* For each $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats})$, we have

$$\mathrm{Hom}_{\mathsf{Cats}}(\mathcal{C}, \mathcal{D}) \overset{\text{def}}{=} \mathrm{Obj}(\mathrm{Fun}(\mathcal{C}, \mathcal{D})).$$

- *Identities.* For each $\mathcal{C} \in \mathrm{Obj}(\mathsf{Cats})$, the unit map

$$\mathbb{1}_{\mathcal{C}}^{\mathsf{Cats}} \colon \mathrm{pt} \longrightarrow \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{C}, \mathcal{C})$$

of Cats at $\mathcal{C}$ is defined by

$$\mathrm{id}_{\mathcal{C}}^{\mathsf{Cats}} \overset{\text{def}}{=} \mathrm{id}_{\mathcal{C}},$$

where $\mathrm{id}_{\mathcal{C}} \colon \mathcal{C} \to \mathcal{C}$ is the identity functor of $\mathcal{C}$ of Example 11.5.1.1.4.

- *Composition.* For each $\mathcal{C}, \mathcal{D}, \mathcal{E} \in \mathrm{Obj}(\mathsf{Cats})$, the composition map

$$\circ_{\mathcal{C},\mathcal{D},\mathcal{E}}^{\mathsf{Cats}} \colon \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{D}, \mathcal{E}) \times \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{C}, \mathcal{D}) \to \mathrm{Hom}_{\mathsf{Cats}}(\mathcal{C}, \mathcal{E})$$

of Cats at $(\mathcal{C}, \mathcal{D}, \mathcal{E})$ is given by

$$G \circ_{\mathcal{C},\mathcal{D},\mathcal{E}}^{\mathsf{Cats}} F \overset{\text{def}}{=} G \circ F,$$

where $G \circ F \colon \mathcal{C} \to \mathcal{E}$ is the composition of F and G of Definition 11.5.1.1.6.

Proposition 11.10.2.1.2 ▶ Properties of the Category Cats

Let $\mathcal{C}$ be a category.

1. *Co/Completeness.* The category Cats is complete and cocomplete.
2. *Cartesian Monoidal Structure.* The quadruple $(\mathsf{Cats}, \times, \mathrm{pt}, \mathsf{Fun})$ is a Cartesian closed monoidal category.

Proof 11.10.2.1.3 ▶ Proof of Proposition 11.10.2.1.2

Item 1: Co/Completeness

Omitted.

Item 2: Cartesian Monoidal Structure

Omitted.

11.10.3 The 2-Category of Categories, Functors, and Natural Transformations

Definition 11.10.3.1.1 ▶ The 2-Category of Categories

The 2-**category of (small) categories, functors, and natural transformations** is the 2-category Cats_2 where

- *Objects.* The objects of Cats_2 are small categories.
- Hom-*Categories.* For each $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathsf{Cats}_2)$, we have

$$\mathrm{Hom}_{\mathsf{Cats}_2}(\mathcal{C}, \mathcal{D}) \stackrel{\text{def}}{=} \mathrm{Fun}(\mathcal{C}, \mathcal{D}).$$

- *Identities.* For each $\mathcal{C} \in \mathrm{Obj}(\mathsf{Cats}_2)$, the unit functor

$$\mathbb{1}^{\mathsf{Cats}_2}_{\mathcal{C}} \colon \mathrm{pt} \to \mathrm{Fun}(\mathcal{C}, \mathcal{C})$$

  of Cats_2 at $\mathcal{C}$ is the functor picking the identity functor $\mathrm{id}_{\mathcal{C}} \colon \mathcal{C} \to \mathcal{C}$ of $\mathcal{C}$.
- *Composition.* For each $\mathcal{C}, \mathcal{D}, \mathcal{E} \in \mathrm{Obj}(\mathsf{Cats}_2)$, the composition bifunctor

$$\circ^{\mathsf{Cats}_2}_{\mathcal{C},\mathcal{D},\mathcal{E}} \colon \mathrm{Hom}_{\mathsf{Cats}_2}(\mathcal{D}, \mathcal{E}) \times \mathrm{Hom}_{\mathsf{Cats}_2}(\mathcal{C}, \mathcal{D}) \to \mathrm{Hom}_{\mathsf{Cats}_2}(\mathcal{C}, \mathcal{E})$$

  of Cats_2 at $(\mathcal{C}, \mathcal{D}, \mathcal{E})$ is the functor where

  - *Action on Objects.* For each object $(G, F) \in \mathrm{Obj}\big(\mathrm{Hom}_{\mathsf{Cats}_2}(\mathcal{D}, \mathcal{E}) \times \mathrm{Hom}_{\mathsf{Cats}_2}(\mathcal{C}, \mathcal{D})\big)$, we have

$$\circ^{\mathsf{Cats}_2}_{\mathcal{C},\mathcal{D},\mathcal{E}}(G, F) \stackrel{\text{def}}{=} G \circ F.$$

  - *Action on Morphisms.* For each morphism $(\beta, \alpha) \colon (K, H) \Longrightarrow (G, F)$ of $\mathrm{Hom}_{\mathsf{Cats}_2}(\mathcal{D}, \mathcal{E}) \times \mathrm{Hom}_{\mathsf{Cats}_2}(\mathcal{C}, \mathcal{D})$, we have

$$\circ^{\mathsf{Cats}_2}_{\mathcal{C},\mathcal{D},\mathcal{E}}(\beta, \alpha) \stackrel{\text{def}}{=} \beta \star \alpha,$$

    where $\beta \star \alpha$ is the horizontal composition of α and β of Definition 11.9.5.1.1.

Let $\mathcal{C}$ be a category.

1. *2-Categorical Co/Completeness.* The 2-category Cats_2 is complete and cocomplete as a 2-category, having all 2-categorical and bicategorical co/limits.

Proof 11.10.3.1.3 ► Proof of Proposition 11.10.3.1.2

Item 1: Co/Completeness

Omitted. ∎

11.10.4 The Category of Groupoids

Definition 11.10.4.1.1 ► The Category of Small Groupoids

The **category of** (**small**) **groupoids** is the full subcategory Grpd of Cats spanned by the groupoids.

11.10.5 The 2-Category of Groupoids

Definition 11.10.5.1.1 ► The 2-Category of Small Groupoids

The **2-category of** (**small**) **groupoids** is the full sub-2-category Grpd_2 of Cats_2 spanned by the groupoids.

Appendices

A Other Chapters

Preliminaries

1. Introduction
2. A Guide to the Literature

Sets

3. Sets
4. Constructions With Sets
5. Monoidal Structures on the Category of Sets
6. Pointed Sets
7. Tensor Products of Pointed Sets

Relations

8. Relations
9. Constructions With Relations
10. Conditions on Relations

Categories

11. Categories

Monoidal Categories

12. Constructions With Monoidal Categories

Bicategories

13. Types of Morphisms in Bicategories

Extra Part

14. Notes

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